Statistics 2 lecture notes 2026–2027
Part I

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Course information

Online resources for this unit are available via Blackboard, where you can find practical information about the unit, tutorial sheets, problem class questions, help with R, and additional learning material. Please check these resources regularly.

Activities

There are many different types of activity for Statistics 2. There are

- Lectures, where we will cover new material and review important concepts.
- Computing labs, where you can ask questions about statistics or statistical computing.
- Tutorials, where you will receive helpful feedback on homework problems.
- Problems classes, where you will go through exam style questions.
- Maths Cafés, where you can discuss problem sheets with undergraduates who have taken the unit before.

For the tutorials, you should attend the tutorial session you are registered for.

Tutorial sheets

There are 7 tutorial sheets. This is a very good way to engage with the material. If you submit the homework you will receive a 1 and if you do not you will recieve a 0. This is because the grades are indicative of submission rather than a grade towards your final mark.

Programming in R

You should engage continuously with programming in R to gain a better understanding of statistical procedures, and practice your programming skills at the same time.

Computer Practicals

The computer practicals are assignments where you will apply concepts from the unit to analyze datasets using R. You are permitted to work alone or in a group of up to two other people. Although they may take some time, this time is often spent reviewing the relevant mathematical concepts and hence is a good way to keep up with the material.

Assessment

80% of the final mark: 2 hour 30 minute exam.

20% of the final mark: 2 computer practical assignments

There are three computer practicals, but the first does not count towards your final mark.

Discussion Board

An important and popular learning resource in past years has been the discussion board, which is a student-led activity. In particular, students can post questions on the board or email them to the relevant lecturer, who will post the question anonymously and a reply.

We expect all students to follow the unit discussion board on Blackboard, as this is the primary way to disseminate information to all students outside of lectures. You can subscribe to the discussion board so that you receive an email every time someone posts. There is also a discussion board where students can talk amongst themselves.

Part I

Estimation

1 Introduction, concepts and notation

This course is concerned with the mathematics underpinning scientific investigation of observed phenomena. The aim is to understand these phenomena and/or make decisions in the presence of *uncertainty*. Various sources of uncertainty are possible: measurements of some quantity are imperfect, our modelling of the phenomenon itself is imperfect, or there are some truly random aspects of the phenomenon. The statistical approach is to *quantify uncertainty using probability*, and involves describing observed data as realizations of random variables. Probability will allow us to take uncertainty into account in our scientific investigations, and to provide some measure of uncertainty about our scientific conclusions.

This entire unit develops further the statistical approach to analyzing data — using probability to model how the data was generated, and also to make appropriate conclusions — that was introduced in Statistics 1.

1.1 A very simple motivating example

Imagine conducting the following experiment. Flip a coin n times, and record the outcomes H (heads) or T (tails) for each flip. One may ask the following chain of questions:

1. If we flip the coin another time, are we more likely to observe heads or tails? Instinctively, assuming that each coin flip has an unknown probability $\theta \in [0,1]$ of resulting in the observation of H, common sense suggests to estimate θ as follows

$$\hat{\theta}_n = \frac{1}{n} \sharp \{\text{observed } H\text{s}\},$$

   (the observed frequency of heads) and to use $\hat{\theta}_n$ to determine the likelihood of observing a head if we toss the coin another time.

   (a) Here the uncertainty comes from the fact that we do not measure θ directly, but observe a finite number n of outcomes from an experiment which provides only partial information about θ.

   (b) If we repeat the experiment (each experiment consists of flipping the coin n times) m times, the estimates we will obtain for all the experiments, say $\hat{\theta}_n^i$ for $i = 1, \ldots, m$, will take a large number of different values.

   (c) For n finite there is therefore no hope to find the "true" value of θ, except maybe by pure luck.

2. Drawing any conclusion based only on the number $\hat{\theta}_n$ is clearly unsatisfactory as this number does not provide us with any information about how confident we are in this value. For example, our intuition tells us that $\hat{\theta}_{100}$ should be more accurate than $\hat{\theta}_{10}$ but neither of these values (called point estimates) provide us with this information.

   (a) One of the goals of this course will be to formalize this notion of uncertainty and to develop tools to quantify our confidence in our estimates.

(b) An important point is that we will supplement "point estimates" with ranges of possible values for θ which are such that we can be certain that the true value of θ will fall in the given range with a probability of our choosing (this is the notion of confidence intervals).

3. If we can repeat the experiment m times, and compute the estimates $\hat{\theta}_n^i$ for $i = 1, \ldots, m$, then we can get an idea of the "typical" values one obtains for n coin flips.

   (a) We expect the distribution of $\hat{\theta}_n^i$ to be spread for small values of n and rather concentrated around a value when n is large, which intuitively reflects two different degrees of uncertainty. This could form the basis for our assessment of uncertainty.

   (b) In numerous real world scenarios, $m = 1$ (we cannot repeat the experiment), and we therefore cannot use this approach. In this course we will develop tools which will allow us to circumvent this practical difficulty and infer the "spread" of estimates around the correct value even in situations where $m = 1$.

4. A natural question is whether the estimator $\hat{\theta}_n$ is the "best estimator" for θ one can hope for given n samples? It turns out that the answer is yes in this particular case and that this can be mathematically proved, at least for a natural performance metric.

5. In more complicated scenarios it may not be obvious to guess what a good estimator for a desired quantity is. For example, assume that you measure the outside temperature on a daily basis with a thermometer that cannot measure temperatures above 16 degrees: whenever $T \geq 16$ the thermometer returns 16 degrees. How do you estimate the average temperature and its variance for a given month, e.g. July? As we shall see, with some modelling, the maximum likelihood principle will provide us with a systematic way of computing estimators in complicated scenarios - it turns out that such maximum likelihood estimators also enjoy nice properties which will be covered in this course.

6. A natural question is whether the coin is a fair coin, that is we would like to test the hypothesis that $\theta = 1/2$? Answering this question is not that obvious.

   (a) When do we decide that $\theta = 1/2$? When $|\hat{\theta}_n - 1/2| \leq 1/10$, or $|\hat{\theta}_n - 1/2| \leq 1/100$ or $|\hat{\theta}_n - 1/2| \leq 1/1000$, etc.? The threshold value might be too large or too small, leading either to unjustified confidence in the conclusion that $\theta = 1/2$ or the systematic rejection of the hypothesis that this is a fair coin.

   (b) A principled way of deciding on the threshold consists of choosing it in a way which ensures that we make the right decision with a given probability (i.e. when indeed $\theta = 1/2$ we want to make sure that we decide that $\theta = 1/2$ with probability, say, 95%. This will be formalized when we consider hypothesis testing.

   (c) Intuitively with $n = 100$ tosses the threshold should be smaller than when $n = 10$ tosses: as we shall see there will be an interplay between our choice of probability of correct decision, the threshold and the number of observed samples.

Numerical illustration: we first "toss a coin" such that $\theta = 1/2$, first $n = 10$ times, and then $n = 100$ times.

```
x10 <- rbinom(10, 1, 0.5) # sample 10 Binomial(1, 0.5) rvs
x10
```

```
##  [1] 1 1 1 1 0 1 1 1 0 0
```

```
x100 <- rbinom(100, 1, 0.5) # sample 100 Binomial(1, 0.5) rvs
x100
```

```
##   [1] 0 1 1 1 1 0 1 1 0 1 1 0 0 0 0 0 1 0 0 1 1 0 0 0 0 1 1 0 1 0 0 1 1 1 1 0 0
##  [38] 0 0 0 1 0 0 1 0 1 1 1 0 0 1 1 1 1 0 1 1 0 1 1 1 0 1 0 0 1 0 0 0 0 0 1 1 0
##  [75] 0 0 1 0 1 0 0 0 0 1 1 0 1 0 1 0 0 0 0 0 1 1 1 1 1 0
```

Now we repeat the experiments above $m = 12$ times and report the calculated frequencies. One way of doing this in R consists of using a for loop.

```
m <- 12
theta.hat.10 <- rep(0,m)
for (i in 1:m) {
  theta.hat.10[i] <- mean(rbinom(10,1,0.5)) # average 10 Binomial(1, 0.5) rvs
}
theta.hat.10
```

```
##  [1] 0.7 0.5 0.3 0.6 0.5 0.4 0.5 0.4 0.4 0.6 0.6 0.6
```

```
hist(theta.hat.10, xlab=expression(hat(theta)[10]),
  main=expression("Histogram of " * hat(theta)[10]))

theta.hat.100 <- rep(0,m)
for (i in 1:m) {
  theta.hat.100[i] <- mean(rbinom(100, 1, 0.5)) # average 100 Binomial(1, 0.5) rvs
}
theta.hat.100
```

```
##  [1] 0.52 0.51 0.50 0.51 0.46 0.47 0.41 0.55 0.51 0.56 0.59 0.50
```

```
hist(theta.hat.100, xlab=expression(hat(theta)[100]),
  main=expression("Histogram of " * hat(theta)[100]))
```

The histogram of these estimates is presented in Figure 1.1. It is possible to replace

```
theta.hat.10 <- rep(0,m)
for (i in 1:m) {
  theta.hat.10[i] <- mean(rbinom(10, 1, 0.5))
}
```

with

```
theta.hat.10 <- sapply(1:m, function(i) mean(rbinom(10, 1, .5)))
```

If we did not know $\theta = 0.5$ we would be more certain that it is close to 0.5 in the second case than in the first. However in practice we cannot repeat the experiment and can only compute one number. One aim of the course will be to develop tools to assess uncertainty about estimates from $m = 1$ experiment only.

1.2 Probabilities, models and random variables.

A sample space Ω is a set of *outcomes*. An *event* is a subset of Ω. A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathcal{F}$ is a set of events, i.e. subsets of Ω, and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a function that assigns to

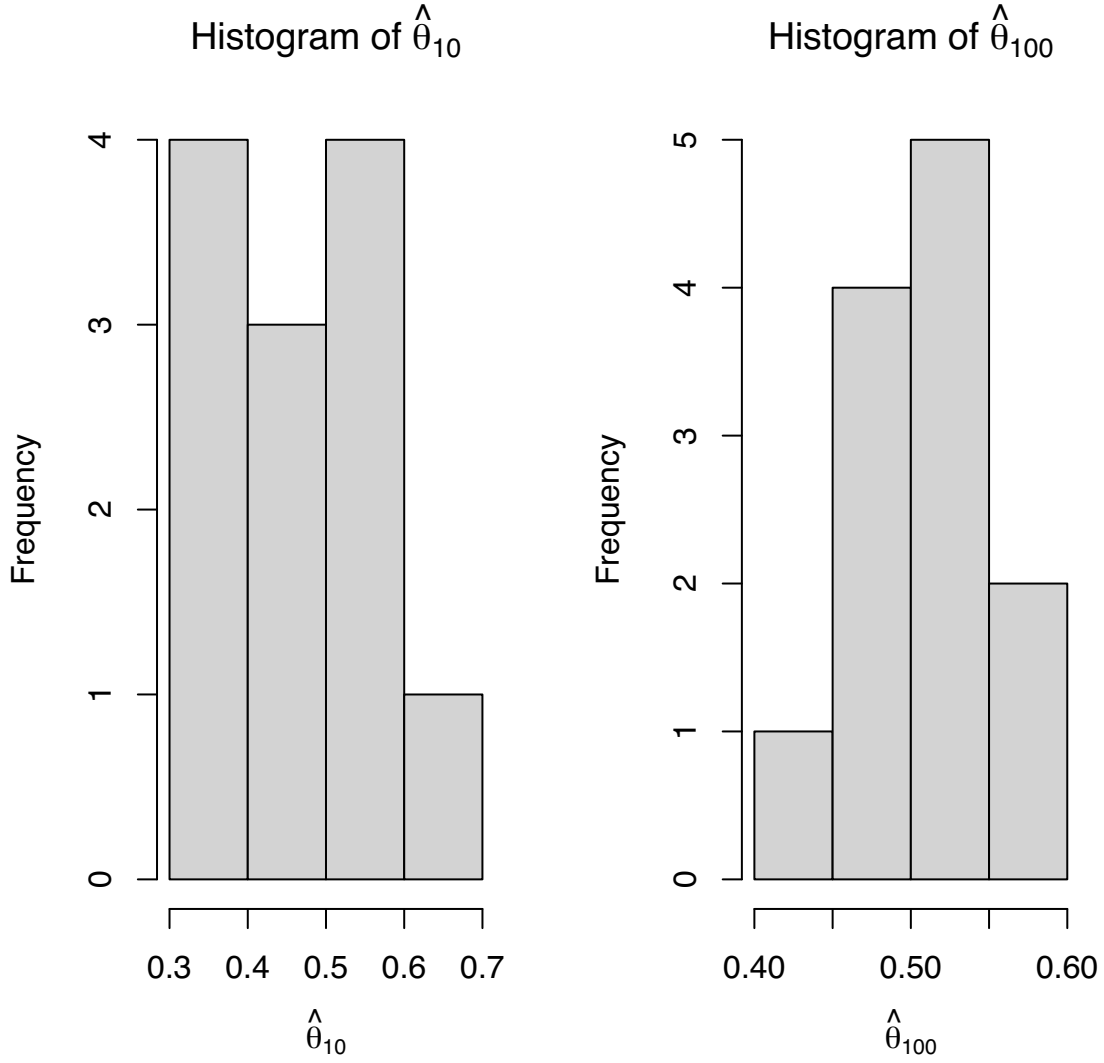


Figure 1.1: Histogram of the estimates for $m = 12$ experiments.

each event $A \in \mathcal{F}$ a *probability*.

There are basic requirements, although we will not dwell on them in this unit:

1. $\Omega \in \mathcal{F}$ and $\mathcal{F}$ is closed under complement and countably many unions/intersections.
2. $\mathbb{P}(\Omega) = 1$ and $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$ for disjoint $A_1, A_2, \ldots$ (countably additive).

Example 1.1. [6-sided die] Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \{A : A \subseteq \Omega\}$ and $\mathbb{P}(A) = |A|/6$.

As this is a unit in Statistics, we are interested in the case where different probabilities may be assigned to events. In particular, we consider a parameterized family of probability functions, which we will denote $\{\mathbb{P}(\cdot; \theta) : \theta \in \Theta\}$. We will typically refer to θ as a *statistical parameter*.

Example 1.2. [1 coin flip] Let $\Omega = \{H, T\}$, $\mathcal{F} = \{\{H\}, \{T\}, \{H, T\}, \varnothing\}$ and we can choose $\mathbb{P}(\{H\}; \theta) = \theta$ and $\mathbb{P}(\{T\}; \theta) = 1 - \theta$ for some $\theta \in \Theta = [0, 1]$. Naturally, $\mathbb{P}(\varnothing; \theta) = 0$ and $\mathbb{P}(\{H, T\}; \theta) = 1$.

Example 1.3. If we toss a coin twice, then we would consider $\Omega = \{(H, H), (H, T), (T, T), (T, H)\}$, and $\mathcal{F}$ consisting of all subsets of Ω, with 16 possible events. An example of an event is $\{(H, H), (H, T)\} \in \mathcal{F}$, which is the event that the first toss is a head.

Definition 1.1. A real-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ is[1] a function $X : \Omega \to \mathbb{R}$.

Probabilities associated with random variables are denoted using a particularly convenient shorthand:

$$\mathbb{P}(X \in A) := \mathbb{P}\left(\{\omega : X(\omega) \in A\}\right), \qquad \mathbb{P}(X = x) := \mathbb{P}(\{\omega : X(\omega) = x\}),$$

for appropriate $A \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. In fact, this type of notation allows us to almost never mention $\mathcal{F}$ again (in this unit!). We rely on the existence of very general $(\Omega, \mathcal{F})$ that allow us to define directly $\mathbb{P}(X \in A)$ for all A of the form $(-\infty, x]$.

We will use the following notation throughout:

- X is a scalar random variable (capitals are used for random variables).
- $\mathbf{X}$ is a vector consisting of scalar random variables, $X_1, X_2, \ldots, X_n$ for some $n \in \mathbb{N} = \{1, 2, \ldots\}$. Sometimes we may use $\mathbf{X}_n$ to emphasize the size of the vector.
- x will be used for a particular realization of X and similarly $\mathbf{x}$ will be used for realizations of $\mathbf{X}$.
- $\theta \in \Theta \subseteq \mathbb{R}^{d_\theta}$, $(d_\theta < \infty)$ will be used to denote a scalar or vector parameter characterizing the probability distributions of the random variables considered.
- $\mathcal{X}$ may be used to denote the support of X, satisfying $\mathbb{P}(X \in \mathcal{X}; \theta) = 1$. More precisely, the support of X is the *smallest* closed set $\mathcal{X}$ such that $\mathbb{P}(X \in \mathcal{X}; \theta) = 1$, but you do not need to worry about this precise definition in this unit.

It will be convenient to use the following generic notation and make use of the notion of probability density for continuous random variables.

- **Discrete case**: assume that X takes its values in a discrete set, e.g. $\mathbb{Z}_+ = \{0, 1, 2, \ldots\}$. Then we will use the generic notation

$$f(x; \theta) := f_X(x; \theta) := \mathbb{P}(X = x; \theta)$$

[1] Note that X should have some additional properties, but this is not important in this unit. In particular, we assume that for any $x \in \mathbb{R}$, the set $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$.

and we will write $X \sim f(\cdot;\theta)$ to say that X is described by f for the given value of the parameter θ. For example, $X \sim \text{Poisson}(\lambda)$ for $\lambda > 0$ will mean that

$$\mathbb{P}(X = x;\lambda) = f(x;\lambda) = f_X(x;\lambda) = \mathbb{I}\{x \in \mathbb{Z}_+\}e^{-\lambda}\frac{\lambda^x}{x!},$$

where for a set A, the indicator function $x \mapsto \mathbb{I}\{x \in A\}$ takes the value 1 if $x \in A$ and 0 otherwise. The cumulative distribution function (CDF, c.d.f. or cdf) is

$$F_X(x;\theta) := \mathbb{P}(X \leq x;\theta).$$

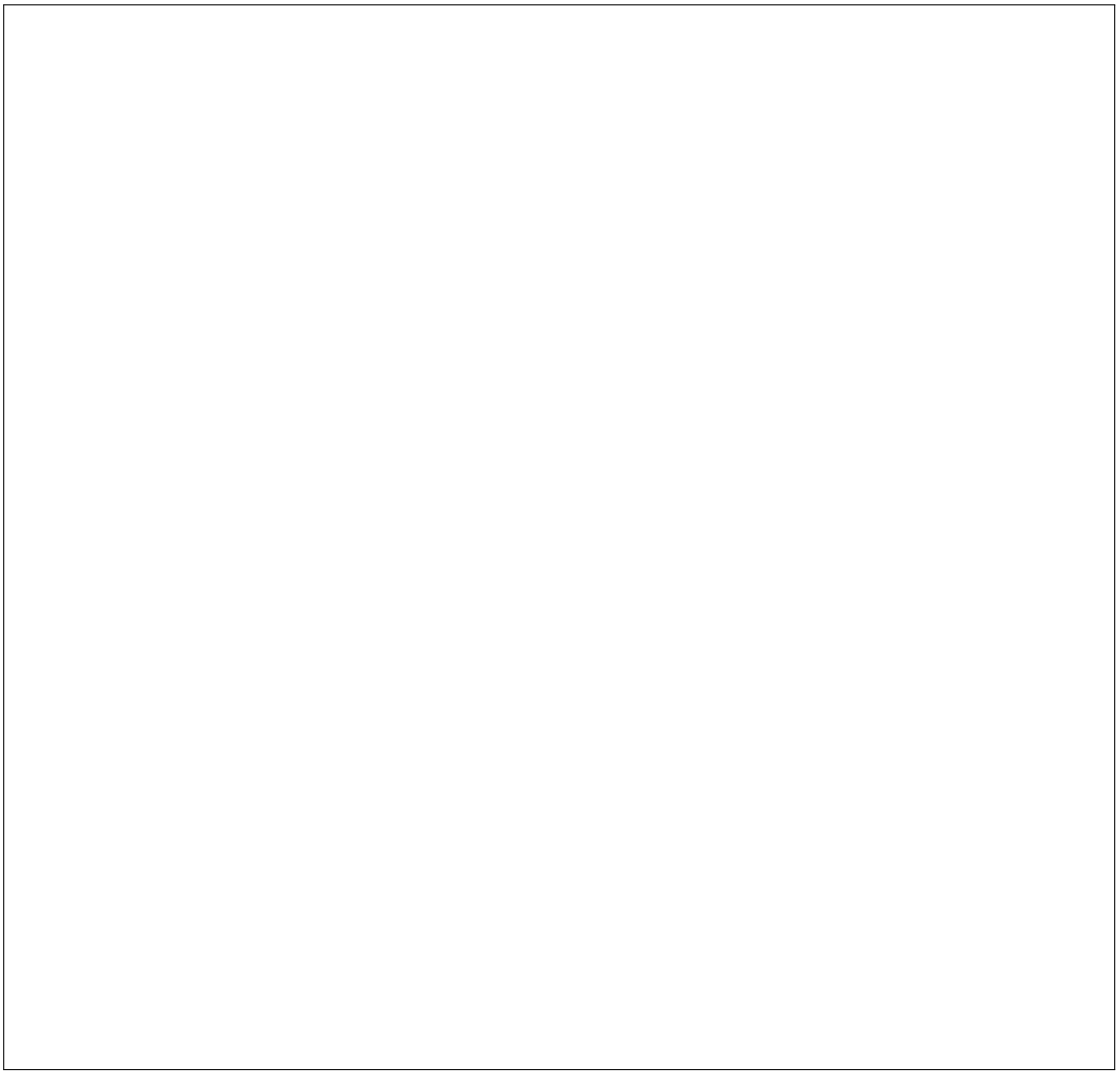
- **Continuous case**: in the continuous case we will make use of the probability density function (PDF, p.d.f., or pdf) of the random variable X which, when defined, is a non-negative function $f(\cdot;\theta) : \mathbb{R} \to [0,\infty)$ such that for any $a, b \in \mathbb{R}$ with $a \leq b$,

$$\int_a^b f(x;\theta)\mathrm{d}x = \mathbb{P}\left(X \in (a,b];\theta\right),$$

and we will also say in this case that $X \sim f(\cdot;\theta)$. In most scenarios encountered in the course the PDF will be defined. If f_X is given, it can be integrated to find $F_X(x;\theta)$. If F_X is given, $f_X(x;\theta)$ can be obtained by differentiating $F_X(x;\theta)$ [w.r.t. x] for a fixed $\theta \in \Theta$. This follows from

$$F_X(x;\theta) = \int_{-\infty}^x f(t;\theta)\mathrm{d}t \Rightarrow F_X'(x;\theta) = \frac{d}{dx}F_X(x;\theta) = f_X(x;\theta).$$

For example if $X \sim \text{Exponential}(\lambda)$ then $f(x;\theta = \lambda) = \mathbb{I}\{x \geq 0\}\lambda\exp(-\lambda x)$, so



The reason for introducing densities is that the probability any given realization of a continuous random variable is 0, so we need something else in order to say anything useful.

For vectors we will emphasize the size of the vector by writing $\mathbf{X} \sim f_n(\cdot;\theta)$ but in the particular case where the components of $\mathbf{X}$ are independent and identically distributed (i.i.d.) we use the notation

$$\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta),$$

to mean that the PDF of $\mathbf{X}$ is

$$f_n(\mathbf{x};\theta) = \prod_{i=1}^{n} f(x_i;\theta).$$

Throughout this course θ will always be finite-dimensional; this means we are talking about "parametric" statistics. If θ was not finite-dimensional one could talk about non-parametric statistics.

Functions of random variables

It is straightforward to consider functions of random variables. Although formally this could be written $Y = g \circ X$, since X is a function, we also often write $Y = g(X)$. One can find, e.g., the CDF of Y by

$F_Y(y) = \mathbb{P}(g(X) \leq y)$. Note that finding the mean of Y does not necessarily require finding the CDF of Y.

Expectation, variance, covariance

The expectation of a random variable is its mean, and is denoted by $\mathbb{E}(X)$. When X is a discrete random variable,

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x f_X(x),$$

and when X is a continuous real-valued random variable,

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x)\, dx.$$

We can easily represent the expectation of a function of a random variable. For a function $g : \mathbb{R} \to \mathbb{R}$, in the discrete case,

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{X}} g(x) f_X(x),$$

and in the continuous case,

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x)\, dx.$$

Expectation is a linear operator:

$$\mathbb{E}[aX + b] = \int_{\mathbb{R}} (ax + b)\, f_X(x) dx = a \int_{\mathbb{R}} x f_X(x) dx + b = a\mathbb{E}[X] + b.$$

The variance of a random variable is defined as $\mathrm{var}(X) := \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$, and satisfies

$$\mathrm{var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}\left[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2\right] = \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2.$$

The nth moment of X is $\mathbb{E}[X^n]$.

The covariance of two random variables defined on the same probability space is

$$\mathrm{cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Observe that $\mathrm{cov}(X, X) = \mathrm{var}(X)$. You should be able to derive that if X and Y are independent then $\mathrm{cov}(X, Y) = 0$, and also that for constants a, b,

$$\mathrm{var}(aX + bY) = a^2 \mathrm{var}(X) + b^2 \mathrm{var}(Y) + 2ab\mathrm{cov}(X, Y).$$

It follows that if $X_1, \ldots, X_n$ are independent random variables and $a_1, \ldots, a_n$ are constants,

$$\mathrm{var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \mathrm{var}(X_i).$$

Independent random variables

The random variables $X_1, \ldots, X_n$ are independent if and only if for any $a_1, \ldots, a_n \in \mathbb{R}$,

$$\mathbb{P}(X_1 \leq a_1, \ldots, X_n \leq a_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq a_i).$$

If the random variables $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.), then

$$\mathbb{P}(X_1 \leq a_1, \ldots, X_n \leq a_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq a_i) = \prod_{i=1}^{n} \mathbb{P}(X_1 \leq a_i).$$

Similarly random variables $X_1, \ldots, X_n$ with a joint PMF/PDF are independent if and only if the joint PMF/PDF factorizes as

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1}(x_1) \cdots f_{X_n}(x_n) = \prod_{i=1}^{n} f_{X_i}(x_i),$$

for any $\mathbf{x} \in \mathbb{R}^n$.

Dependence on a statistical parameter

We will use the notation $\mathbb{E}[X; \theta]$ and $\mathrm{var}(X; \theta)$ to emphasize dependence of an expectation or variance on θ. For example, in the continuous case,

$$\mathbb{E}\left[X; \theta\right] = \int_{\mathbb{R}} x f_X(x; \theta)\, dx.$$

1.3 Frequentist and subjective interpretations of probabilities

There are two dominant approaches to statistical inference, "frequentist" and "Bayesian" (named after Rev. T. Bayes) which differ in the way they interpret probabilities when it comes to modelling the world. We will focus on the frequentist approach for the first two parts of the unit and then turn to the Bayesian approach for the third and last part.

The frequentist approach relies on three postulates [see, e.g., Wasserman p. 175]

1. Probability refers to the limiting relative frequencies of events. Probabilities are objective properties of the world.
2. The parameters of the model are fixed constants, even though they are unknown. Since they are not random, we cannot make probabilistic statements about them.
3. Statistical procedures should be assessed (and designed) on their long-term frequency properties, that is as $n \to \infty$ or at least when n is "large" [e.g. 95% confidence intervals should contain the true parameter with a limiting frequency of 95%].

The Bayesian approach, in contrast, postulates that probabilities are simply a way of expressing our subjective beliefs given the evidence. Inference is based on specifying a prior distribution (encapsulating our beliefs) for the parameter before observing data, and updating our beliefs after observing the data. This amounts to treating the unknown parameter mathematically as if it were a random variable.

As you can imagine, there is fundamental disagreement between these two approaches, but both are widely used and widely useful.

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2 Method of moments estimators

Definition 2.1. (Moments). For $k \in \mathbb{N} = \{1, 2, 3, ...\}$, the kth moment of X is $\mathbb{E}[X^k]$.

For computations, it is convenient to use the fact that we can write the expectaion of a function of a random variable by intergating / summing that function against the PDF/PMF of that random variable. That is, for a continuous random variable X,

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f_X(x) \mathrm{d}x,$$

while for a discrete random variable

$$\mathbb{E}[X^k] = \sum_{x \in S_X} x^k p_X(x).$$

The first moment is the mean or expectation of X.

Definition 2.2. (Empirical moments). Let $X_1, ..., X_n$ be independent random variables with the same distributions as X. For $k \in \mathbb{N}$, the kth empirical moment is

$$M_k = \frac{1}{n} \sum_{i=1}^{n} X_i^k.$$

The realisation of the kth empirical moment is

$$m_k = \frac{1}{n} \sum_{i=1}^{n} x_i^k,$$

where x_i is a realisation of X_i for $i \in \{1, ..., n\}$.

The method of moments estimator is defined by equating theoretical moments with empirical moments.

Definition 2.3. (Methods of Moments estimator). Let $X_1, ..., X_n$ be independent random variables with statistical parameter $\theta \in \mathbb{R}^k$. The method of moments estimator $\hat{\theta}_{MoM}$ is the solution to the simultaneous equations

$$\begin{aligned} \mathbb{E}[X; \hat{\theta}_{MoM}] &= M_1 \\ \mathbb{E}[X^2; \hat{\theta}_{MoM}] &= M_2 \\ &\vdots \\ \mathbb{E}[X^k; \hat{\theta}_{MoM}] &= M_k \end{aligned}$$

If these definitions are not familar to you, do go back and look over last years material.

In this unit we will focus our attention more on a second type of estimator - the maximum likelihood estimator - that you also came across in Statistics 1.

3 The (observed) likelihood function

We define the notion of the (observed) likelihood function. The likelihood function is defined by taking into account observed data.

Throughout these notes log denotes the natural logarithm.

Definition 3.1. Assume that $\mathbf{X} \sim f_n(\cdot;\theta^*)$ for some unknown $\theta^* \in \Theta$ and that we have observed a particular realization $\mathbf{x}$. For this scenario,

1. an (observed) likelihood function is any function $L(\cdot;\mathbf{x}) : \Theta \to [0,\infty)$ proportional to the probability mass function / probability density (with positive constant of proportionality) of the observed realization $\mathbf{x}$, that is such that for some $C > 0$

$$L(\theta;\mathbf{x}) = C f_n(\mathbf{x};\theta)$$

   for all $\theta \in \Theta$.

2. a log-likelihood, $\ell(\theta;\mathbf{x})$ is any function $\ell(\cdot;\mathbf{x})$ such that $\ell(\theta;\mathbf{x}) = \log f_n(\mathbf{x};\theta) + C$ for some constant $C \in \mathbb{R}$.

Remark 3.1. As above, θ will usually be used for generic values of the parameter, while θ^* will be used for the true, unknown value of the parameter defining the distribution from which the observations are assumed to come from.

Remark 3.2. We will often talk about *the* likelihood or *the* log-likelihood since the constants C above turn out to be unimportant in what follows. Many authors will write $L(\theta;\mathbf{x}) := f_n(\mathbf{x};\theta)$ and $\ell(\theta;x) := \log f_n(\mathbf{x};\theta)$, which is also perfectly acceptable.

Remark 3.3. Notice the swap between the parameter and the observations. The likelihood and log-likelihood functions are functions of θ, defined in terms of an observed $\mathbf{x}$. The likelihood function is not a probability density function and does not sum/integrate to one in general.

Example 3.1. Consider the following situations

- assume that $X \sim \text{Geometric}(p)$, $\mathcal{X} = \mathbb{N}$, $\theta = p \in \Theta = (0,1)$ and for $x \geq 1$

$$L(p;x) \propto p(1-p)^{x-1} \quad ,$$

- assume that $\mathbf{X} \overset{\text{iid}}{\sim} \text{Geometric}(p)$, then provided that $x_i \geq 1$ for all $i \in \{1,\ldots,n\}$,

$$\begin{aligned} L(p;\mathbf{x}) &\propto f_n(\mathbf{x};p), \\ &= \prod_{i=1}^n f(x_i;p) = \prod_{i=1}^n p(1-p)^{x_i-1} \quad , \end{aligned}$$

- assume that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu,\sigma^2)$, $\theta = (\mu,\sigma^2) \in \Theta = \mathbb{R} \times \mathbb{R}_+$ and the likelihood is

$$L(\mu,\sigma^2;\mathbf{x}) \propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_i-\mu)^2\right) \quad .$$

Lemma 3.1 (Multidimensional transforms)**.** *Suppose* $\mathbf{X}$ *is a continuous random vector in* $\mathbb{R}^n$ *with PDF* $f_{\mathbf{X}}$ *and* g *is a continuously differentiable bijection from* $\mathbb{R}^n$ *to* $\mathbb{R}^n$ *with inverse* h*. Then* $\mathbf{Y} = g(\mathbf{X})$ *is a continuous random vector and its PDF is*

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(h(\mathbf{y}))J_h(\mathbf{y}),$$

where J_h is the absolute value of the determinant of the Jacobian matrix

$$J_h = \left|\det\left(\frac{\partial h}{\partial \mathbf{y}}\right)\right| = \left|\det\begin{pmatrix} \frac{\partial h_1}{\partial y_1} & \cdots & \frac{\partial h_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{\partial y_1} & \cdots & \frac{\partial h_n}{\partial y_n} \end{pmatrix}\right|.$$

Proposition 3.1 (Invariance of the likelihood function by bijective transformation of the observations independent of θ)**.** *For g not depending on θ, bijective (and continuously differentiable in the continuous case), let $\mathbf{Y} = g(\mathbf{X})$. Then $\mathbf{Y}$ is a random vector/variable with PDF/PMF*

$$f_{\mathbf{Y}}(\mathbf{y};\theta) \propto f_{\mathbf{X}}(g^{-1}(\mathbf{y});\theta) \quad .$$

Hence, if $\mathbf{y} = g(\mathbf{x})$ then $L_{\mathbf{Y}}(\theta;\mathbf{y}) \propto L_{\mathbf{X}}(\theta;\mathbf{x})$.

Proof.

Remark 3.4. If inference is conducted based solely on the likelihood function, then inferences will be unchanged by bijective (and continuously differentiable in the continuous case) transformations of the observations independent of θ.

Example 3.2. A non-negative random variable Y is said to be log-normal with parameters $\theta = (\mu, \sigma^2)$ if $Y = e^X$ where $X \sim \mathcal{N}(\mu, \sigma^2)$. From above, since $y = g(x) = \exp(x)$ satisfies the conditions of the proposition, we conclude that likelihood inference about the parameters $\theta = (\mu, \sigma^2)$ based on the realization $\mathbf{y} = (y_1, \ldots, y_n)$ of i.i.d. Log-normal(μ, σ^2) random variables is equivalent to likelihood inference for θ based on the transformed observations $\mathbf{x} = (x_1, \ldots, x_n) = (\log(y_1), \ldots, \log(y_n))$ of i.i.d. $\mathcal{N}(\mu, \sigma^2)$ random variables. If you would like to verify this directly, you can consider the PDFs

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\},$$

and

$$f_Y(y; \mu, \sigma^2) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2}(\ln(y)-\mu)^2\right\} \mathbb{I}(y > 0).$$

Exercise 3.1. Show without using Lemma 3.1 that if X is a continuous, real-valued random variable with PDF f_X and g is a strictly monotone, differentiable function then the PDF of $Y = g(X)$ is

$$f_Y(y) = \left|\frac{1}{g'(g^{-1}(y))}\right| f_X(g^{-1}(y)).$$

You may find it helpful to recall the fact that

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{g'(g^{-1}(y))}.$$

4 Maximum likelihood estimates - invariance property

4.1 Definition

Suppose that we have observed $\mathbf{x}$, assumed to be a realization of a random vector with PDF/PMF $f_n(\cdot;\theta^*)$ where θ^* is unknown and to be determined. A natural choice of estimate of the true θ^* consists of choosing $\hat{\theta}$ in such a way that it maximizes the probability/probability density of observing $\mathbf{x}$. Hence the name maximum likelihood estimate.

Definition 4.1. Assume that $\mathbf{X} \sim f_n(\cdot;\theta)$ and that we observe a realization $\mathbf{x}$ of $\mathbf{X}$. A maximum likelihood estimate (MLE) is a value $\hat{\theta} \in \Theta$ of the parameter such that for any $\theta \in \Theta$

$$f_n(\mathbf{x};\hat{\theta}) \geq f_n(\mathbf{x};\theta)$$

or equivalently

$$L(\hat{\theta};\mathbf{x}) \geq L(\theta;\mathbf{x})$$

or (since log is strictly increasing)

$$\ell(\hat{\theta};\mathbf{x}) \geq \ell(\theta;\mathbf{x}).$$

Remark 4.1. In general $\hat{\theta}(\mathbf{x}) := \arg\max_{\theta\in\Theta} L(\theta;\mathbf{x})$ is a set, that is the MLE may not be unique, and any $\hat{\theta} \in \hat{\theta}(\mathbf{x})$ is a MLE estimate.

Example 4.1 (Uniform with unknown range)**.** Consider $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{U}(0,\theta)$ for $\theta > 0$ (the uniform distribution on $[0,\theta]$, with density $\mathbb{I}\{x \in [0,\theta]\}/\theta$). The likelihood of θ given an observation $\mathbf{x}$ is proportional to

$$L(\theta;\mathbf{x}) \propto f_n(\mathbf{x};\theta) = \prod_{i=1}^{n} \frac{1}{\theta}\mathbb{I}\{x_i \in [0,\theta]\} = \frac{1}{\theta^n}\prod_{i=1}^{n}\mathbb{I}\{x_i \in [0,\theta]\}.$$

The first term is a decreasing function of θ, so θ should be taken as small as possible, but the product of indicator functions is 0 as soon as $\theta < \max\{x_1,\ldots,x_n\}$. So the MLE is precisely $\hat{\theta} = \max\{x_1,\ldots,x_n\}$.

Note that if we consider the probability density $\mathbb{I}\{x \in [0,\theta)\}/\theta$ (now $x = \theta$ has density zero but the density still corresponds to the same uniform probability distribution), then the MLE does not exist! We will avoid such contrived examples in what follows, but you should remember that the MLE does not necessarily exist.

Remark 4.2. In some situations (think of the normal distribution) θ might be a vector, i.e. have several components, for example, $\theta = (\theta_1,\theta_2) \in \Theta_1 \times \Theta_2$, and we assume that $\mathbf{x}$ gives a likelihood $L(\theta;\mathbf{x}) \propto f_n(\mathbf{x};\theta = (\theta_1,\theta_2))$. Assume that we are interested in inferring θ_1^* but not necessarily θ_2^*. In this case we will say that $\hat{\theta}_1$ is a MLE of θ_1 if there exists a MLE $\hat{\theta}$ of θ such that $\hat{\theta} = (\hat{\theta}_1,\hat{\theta}_2)$. For example assume that $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu,\sigma^2)$ and we are asked the question "what is the MLE of μ?". The above says that we should find the MLE $(\hat{\mu},\hat{\sigma}^2)$ of (μ,σ^2) and return $\hat{\mu}$.

Maximum likelihood estimators are intuitively sensible. Such estimators also enjoy numerous interesting theoretical properties.

4.2 Invariance property of MLE

Now we might wonder if inference about θ depends on the parameterization of the probability distribution involved. For example in the case of the exponential distribution we could choose either of the following parameterizations

$$f(x;\theta) = \mathbb{I}\{x \geq 0\}\theta\exp(-\theta x),$$

or

$$\tilde{f}(x;\tau) = \mathbb{I}\{x \geq 0\}\tau^{-1}\exp(-x/\tau),$$

that is $\tau = g(\theta) = 1/\theta$ with here $g : (0,\infty) \to (0,\infty)$ a bijection in the present case and hence g^{-1} is well defined. It would be sensible to have the property that if $\hat{\theta}$ is a MLE of θ^* then $\hat{\tau} := g(\hat{\theta})$ is also a MLE of τ^*.

Remark 4.3. One cannot use the same letters for the two densities above since for a given x the likelihood functions $\theta\exp(-\theta x)$ and $1/\tau \times \exp(-x/\tau)$ are not identical (draw them as a function of θ and τ if you are not convinced!), hence the distinction between f and $\tilde{f}$.

Theorem 4.1 (Invariance under bijective reparameterization)**.** *Let $g : \Theta \to G$ be a bijective transformation of the statistical parameter θ. Let $\mathbf{X} \sim f(\cdot;\theta) = \tilde{f}(\cdot; g(\theta))$ for some θ, and $\mathbf{x}$ an observation/realization of $\mathbf{X}$. If $\hat{\theta}$ is an MLE of θ then $\hat{\tau} = g(\hat{\theta})$ is an MLE of τ.*

Proof.

There may be reparameterizations that are not bijective, in particular if different values of θ map to the same value of τ. In this case it is helpful to consider the *induced likelihood*:

$$L^*(\tau; \mathbf{x}) := \max_{\theta \in G_\tau} L(\theta; \mathbf{x}), \qquad G_\tau := \{\theta : g(\theta) = \tau\},$$

which reduces to the likelihood when g is bijective. Now, if $\hat{\theta}$ is an MLE for θ then we can see that $\hat{\tau} = g(\hat{\theta})$ maximizes the *induced* likelihood for τ. Indeed, for any $\tau \in G$, we have

$$L^*(\tau; \mathbf{x}) = \max_{\theta \in G_\tau} L(\theta; \mathbf{x}) \leq \max_{\theta \in \Theta} L(\theta; \mathbf{x}) = L(\hat{\theta}; \mathbf{x}) \leq L^*(g(\hat{\theta}); \mathbf{x}),$$

the last inequality follows from the fact that $\hat{\theta} \in G_{g(\hat{\theta})}$.

Remark 4.4. Conceptually, you can think of maximizing the induced likelihood as bypassing the introduction of auxiliary parameters that would make an extension of g a bijection, and then relying on the definition of the MLE in the presence of "nuisance parameters" given in Remark 4.2. Hence an induced MLE can also be an MLE for an extension of the reparameterized model.

Example 4.2. Suppose that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ with MLE $\hat{\mu}$. Let $\tau := g(\mu) = \mu^2$. Even though g is not bijective (since we lose the sign information), we can still conclude that $\hat{\mu}^2$ is an MLE of τ.

Example 4.3. Suppose that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with MLE $(\hat{\mu}, \hat{\sigma}^2)$. Let $\tau = \sqrt{\sigma^2}/\mu$ be the coefficient of variation. The function $(\mu, \sigma^2) \mapsto \sqrt{\sigma^2}/\mu$ is not bijective and the first theorem cannot be applied. However we can still deduce that $\sqrt{\hat{\sigma}^2}/\hat{\mu}$ is an MLE of τ.

5 Determining MLEs: the tractable case

5.1 Differentiable likelihood in the continuous case - the multivariate scenario

When $L(\theta; \mathbf{x})$ is differentiable in the interior of Θ one can consider finding the extrema of the likelihood function by identifying the zeros of the gradient of the likelihood. That is, the set of θs such that $\nabla L(\theta; \mathbf{x}) = 0$ or $\nabla \ell(\theta; \mathbf{x}) = 0$ (the latter turns out to lead to simpler equations in many situations). We then have to check which solutions maximize the likelihood, if any. Remember that

$$\nabla \ell(\theta; \mathbf{x}) := \left(\frac{\partial \ell(\theta; \mathbf{x})}{\partial \theta_1}, \ldots, \frac{\partial \ell(\theta; \mathbf{x})}{\partial \theta_d} \right)$$

and that it is also required to examine the Hessian $\nabla^2 \ell(\theta; \mathbf{x})$ at the zero(s) to determine which zero(s) correspond to local maxima. Specifically, we should check that the Hessian is negative definite[2].

Example 5.1 (Normal distribution)**.** Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$.

[2]Recall that a matrix A is negative definite if $x^T A x < 0$ for all $x \neq 0$.

5.2 An interesting example: ML estimation for capture recapture models

We consider here a somewhat substantial case, the capture/recapture model, a technique used to estimate the size of a population. This is a good illustration of the power of statistical thinking in the presence of uncertainty, in a non-trivial situation.

Example 5.2 (Capture/recapture)**.** Given a population of unknown size n, capture $t \leq n$ animals at random, tag them and release them, then

1. Wait for a while (e.g. to let them spread out spatially).
2. Capture c animals uniformly at random and count how many of those are tagged (say $x \leq \min\{t, c\} \leq n$).

   The parameter to estimate here is $\theta = n$, the unknown population size, assuming that the probability of capturing a particular individual is uniform.

The main intuition for the "experiment" and a resulting estimate is:

- We know that there are t tagged animals.
- If x of the c captured animals are tagged then x/c is the proportion of tagged animals among the captured animals.
- It is reasonable to think of this as an *estimate* of the proportion of tagged animals among all animals (since the capturing is uniform), i.e. t/n.
- By equating $t/n \approx x/c$ we obtain $n \approx ct/x$ and an integer-valued estimate could be $n = [ct/x]$ (the closest integer to ct/x).

Beyond just intuition, one can justify this estimate using the maximum likelihood framework.

We first work out the associated probability distribution for X: there are

- $\binom{t}{x}$ ways of choosing x individuals among the tagged ones,
- $\binom{n-t}{c-x}$ ways of choosing the remaining animals among the non-tagged ones,
- $\binom{n}{c}$ ways of choosing c animals in a population of n individuals.

Consequently the probability of catching x tagged animals is

$$f(x;n) = \frac{\binom{t}{x}\binom{n-t}{c-x}}{\binom{n}{c}} \quad .$$

In other words $X \sim \text{Hypergeometric}(t, n, c)$, but here t and c are known (and part of the *design* of the experiment). We now look for an MLE of the population size.

Here is a small program to plot the likelihood in the case of a capture-recapture model, for the chickpeas example, assuming that $t = 97$ and $x = 7$ of the $c = 100$ re-captured chickpeas are tagged. We use the fact that

$$f(x; n) = \frac{\binom{t}{x}\binom{n-t}{c-x}}{\binom{n}{c}}$$

is a hypergeometric distribution, that is $X \sim \text{Hypergeometric}(t, n, c)$, where t and c are known (check Wikipedia for more information about the hypergeometric distribution).

```
tagged <- 97
captured <- 100
x <- 7
nhat <- floor(tagged*captured/x)
n <- seq(round(.9*nhat), round(1.1*nhat), by=1)
likelihood <- dhyper(x, tagged, n-tagged, captured) # different parameterization in R
plot(n, likelihood, pch=".")
lines(c(nhat, nhat), c(max(likelihood), min(likelihood)), col=2)
```

```
nhat


## [1] 1385
```

The likelihood function is presented in Fig. 5.1.

Clearly the estimator $\hat{n}(X) = \lfloor ct/X \rfloor$ is a random variable as it depends on the number of re-captured animals which are found to be tagged, and the re-capture is a random process. It is interesting to

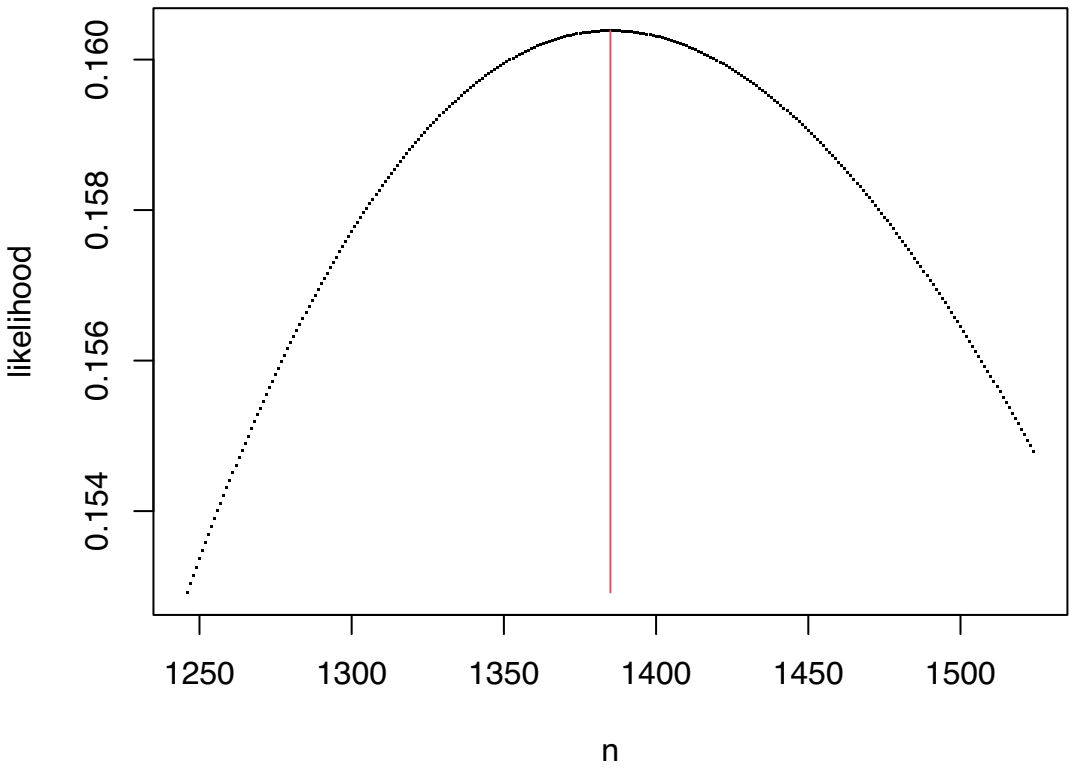


Figure 5.1: Likelihood function for the capture-recapture example.

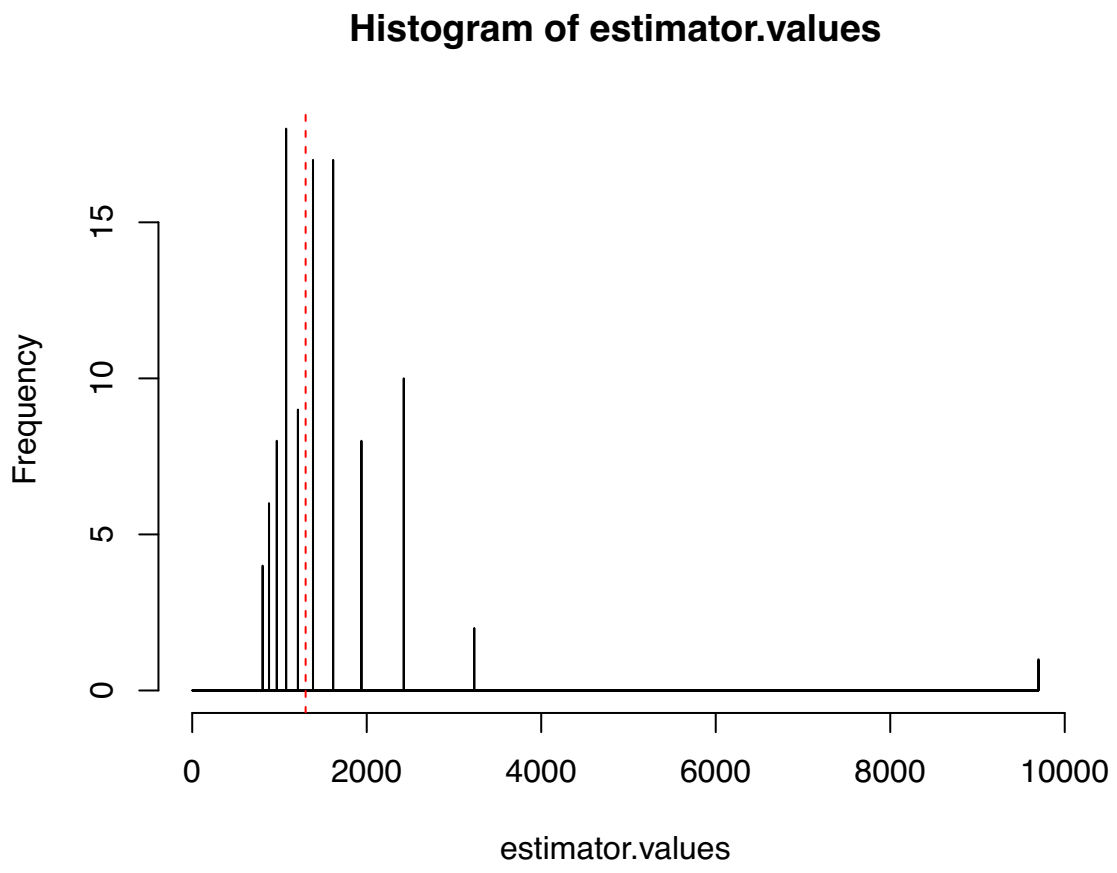


Figure 5.2: Histogram of the estimators of the population size.

characterize the distribution of this random variable as a means to ascertain the quality of the estimator $\hat{n}(X) = \lfloor ct/X \rfloor$.

To fix ideas, let us assume that $n^* = 1300$ (this is the true population size, which is of course unknown in practice) that $t = 97$, $c = 100$ and that we repeat the re-capture $m = 100$ times ("releasing" the peas after each capture). We therefore observe x^i for $i = 1, \ldots, m = 100$ and compute $\hat{n}(x^i)$ for $i = 1, \ldots, m = 100$. One can then plot the histogram (or empirical distribution) of $\hat{n}(x^i)$ for $i = 1, \ldots, m = 100$,

```
nstar <- 1300
tagged <- 97
captured <- 100
# sample m=100 realizations of the Hypergeometric random variable
Xs <- rhyper(100, tagged, nstar-tagged, captured)
estimator.values <- floor(tagged * captured / Xs)
hist(estimator.values, seq(1,max(estimator.values)))
abline(v=nstar, col='red', lty='dashed')
```

The histogram of $\hat{n}(x^i)$ for $i = 1, \ldots, m = 100$ is presented on Fig. 5.2

From this histogram one can for example get an idea of the answer to the statement: given that $n^* = 1300$, what is the probability that the ML estimator $\hat{n}(X)$ is such that $|\hat{n}(X) - n^*| < 100$? In fact, the result of such an analysis will be arbitrarily precise as m increases, but in the present case it is possible to calculate the limiting distribution as $m \to \infty$ exactly, by finding the PMF of the distribution of $\hat{n}(X)$. We have,

$$\mathbb{P}(\hat{n}(X) = n; n^*) = \mathbb{P}(\lfloor ct/X \rfloor = n; n^*) = \sum_{x=0}^{\infty} \mathbb{I}\{\lfloor ct/x \rfloor = n\} f(x; n^*),$$

that is we look for all the values of x such that $\lfloor ct/x \rfloor = n$ and add the corresponding probabilities of observing those x's. One can do this using R,

```
nstar <- 1300
tagged <- 97
captured <- 100

xvalues <- seq(1, min(c(tagged, captured))) # all possible non-zero x values
estimator.values <- seq(1, tagged * captured, by=1) # maximum possible finite value
nhatvalues <- floor(captured*tagged/xvalues)
pmf <- rep(0, length(estimator.values))

for (n in estimator.values){
  # nhatvalues==n returns the indices such that nhatvalues==n
  # xvalues[nhatvalues==n] returns the values of X such that the estimator is n
  pmf[n] <- sum(dhyper(xvalues[nhatvalues==n], tagged, nstar-tagged, captured))
}

plot(estimator.values, pmf, type = 'o')
abline(v=nstar, col='red', lty='dashed')
```

and compute the probability that $\hat{n}(X)$ is such that $|\hat{n}(X) - n^*| < 100$

```
sum(pmf[abs(estimator.values-nstar)<100])

## [1] 0.3062317
```

or within 200 of n^*

```
sum(pmf[abs(estimator.values-nstar)<200])

## [1] 0.3062317
```

or within 50 of n^*

```
sum(pmf[abs(estimator.values-nstar)<50])

## [1] 0
```

Naturally in practice we do not know the value of n^*. We will see how to circumvent this problem, for example by testing the assumption that $n^* = 1300$ etc.

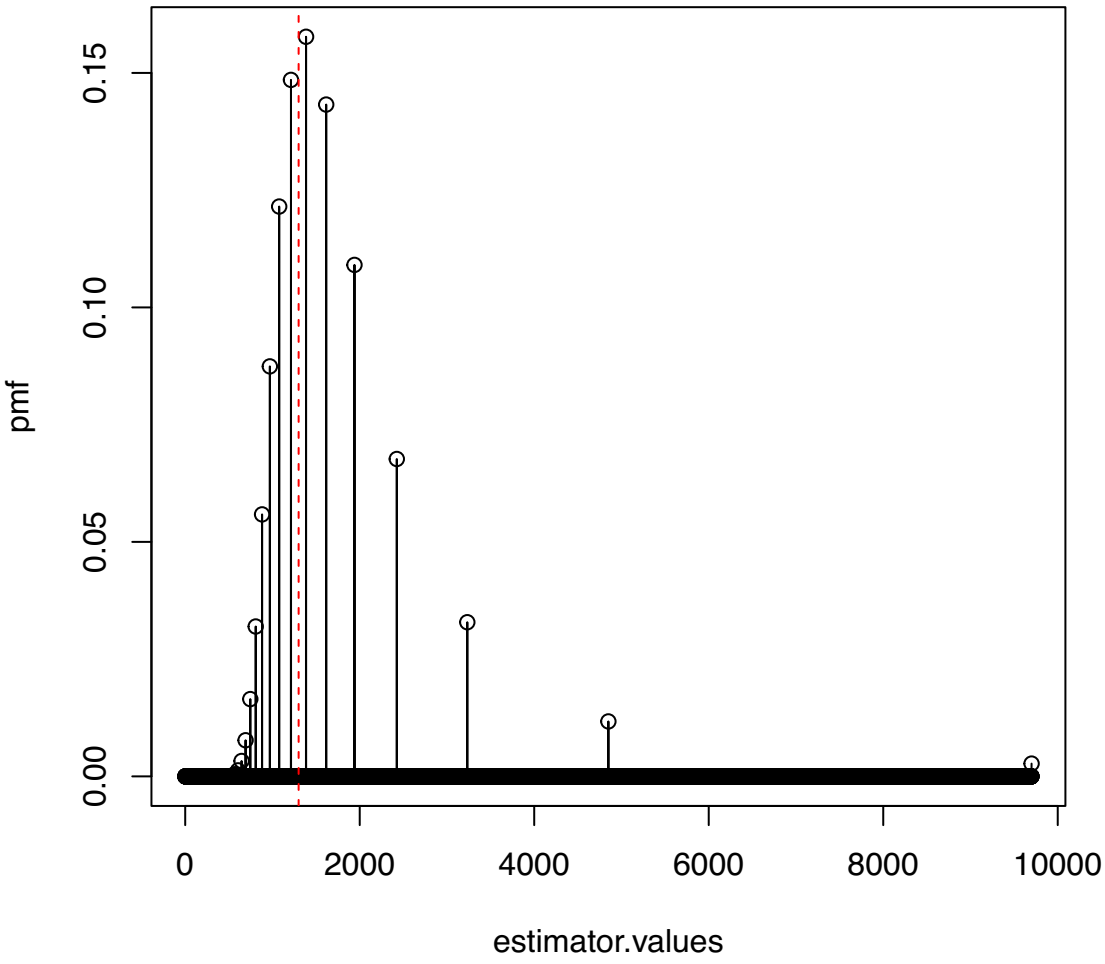


Figure 5.3: Sampling distribution of the estimator of the population size.

6 Statistics and estimators

We review estimators; some of the content here should already be known to you.

Definition 6.1 (Statistic)**.** Given some data $\mathbf{x}$, a statistic is a function of the data $T(\mathbf{x})$.

A statistic cannot depend on an unknown statistical parameter.

Definition 6.2 (Estimate)**.** Assume that $\mathbf{X} \sim f_n(\cdot;\theta^*)$ with $\theta^* \in \Theta$ and $\mathbf{x}$ is a realization of $\mathbf{X}$. An estimate of θ^* is a statistic $T(\mathbf{x})$ that is intended to approximate θ^*. In many cases, the notation $\hat{\theta}$ is used in place of T, i.e. we compute $\hat{\theta}(\mathbf{x})$. Notice that an estimate is a value.

Definition 6.3 (Estimator)**.** Assume that $\mathbf{X} \sim f_n(\cdot;\theta^*)$ with $\theta^* \in \Theta$ and $\mathbf{x}$ is a realization of $\mathbf{X}$. If $T(\mathbf{x})$ [or $\hat{\theta}(\mathbf{x})$] is an estimate of θ^* then we call T [or $\hat{\theta}$] an estimator. We can also call $T(\mathbf{X})$ or $\hat{\theta}(\mathbf{X})$ an estimator. Notice that an estimator is a function / random variable.

In most situations we would like the estimate to take values in Θ. An example of an estimate is a ML estimate. An example of an estimator is a ML estimator. In this unit, we will focus on ML estimates/estimators but others are possible, e.g. method of moments estimates/estimators.

Definition 6.4 (Distribution of an estimator)**.** Assume that $\mathbf{X} \sim f_n(\cdot;\theta^*)$ with $\theta^* \in \Theta \subseteq \mathbb{R}$. If $\hat{\theta}(\mathbf{X})$ is a real-valued random variable, one can write its CDF as

$$F_{\hat{\theta}(X)}(t;\theta^*) = \mathbb{P}(\hat{\theta}(\mathbf{X}) \leq t;\theta^*) = \int_{\mathcal{X}^n} \mathbb{I}\{\hat{\theta}(\mathbf{x}) \leq t\} f_n(\mathbf{x};\theta^*) d\mathbf{x},$$

for $t \in \Theta$. Notice that the CDF of $\hat{\theta}(\mathbf{X})$ depends on the random vector $\mathbf{X}$, whose distribution depends on the particular value of θ^*.

Remark 6.1. A fundamental point is that the distribution of the estimator depends on the assumed "true", but unknown, value θ^* of the statistical parameter θ. It should therefore not be surprising if in what follows the following workarounds will be used:

1. define/investigate properties of the estimator's CDF $\mathbb{P}(\hat{\theta}(\mathbf{X}) \leq t;\theta)$ that are uniform in $\theta \in \Theta$ (e.g. it is uniformly bounded by some known quantity),
2. quantities whose distributions do not depend on θ,
3. if we focus on a particular possible value of θ, say θ_0, one can adopt a reasoning of the type : "if $\mathbf{x}$ is a realization of $\mathbf{X} \sim f_n(\cdot;\theta_0)$ then the CDF of the estimator is $\mathbb{P}(\hat{\theta}(\mathbf{X}) \leq t;\theta_0)$ and we can think about whether $\hat{\theta}(\mathbf{x})$ is likely to be close to θ_0."

Remark 6.2. It is natural to consider these distributions in order to characterize how good an estimator $\hat{\theta}_1(\mathbf{X})$ is compared to an estimator $\hat{\theta}_2(\mathbf{X})$. In general one can, in principle, ask questions such as:

Given that the true probability distribution/probability density of the observations was $f(\cdot;\theta^*)$,

- what is the probability that $\mathbb{P}(|\hat{\theta}(\mathbf{X}) - \theta^*| > \epsilon;\theta^*)$ for a given ϵ?
- or, what is a $\delta > 0$ such that $\mathbb{P}(|\hat{\theta}(\mathbf{X}) - \theta^*| > \delta;\theta^*) = 0.05$?

These distributions are unfortunately usually intractable, and the use of the tools introduced in Year 1 may turn out to be inadequate. The classical "trick" consists of exploiting the fact that as the size of the sample $\mathbf{X}_n$ increases, the distribution of $\hat{\theta}_n(\mathbf{X}_n)$ (or a simple transformation of it) may converge to a well-known distribution: in this unit it will typically be the normal distribution. Before exploring this we review some classical notions regarding estimators.

Definition 6.5 (Unbiased estimator)**.** An estimator $\hat{\theta}$ of θ for the model $\mathbf{X} \sim f_n(\cdot;\theta)$ is unbiased if for any $\theta \in \Theta$,

$$\mathbb{E}(\hat{\theta}(\mathbf{X});\theta) = \theta \quad .$$

Remark 6.3. Notice the important fact that this property holds for *any* $\theta \in \Theta$.

Remark 6.4. Although appealing ("on average the estimator gives the right answer") this property does not provide us with a good measure of performance of the estimator since it does not provide us with any measure of discrepancy between the true value θ^* and $\hat{\theta}$.

A popular, and sensible, way of quantifying how good an estimator is involves calculating its mean square error (MSE).

Definition 6.6 (Mean square error)**.** The mean square error of an estimator $\hat{\theta}(\cdot)$ for $\mathbf{X} \sim f_n(\cdot;\theta^*)$ is

$$\mathrm{MSE}(\hat{\theta};\theta^*) := \mathbb{E}\left[(\hat{\theta}(\mathbf{X}) - \theta^*)^2;\theta^*\right] \quad .$$

This is the mean of the squared error associated with the random variable $\hat{\theta}(\mathbf{X})$.

Remark 6.5. The MSE combines the notion of variance and bias. The bias-variance decomposition of the MSE is

$$\mathbb{E}\left[(\hat{\theta}(\mathbf{X}) - \theta^*)^2;\theta^*\right] = \mathbb{E}\left[\left\{\hat{\theta}(\mathbf{X}) - \mathbb{E}(\hat{\theta}(\mathbf{X});\theta^*)\right\}^2;\theta^*\right] + \left(\mathbb{E}[\hat{\theta}(\mathbf{X}) - \theta^*;\theta^*]\right)^2 \quad ,$$

or

$$\mathrm{MSE}(\hat{\theta};\theta^*) = \mathrm{var}(\hat{\theta};\theta^*) + \mathrm{bias}(\hat{\theta};\theta^*)^2,$$

where

$$\mathrm{bias}(\hat{\theta};\theta^*) = \mathbb{E}[\hat{\theta}(\mathbf{X}) - \theta^*;\theta^*] = \mathbb{E}[\hat{\theta}(\mathbf{X});\theta^*] - \theta^*.$$

It is a useful exercise for you to prove this.

You may write $\mathrm{MSE}(\hat{\theta}(\mathbf{X});\theta^*)$, $\mathrm{var}(\hat{\theta}(\mathbf{X});\theta^*)$ and $\mathrm{bias}(\hat{\theta}(\mathbf{X});\theta^*)$ to emphasize that we are concerned with the properties of the random variable $\hat{\theta}(\mathbf{X})$ when $\mathbf{X} \sim f_n(\cdot;\theta^*)$.

Example 6.1. Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \mathrm{Poisson}(\lambda^*)$ and that we are using the sample mean $\hat{\lambda}(\mathbf{x}) := n^{-1}(x_1 + \ldots + x_n)$, as an estimate of λ^* (this is reasonable since for any $\lambda \in \Theta$, $\mathbb{E}(X_1;\lambda) = \lambda$ and it is also the ML estimate in this case). The corresponding estimator $\hat{\lambda}(\mathbf{X})$ is unbiased since for any $\lambda \in \Theta$,

This latter result might appear to be of limited practical use: what value of λ should one use above? (λ^* is unknown). However it provides us with some useful insights:

- if we assume that the true value was λ_0 then we get an idea of the precision of our estimator for n samples,
- if we know $\lambda \leq C$ for some $C > 0$ we can use C/n as a (conservative) upper bound for the MSE,
- we learn that we will require more samples to achieve a given precision for larger λ,
- if we consider the sequence of estimators $\{\hat{\lambda}_n : \mathcal{X}^n \to \mathbb{R}\}$ the result above tells us that for any $\lambda \in \Theta$ (and in particular the unknown λ^*) $\mathrm{MSE}(\hat{\lambda}_n; \lambda) \to 0$ as $n \to \infty$, e.g. if we consider more and more data, then the estimator gets closer and closer to λ^* (in mean square). This is called mean square/quadratic mean consistency, an asymptotic property of the estimator.

7 Probabilistic convergence

We are interested in future lectures in investigating the properties of sequences of maximum likelihood estimators $\hat{\theta}_n(X_1, \ldots, X_n)$ as the number of observations n increases. In particular we would like to say something about their distribution. For example, if $X_1, X_2, \ldots$ are i.i.d. $\mathcal{N}(\theta^*, 1)$ random variables then the ML estimator is

$$\hat{\theta}_n(X_1, \ldots, X_n) = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

and in this particular case it can be shown that $\bar{X}_n \sim \mathcal{N}(\theta^*, n^{-1})$. We are able to deduce from this that $\bar{X}_n$ becomes very close to θ^* with high probability as n grows, and we are also able to deduce probabilities that $\bar{X}_n \in [a, b]$ for constants $a < b$.

In general, the distribution of ML and other estimators is rarely easy to describe. Nevertheless, we will strive to be able to say that $\hat{\theta}_n(X_1, \ldots, X_n)$ becomes close in an appropriate sense to θ^* and also approximate in an appropriate sense the probability that $\hat{\theta}_n(X_1, \ldots, X_n)$ takes values in particular sets for large n. In order to do this, we need to develop and work with suitable notions of convergence for random variables.

7.1 Two useful inequalities

Lemma 7.1 (Markov's inequality)**.** *For a non-negative random variable X and constant $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. [Continuous X] We have

$$a\mathbb{P}(X \geq a) = a\int_a^\infty f_X(x)\,dx \leq \int_a^\infty x f_X(x)\,dx \leq \int_0^\infty x f_X(x)\,dx = \mathbb{E}[X]. \qquad \square$$

Corollary (Chebyshev's inequality)**.** *Let $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{var}(X)$. Then for any $k > 0$,*

$$\mathbb{P}\left(|X - \mu| \geq k\right) \leq \frac{\sigma^2}{k^2}.$$

Proof. We have

$$\mathbb{P}\left(|X - \mu| \geq k\right) = \mathbb{P}\left(|X - \mu|^2 \geq k^2\right) \leq \frac{\mathbb{E}\left[(X-\mu)^2\right]}{k^2} = \frac{\sigma^2}{k^2}. \qquad \square$$

7.2 Types of probabilistic convergence

A deterministic sequence of real valued vectors $z_1, z_2, \ldots$ is said to converge to a limit ℓ if for any $\epsilon > 0$ there exists n_0 such that for any $n \geq n_0$, $|z_n - \ell| \leq \epsilon$. We now review how this familiar notion can be extended to the case where $Z_1, Z_2, \ldots$ is a sequence of random variables taking values in Z with probability distribution $\mathbb{P}(\cdot)$. We will focus here on convergence in probability and convergence in distribution. There are other types of convergence for random variables which we will not need for this course.

Definition 7.1 (Convergence in probability)**.** We say that $\{Z_n\}$ converges in probability to the random variable Z if for any $\epsilon > 0$

$$\lim_{n\to\infty} \mathbb{P}(|Z_n - Z| > \epsilon) = 0 \quad ,$$

This is traditionally denoted $Z_n \to_{\mathbb{P}} Z$. The random variables $\{Z_n\}$ and Z should be defined on the same probability space.

Definition 7.2 (Convergence in distribution)**.** We say that $\{Z_n\}$ converges in distribution to a random variable Z if for any $z \in \mathsf{Z}$ where $\mathbb{P}(Z \leq z)$ is continuous, $\lim_{n\to\infty} \mathbb{P}(Z_n \leq z) = \mathbb{P}(Z \leq z)$. This is traditionally denoted $Z_n \to_{\mathcal{D}} Z$. This is equivalent to saying that for all z at which F_Z is continuous,

$$\lim_{n\to\infty} F_{Z_n}(z) = F_Z(z).$$

The random variables need not be defined on the same probability space [the definition concerns only their CDFs].

Example 7.1 (Weak Law of Large Numbers)**.** If $\{X_n\}$ are i.i.d. and $\mathbb{E}(X_1) = \mu$ is finite then

$$Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i \to_{\mathbb{P}} \mu.$$

Example 7.2 (Central Limit Theorem)**.** If in addition, $\mathrm{var}(X_1) = \sigma^2 < \infty$ then

$$\frac{\sqrt{n}(Z_n - \mu)}{\sigma} \to_{\mathcal{D}} Z \sim \mathcal{N}(0, 1).$$

Convergence in distribution is a weaker mode of convergence than convergence in probability.

Theorem 7.1. *Convergence in probability implies convergence in distribution, but the opposite is not generally true.*

Proposition 7.1. *Convergence in distribution to a constant and convergence in probability to a constant are equivalent.*

Example 7.3. Let X be a Bernoulli($\frac{1}{2}$) random variable and $X_1, X_2, \ldots$ a sequence of random variables defined by $X_n := (1 - X) + \frac{1}{n}$. Then $X_n \to_{\mathcal{D}} X$.

To see this, we have

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{2} & x \in [0, 1) \\ 1 & x \geq 1. \end{cases} \qquad F_{X_n}(x) = \begin{cases} 0 & x < \frac{1}{n}, \\ \frac{1}{2} & x \in [\frac{1}{n}, 1 + \frac{1}{n}), \\ 1 & x \geq 1 + \frac{1}{n}. \end{cases}$$

Clearly $F_{X_n}(x) \to F_X(x)$ at all points at which F_X is continuous, i.e. $x \in \mathbb{R} \setminus \{0, 1\}$.

Exercise. Show that X_n does not converge in probability to X.

We do not prove results here (the proofs can be found in standard textbooks on probability). The following classical results will be useful in what follows. The first extends an important property of continuous functions and convergent sequences to the stochastic scenario.

Theorem 7.2 (Continuous Mapping Theorem)**.** *Let $g : \mathsf{Z} \to G$ be continuous. Then*

1. If $Z_n \to_{\mathbb{P}} Z$, then $g(Z_n) \to_{\mathbb{P}} g(Z)$.

2. If $Z_n \to_{\mathcal{D}} Z$, then $g(Z_n) \to_{\mathcal{D}} g(Z)$.

Example 7.4. If $X_n \to_{\mathcal{D}} X \sim \mathcal{N}(0,1)$ then $X_n^2 \to_{\mathcal{D}} X^2 \sim \chi^2(1)$.

The second is very useful when performing a variety of statistical manipulations. We will use this and the Continuous Mapping Theorem above in an example later.

Theorem 7.3 (Slutsky's theorem)**.** *If $Y_n \to_{\mathcal{D}} Y$ and $Z_n \to_{\mathcal{D}} c$, with $c \in \mathbb{R}$ a constant, then*

1. $Y_n + Z_n \to_{\mathcal{D}} Y + c$.
2. $Y_n Z_n \to_{\mathcal{D}} Yc$.
3. $Y_n / Z_n \to_{\mathcal{D}} Y/c$ if $c \neq 0$.

Convergence in probability is not always easy to establish. Markov's inequality can be helpful.

A sufficient (but not necessary) condition for convergence in probability is convergence in "quadratic mean" or "mean square".

Definition 7.3 (Convergence in quadratic mean)**.** We say that $\{Z_n\}$ converges in quadratic mean to the random variable Z if $\lim_{n\to\infty} \mathbb{E}\left[(Z_n - Z)^2\right] = 0$. This is traditionally denoted $Z_n \to_{\text{qm}} Z$.

Proposition. *If $Z_n \to_{\text{qm}} Z$ then $Z_n \to_{\mathbb{P}} Z$.*

Proof. Fix any $\epsilon > 0$. By Markov's inequality,

$$\mathbb{P}(|Z_n - Z| > \epsilon) = \mathbb{P}(|Z_n - Z|^2 > \epsilon^2) \leq \frac{\mathbb{E}\left[(Z_n - Z)^2\right]}{\epsilon^2} \to 0,$$

since $Z_n \to_{\text{qm}} Z$. Hence $Z_n \to_{\mathbb{P}} Z$. □

Consequently if we can show that $\lim_{n\to\infty} \mathbb{E}(|Z_n - Z|^2) = 0$ then we have convergence in probability. It is sometimes easier to work with the MSE than with probabilities directly.

Convergence in probability or in distribution does not imply that the random variables inherit all properties of their limiting random variable. In particular, they do not imply the convergence of moments.

Example. Let $\{B_n\}$ be defined by $B_n \sim \text{Bernoulli}(\frac{1}{n})$ and define $X_n := nB_n$. Then for $\epsilon \in (0,1)$,

$$\mathbb{P}(|X_n| > \epsilon) = \mathbb{P}(B_n = 1) = \frac{1}{n} \to 0,$$

so $X_n \to_{\mathbb{P}} 0$. However, for each $n > 0$, $\mathbb{E}[X_n] = n \cdot \frac{1}{n} = 1$, i.e. the mean of X_n does not converge to 0.

7.3 Probabilistic convergence and estimators

Definition 7.4 (Consistency of a sequence of estimators)**.** A sequence of estimators $\{\hat{\theta}_n(\cdot) : \mathcal{X}^n \to \Theta\}$ is *consistent* if for all $\theta \in \Theta$, with $\mathbf{X}_n \sim f_n(\cdot;\theta)$,

$$\hat{\theta}_n(\mathbf{X}_n) \to_{\mathbb{P}(\cdot;\theta)} \theta \quad .$$

Remark 7.1. Multiple comments can be made:

1. In numerous situations one will talk about the consistency of *the* estimator: e.g. for the MLE, but also for the mean, etc. This implicitly refers to the corresponding sequence of MLEs, sequence of means, etc.

2. Note the $\mathbb{P}(\cdot;\theta)$ in the limit above, and in particular the dependence on θ which is often omitted in practice - you should however not forget what the symbols actually mean, since this will be crucial in future lectures.

3. Quadratic mean / mean square convergence imply consistency. That is, if the MSE of the estimator converges to 0, the estimator is consistent.

Example 7.5 (Back to flipping coins)**.** Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta^*)$ for some $\theta^* \in \Theta = [0,1]$. The ML estimator and the method of moments estimator are the sample mean:

$$\hat{\theta}_n(X_1,\ldots,X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

1. Consistency of $\hat{\theta}_n$ (more formally $\{\hat{\theta}_n\}$) is provided by the WLLN, since $\mathbb{E}[X_1] = \theta^*$.

2. Now we look at a crude way to "capture" θ^* in a random interval with high probability. We will write $\hat{\theta}_n = \hat{\theta}_n(X_1,\ldots,X_n)$ to keep things clear. We notice that

   $$\mathbb{E}[\hat{\theta}_n;\theta^*] = \theta^*$$

   and

   $$\text{var}\left(\hat{\theta}_n;\theta^*\right) = \frac{\theta^*(1-\theta^*)}{n}.$$

   We can use Chebyshev's inequality

   $$\mathbb{P}\left(\left|\hat{\theta}_n - \theta^*\right| \geq \epsilon;\theta^*\right) \leq \frac{\theta^*(1-\theta^*)}{n\epsilon^2}.$$

   We don't know θ^*, but we can deduce that $\theta^*(1-\theta^*) \leq 1/4$ and so

   $$\mathbb{P}\left(\left|\hat{\theta}_n - \theta^*\right| \geq \epsilon;\theta^*\right) \leq \frac{1}{4n\epsilon^2}.$$

   To have a probability bound of our choosing we can substitute $\alpha = 1/(4n\epsilon^2)$ to obtain

   $$\mathbb{P}\left(\left|\hat{\theta}_n - \theta^*\right| \geq \frac{1}{2\sqrt{n\alpha}};\theta^*\right) \leq \alpha,$$

   so that

   $$\mathbb{P}\left(\hat{\theta}_n - \frac{1}{2\sqrt{n\alpha}} < \theta^* < \hat{\theta}_n + \frac{1}{2\sqrt{n\alpha}};\theta^*\right) \geq 1-\alpha.$$

   That is, the random interval $(\hat{\theta}_n - \frac{1}{2\sqrt{n\alpha}}, \hat{\theta}_n + \frac{1}{2\sqrt{n\alpha}})$ captures θ^* with probability $1-\alpha$. We observe that the width of this interval decreases as n increases and increases as α decreases, which may be intuitive for you.

3. We can do better than the crude bound above. We will show that for $W \sim \mathcal{N}(0,1)$

   $$\frac{\sqrt{n}(\hat{\theta}_n - \theta^*)}{\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)}} \rightarrow_{\mathcal{D}(\cdot;\theta^*)} W.$$

We note that $\hat{\theta}_n \to_{\mathbb{P}} \theta^*$ by the WLLN. In addition, $\mathrm{var}(X_1) = \theta^*(1-\theta^*)$ so the CLT gives

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta^*)}{\sqrt{\theta^*(1-\theta^*)}} \to_{\mathcal{D}(\cdot;\theta^*)} W.$$

Now let

$$Y_n = \frac{\sqrt{n}(\hat{\theta}_n - \theta^*)}{\sqrt{\theta^*(1-\theta^*)}}, \qquad Z_n = \frac{\sqrt{\theta^*(1-\theta^*)}}{\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)}},$$

and the Continuous Mapping Theorem tells us $Z_n \to_{\mathbb{P}(\cdot;\theta^*)} 1$ [the function $x \mapsto \sqrt{\theta^*(1-\theta^*)}/\sqrt{x(1-x)}$ is continuous]. Hence, by Slutsky's Theorem we obtain

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta^*)}{\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)}} = Y_n Z_n \to_{\mathcal{D}(\cdot;\theta^*)} W.$$

This allows us to produce the random interval [think about what convergence in distribution means]

$$\left(\hat{\theta}_n - z_{\alpha/2}\sqrt{\frac{\hat{\theta}_n(1-\hat{\theta}_n)}{n}}, \hat{\theta}_n + z_{\alpha/2}\sqrt{\frac{\hat{\theta}_n(1-\hat{\theta}_n)}{n}}\right),$$

which asymptotically (in n) captures θ^* with probability $1-\alpha$, where we have used the notation $z_\alpha = \Phi^{-1}(1-\alpha)$ so that $\Phi(z_\alpha) = 1-\alpha$, where Φ is the CDF of a standard normal random variable.

Exercise. Simulate the different random intervals multiple times in `R` for different values of n, and compare their lengths and the frequency with which they capture θ^*.

For the crude interval we have used a rough bound on the cumulative distribution function of $\left|\hat{\theta}_n - \theta^*\right|$ in order to construct a confidence interval for θ^*. We have also produced a second, asymptotically exact confidence interval, but with considerable effort and a heavy reliance on the fact that $\hat{\theta}_n$ in this case is an average of i.i.d. random variables.

We will now aim to construct a simple and tractable approximation of the distribution of $\hat{\theta}_n - \theta^*$ to produce confidence intervals of the second type more generally and systematically when $\hat{\theta}_n$ is a maximum likelihood estimator, and *not necessarily an average of i.i.d. random variables.*

8 The Fisher information

In the next few lectures we are going to cover the following two fundamental important points: given that $\mathbf{X}_n \overset{\text{iid}}{\sim} f(\cdot;\theta^*)$ we are going to establish that for models that are sufficiently regular:

1. there exists a lower bound on the achievable performance of *any* estimator of θ^*, where performance is measured in terms of the variance, that is we will show that for any unbiased estimator of θ^* we have
$$\text{var}\left(\hat{\theta}_n(\mathbf{X}_n);\theta^*\right) \geq \frac{1}{nI(\theta^*)},$$
for a quantity $I(\theta^*)$ that will be defined.

2. a scaled and centered sequence of ML estimators $\{\hat{\theta}_n(\mathbf{X}_n)\}$ become asymptotically normal as $n \to \infty$ (i.e. as the size of the observation vector increases)
$$\sqrt{nI(\theta^*)}(\hat{\theta}_n(\mathbf{X}_n) - \theta^*) \to_{\mathcal{D}(\cdot;\theta^*)} Z$$
where $Z \sim \mathcal{N}(0,1)$.

The interest of such results should be self-evident. The first result allows one to define a notion of optimality for the estimation of θ^* for a given model, and hence to measure how good a given estimator is in comparison to the lower bound. The second result suggests that when n is sufficiently large the distribution of
$$\sqrt{nI(\theta^*)}(\hat{\theta}_n(\mathbf{X}_n) - \theta^*)$$
becomes very close to a standard normal. Loosely speaking, $\hat{\theta}_n(\mathbf{X}_n)$ will approximately be distributed like a $\mathcal{N}(\theta^*, 1/(nI(\theta^*)))$ random variable and the variance of the normal random variable is exactly the same as in the lower bound, indicating that $\hat{\theta}_n(\mathbf{X}_n)$ is asymptotically optimal in a particular sense. Moreover, we can use the limiting distribution to approximate the sampling distribution of $\hat{\theta}_n(\mathbf{X}_n)$ and hence quantify (approximately) the uncertainty associated with our estimator in cases where direct calculations are not possible.

In order to establish the important properties above we will require some properties of the likelihood and its first and second order derivatives with respect to θ. Throughout this section we will focus on the situation where $\mathbf{X}_n \overset{\text{iid}}{\sim} f(\cdot;\theta)$ for $\theta \in \Theta \subseteq \mathbb{R}$ for simplicity, but it is important to remember that the results we are going to establish in this particular set up extend to more general scenarios, where the components of $\mathbf{X}_n$ are not independent but satisfy specific dependence properties, and to the situation where θ is multivariate.

(**A1**) **(Fisher information regularity conditions)**

Θ is an open interval of $\mathbb{R}$ and the probability density/probability mass function $f(x;\theta)$ satisfies the following properties

1. both $L'(\theta;x) = \frac{df(x;\theta)}{d\theta}$ and $L''(\theta;x) = \frac{d^2 f(x;\theta)}{d\theta^2}$ exist for any $x \in \mathcal{X}$,
2. the set
$$S := \{x \in \mathcal{X} \,:\, f(x;\theta) > 0\},$$
does not depend on $\theta \in \Theta$ (hence the use of S, not S_θ),
3. one has the identity
$$\int_S \frac{d}{d\theta} f(x;\theta)dx = \frac{d}{d\theta}\int_S f(x;\theta)dx = 0 \quad .$$

Remark 8.1. Notice that the second condition is not satisfied for $X \sim \text{Uniform}(0, \theta)$, and we will see in Problems Class 2 that the asymptotic distribution of the corresponding ML estimator is *not* normal.

Definition 8.1 (The score function)**.** Let $\ell(\theta; x) = \log f(x; \theta)$. The score function, defined on $\Theta \times \mathcal{X}$, is

$$\ell'(\theta; x) := \frac{d}{d\theta}\ell(\theta; x) = \frac{\frac{d}{d\theta}f(x; \theta)}{f(x; \theta)} \quad .$$

The score measures the sensitivity of the likelihood function to changes in θ.

Remark 8.2. Note that under the FIRCs we have the property that for any $\theta \in \Theta$,

Example 8.1. Assume that $X \sim \text{Poisson}(\theta)$, that is $f(x; \theta) = \exp(-\theta)\frac{\theta^x}{x!}\mathbb{I}\{x \in \mathbb{N}\}$.

Example 8.2. Assume that $X \sim \mathcal{N}(\mu, 1)$, i.e. $f(x; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(x - \mu)^2\right\}$.

Definition 8.2 (Fisher information)**.** Let $X \sim f(\cdot;\theta)$ and $\ell(\theta;x) := \log f(x;\theta)$. The Fisher information is, for any $\theta \in \Theta$,

$$I(\theta) := \mathbb{E}(\ell'(\theta;X)^2;\theta) \geq 0.$$

Remark 8.3. Some remarks:

1. the Fisher information is a function of θ.
2. we will see that $I(\theta)$ can be thought of as being the average "information" brought by a single observation X about θ, assuming that $X \sim f(\cdot;\theta)$,
3. since for any $\theta \in \Theta$, $\mathbb{E}(\ell'(\theta;X);\theta) = 0$ we have the equivalent definition

$$I(\theta) = \mathrm{var}(\ell'(\theta;X);\theta) \quad .$$

Example 8.3. For $X \sim \mathrm{Poisson}(\theta)$,

Another example,

Example 8.4. For $X \sim \mathcal{N}(\mu, 1)$

$$I(\mu) = \mathrm{var}(X - \mu;\mu) = \mathrm{var}(X;\mu) = 1.$$

The following theorem establishes a convenient alternative expression for the Fisher information, and also the interpretation that it measures the expected rate of change of the score function around θ when $X \sim f(\cdot;\theta)$. If $I(\theta)$ is large, we expect the likelihood function to be highly peaked around θ while if $I(\theta)$ is small we expect it to be relatively flatter.

Theorem 8.1. *Assume that $f(x;\theta)$ satisfies (A1) and that in addition the following holds for any $\theta \in \Theta$*

$$\int_{\mathcal{X}} \frac{d^2}{d\theta^2} f(x;\theta)dx = \frac{d}{d\theta}\int_{\mathcal{X}} \frac{d}{d\theta} f(x;\theta)dx \quad (=0) \quad .$$

Then $I(\theta)$ can be rewritten for any $\theta \in \Theta$ as

$$I(\theta) = -\mathbb{E}\left(\frac{d^2}{d\theta^2}\ell(\theta;X);\theta\right) \quad .$$

Proof. We consider directly

$$\begin{aligned}\frac{d^2}{d\theta^2}\ell(\theta;x) &= \frac{d}{d\theta}\frac{\frac{d}{d\theta}f(x;\theta)}{f(x;\theta)} \\ &= \frac{\frac{d^2}{d\theta^2}f(x;\theta)}{f(x;\theta)} - \frac{\left[\frac{d}{d\theta}f(x;\theta)\right]^2}{f(x;\theta)^2}.\end{aligned}$$

Consequently,

9 Efficiency and the Cramer–Rao bound

Here we consider the situation with n observed realizations of i.i.d. random variables $X_1, \ldots, X_n$. That is, $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta)$ for some $\theta \in \Theta$ (assumed again scalar). The score function is

$$\ell_n'(\theta;\mathbf{x}) := \frac{d}{d\theta}\ell_n(\theta;\mathbf{x}),$$

where $\ell_n(\theta;\mathbf{x}) := \log f_n(\mathbf{x};\theta)$. The Fisher information is

$$I_n(\theta) := \mathbb{E}\left[\ell_n'(\theta;\mathbf{X})^2;\theta\right] = \operatorname{var}\big(\ell_n'(\theta;\mathbf{X});\theta\big) \quad .$$

Theorem 9.1. *Consider the situation where for any $\theta \in \Theta$, $f_n(\mathbf{x};\theta) = \prod_{i=1}^n f(x_i;\theta)$. Then for any $\theta \in \Theta$, $I_n(\theta) = nI(\theta)$*

Proof.

Now we are in a position to give some answers to the following question: assume that $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta^*)$ and that $\hat{\theta}(\mathbf{X})$ is an estimator (not necessarily the ML estimator) of θ. What is the best estimator we can

hope for, where best is measured in terms of variance, i.e. $\mathrm{var}(\hat{\theta}(\mathbf{X});\theta)$? The Cramer–Rao inequality gives us an answer. First, some preliminary results:

Lemma 9.1 (Cauchy–Schwarz inequality for expectations)**.** *Let X and Y be real-valued random variables on the same probability space. Then,*

$$\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

Proof. If $\mathbb{E}[Y^2] = 0$ then $\mathbb{P}(Y = 0) = 1$ so $\mathbb{E}[XY] = 0$ and the statement holds. So assume $\mathbb{E}[Y^2] > 0$ and let $\lambda = \mathbb{E}[XY]/\mathbb{E}[Y^2]$. Then

$$\begin{aligned} 0 &\leq \mathbb{E}[(X - \lambda Y)^2] \\ &= \mathbb{E}\left[X^2\right] - 2\lambda\mathbb{E}[XY] + \lambda^2\mathbb{E}[Y^2] \\ &= \mathbb{E}\left[X^2\right] - 2\frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} + \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} \\ &= \mathbb{E}\left[X^2\right] - \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]}, \end{aligned}$$

so indeed $\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$. □

Corollary 9.1 (Covariance inequality)**.** *Let W and Z be real-valued random variables on the same probability space. Then,*

$$\mathrm{cov}(W, Z)^2 \leq \mathrm{var}(W)\mathrm{var}(Z).$$

Proof. Let $X = W - \mathbb{E}[W]$ and $Y = Z - \mathbb{E}[Z]$ to obtain $\mathbb{E}[XY] = \mathrm{cov}(W, Z)$, $\mathbb{E}[X^2] = \mathrm{var}(W)$ and $\mathbb{E}[Y^2] = \mathrm{var}(Z)$, and apply the Lemma. □

Remark 9.1. This is why correlation — $\mathrm{corr}(W, Z) = \mathrm{cov}(W, Z)/\sqrt{\mathrm{var}(W)\mathrm{var}(Z)}$ — is valued in $[-1, 1]$.

Theorem 9.2 (Cramer-Rao inequality, in the case of a scalar parameter)**.** *Let $\mathbf{X}_n \overset{\text{iid}}{\sim} f(\cdot;\theta)$ and assume (A1). Let $\hat{\theta}_n(\cdot)$ be an estimator of θ with expectation $m(\theta) := \mathbb{E}(\hat{\theta}_n(\mathbf{X}_n);\theta)$, and such that for any $\theta \in \Theta$,*

$$\frac{d}{d\theta}\int \hat{\theta}_n(\mathbf{x}) f_n(\mathbf{x};\theta) d\mathbf{x} = \int \hat{\theta}_n(\mathbf{x}) \frac{d}{d\theta} f_n(\mathbf{x};\theta) d\mathbf{x}.$$

Then for any $\theta \in \Theta$, we have

$$\mathrm{var}\left(\hat{\theta}_n(\mathbf{X});\theta\right) \geq \frac{[m'(\theta)]^2}{nI(\theta)} \quad .$$

If $\hat{\theta}_n(\mathbf{X}_n)$ is an unbiased estimator, i.e. $m(\theta) = \theta$, then the inequality simplifies to

$$\mathrm{var}\left(\hat{\theta}_n(\mathbf{X}_n);\theta\right) = \mathrm{MSE}(\hat{\theta}_n(\mathbf{X}_n);\theta) \geq \frac{1}{nI(\theta)} \quad .$$

Proof.

Definition 9.1 (Efficient estimator)**.** An estimator is said to be efficient when its variance is equal to the Cramer–Rao lower bound for any value of θ^*.

Example 9.1 (More coin flipping)**.** Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ with $\theta \in [0,1]$, which corresponds to flipping a coin n times and considering for each flip the random variable $X : \{H, T\} \to \{0, 1\}$ such that $X(H) = 1$ and $X(T) = 0$ with probability distribution such that $\mathbb{P}(X = 1; \theta) = \theta$ and $\mathbb{P}(X = 0; \theta) = 1 - \theta$. We consider the intuitive estimator of θ

$$\hat{\theta}_n := \hat{\theta}_n(\mathbf{X}_n) := \frac{1}{n} \sum_{i=1}^{n} X_i.$$

This estimator is unbiased for any $n \in \mathbb{N}$ and its variance is

$$\text{var}(\hat{\theta}_n; \theta) = \frac{\text{var}(X_1; \theta)}{n} = \frac{\mathbb{E}(X_1^2; \theta) - \mathbb{E}(X_1; \theta)^2}{n} = \frac{\theta - \theta^2}{n} = \frac{\theta(1-\theta)}{n}.$$

Now we consider the Cramer–Rao bound.

Example 9.2. Continuing the example above, we can consider alternative estimators, such as

$$\tilde{\theta}_n = \frac{\sum_{i=1}^n \alpha_i X_i}{n},$$

with $\alpha_1, \ldots, \alpha_n$ fixed constants satisfying $\sum_{i=1}^n \alpha_i = n$. Such estimators are also unbiased but their variance is

$$\operatorname{var}\left(\tilde{\theta}_n; \theta\right) = \frac{\operatorname{var}(X_1; \theta) \sum_{i=1}^n \alpha_i^2}{n^2},$$

and it can be shown that taking $\alpha_1 = \cdots = \alpha_n = 1$ is optimal for minimizing the variance.

Indeed, the standard Cauchy–Schwarz inequality says that for vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$,

$$\left(\sum_{i=1}^n u_i v_i\right)^2 = \|\mathbf{u} \cdot \mathbf{v}\|_2^2 \leq \|\mathbf{u}\|_2^2 \|\mathbf{v}\|_2^2 = \left(\sum_{i=1}^n u_i^2\right)\left(\sum_{i=1}^n v_i^2\right).$$

Hence, taking $u_i = \alpha_i$ and $v_i = 1$ for $i \in \{1, \ldots, n\}$ we have

$$\left(\sum_{i=1}^{n} \alpha_i \right)^2 \leq n \left(\sum_{i=1}^{n} \alpha_i^2 \right),$$

and using $\sum_{i=1}^{n} \alpha_i = n$, we obtain

$$\sum_{i=1}^{n} \alpha_i^2 \geq n,$$

and we notice that equality occurs when $\alpha_1 = \cdots = \alpha_n = 1$ which is therefore the optimal choice in terms of variance. Other choices of the α_i are not efficient, as you can see by trying out some values.

10 Asymptotic distribution of the MLE

In this lecture we prove that sequences of ML estimators, appropriately centred and scaled, are asymptotically normally distributed under some regularity assumptions.

Theorem 10.1. *Suppose that* $\mathbf{X}_n \overset{\text{iid}}{\sim} f(\cdot;\theta^*)$ *for some* $\theta^* \in \Theta$ *and assume that*

1. the sequence of ML estimators $\{\hat{\theta}_n(\mathbf{X}_n)\}$ is consistent,
2. (A1) holds and $I(\theta^*) = -\mathbb{E}\left[\ell''(\theta^*;X);\theta^*\right] > 0$,
3. there exists $C(\cdot) : \mathcal{X} \to [0,\infty)$ such that $\mathbb{E}(C(X_1);\theta^*) < \infty$, $\Xi \subset \Theta$ an open set containing θ^* and $\Delta(\cdot) : \Xi \to [0,\infty)$ continuous at 0 such that $\Delta(0) = 0$, such that for any $\theta, \theta', x \in \Xi^2 \times \mathcal{X}$

$$|\ell''(\theta;x) - \ell''(\theta';x)| \leq C(x)\Delta(\theta - \theta') \quad .$$

Then for any $\theta^* \in \Theta$

$$\sqrt{nI(\theta^*)}(\hat{\theta}_n(\mathbf{X}_n) - \theta^*) \to_{\mathcal{D}(\cdot;\theta^*)} Z$$

where $Z \sim \mathcal{N}(0,1)$.

Lemma 10.1. *Under the conditions of the Theorem, with* $\hat{\theta}_n = \hat{\theta}_n(\mathbf{X})$ *the ML estimator,*

$$\ell_n'(\hat{\theta}_n;\mathbf{X}) = \ell_n'(\theta^*;\mathbf{X}) + (\hat{\theta}_n - \theta^*)\left\{\ell_n''(\theta^*;\mathbf{X}) + R_n\right\},$$

where $\frac{1}{n}R_n \to_{\mathbb{P}(\cdot;\theta^*)} 0$.

Proof of Theorem 10.1.

The following proof sketch is for information and not examinable.

Sketch of proof of Lemma 10.1. By the regularity conditions and the mean value theorem

$$\frac{\ell_n'(\theta;\mathbf{x}) - \ell_n'(\theta^*;\mathbf{x})}{\theta - \theta^*} = \ell_n''(\tilde{\theta};\mathbf{x}),$$

for some $\tilde{\theta} \in (\theta, \theta^*)$. Hence, we deduce that

$$\begin{aligned}
\ell_n'(\theta;\mathbf{x}) - \ell_n'(\theta^*;\mathbf{x}) &= (\theta - \theta^*)\ell_n''(\tilde{\theta};\mathbf{x}) \\
&= (\theta - \theta^*)\left\{\ell_n''(\theta^*;\mathbf{x}) + \left[\ell_n''(\tilde{\theta};\mathbf{x}) - \ell_n''(\theta^*;\mathbf{x})\right]\right\} \\
&= (\theta - \theta^*)\left\{\ell_n''(\theta^*;\mathbf{x}) + R_n(\theta, \theta^*, \mathbf{x})\right\}.
\end{aligned}$$

Now we replace θ with the ML estimator $\hat{\theta}_n = \hat{\theta}_n(\mathbf{X})$. We find

$$\ell'(\hat{\theta}_n;\mathbf{X}) = \ell_n'(\theta^*;\mathbf{X}) + (\hat{\theta}_n - \theta^*)\left\{\ell_n''(\theta^*;\mathbf{X}) + R_n(\hat{\theta}_n, \theta^*, \mathbf{X})\right\},$$

and we need to analyze R_n. Since $\hat{\theta}_n \to_{\mathbb{P}(\cdot;\theta^*)} \theta^*$ we can take n large enough that $\mathbb{P}\left(\hat{\theta}_n \in \Xi; \theta^*\right)$ with arbitrarily high probability. On the event $\{\hat{\theta}_n \in \Xi\}$, we have also that $\{\tilde{\theta}_n \in \Xi\}$ since $\tilde{\theta}_n \in (\hat{\theta}_n, \theta^*)$ and,

$$\begin{aligned}
\left|\frac{1}{n}R_n\right| &= \frac{1}{n}\left|\ell_n''(\tilde{\theta}_n;\mathbf{X}) - \ell_n''(\theta^*;\mathbf{X})\right| \\
&= \frac{1}{n}\left|\sum_{i=1}^n \ell''(\tilde{\theta}_n;X_i) - \ell''(\theta^*;X_i)\right| \\
&\leq \frac{1}{n}\sum_{i=1}^n \left|\ell''(\tilde{\theta}_n;X_i) - \ell''(\theta^*;X_i)\right| \\
&\leq \Delta(\tilde{\theta}_n - \theta^*)\left\{\frac{1}{n}\sum_{i=1}^n C(X_i)\right\},
\end{aligned}$$

from the smoothness condition on ℓ''. From the WLLN,

$$\frac{1}{n}\sum_{i=1}^n C(X_i) \to_{\mathbb{P}(\cdot;\theta^*)} \mathbb{E}\left[C(X_1);\theta^*\right] < \infty$$

and from the consistency of $\{\hat{\theta}_n\}$ and $\{\tilde{\theta}_n\}$ and continuity of $\Delta(\cdot)$ we have by the Continuous Mapping Theorem

$$\Delta\left(\tilde{\theta}_n - \theta^*\right) \to_{\mathbb{P}(\cdot;\theta^*)} 0.$$

Hence, $\frac{1}{n}R_n \to_{\mathbb{P}(\cdot;\theta^*)} 0$. □

Corollary 10.1. *Under the conditions of the theorem the ML estimator is "asymptotically efficient".*

Remark 10.1. One should be careful about what we mean here by asymptotic efficiency. Specifically, we say that a sequence of estimators $\{\hat{\theta}_n(\mathbf{X})\}$ is *asymptotically efficient* if either its mean-squared error converges to the Cramer–Rao lower bound for all $\theta \in \Theta$ as $n \to \infty$,

$$n\text{MSE}(\hat{\theta}_n(\mathbf{X}_n); \theta) \to \frac{1}{I(\theta)},$$

or $\hat{\theta}_n$ is asymptotically normally distributed in the sense of Theorem 10.1, i.e. for all $\theta \in \Theta$,

$$\sqrt{nI(\theta)}(\hat{\theta}_n(\mathbf{X}_n) - \theta) \to_{\mathcal{D}(\cdot;\theta)} Z.$$

Notice that the variance of $Z/\sqrt{nI(\theta^*)}$ is exactly $1/[nI(\theta)]$. We are concerned with the shape of the distribution, which captures "variability" around θ^*, and not necessarily the actual variance or MSE of $\hat{\theta}_n(\mathbf{X}_n)$.

In many cases there is no distinction, but there can be, as we shall see later.

Definition 10.1. In this unit we will refer to statistical models satisfying the conditions of Theorem 10.1 as *regular statistical models*.

11 Confidence sets around the ML estimator

In the previous section we have established that the ML estimator is asymptotically normal and asymptotically efficient for regular statistical models. This justifies in some sense its use when the number of observations is sufficiently large. Now we would like to quantify uncertainty about the maximum likelihood estimator $\hat{\theta}_n$ in a systematic way. Confidence intervals are a way of addressing this issue. We primarily focus on the case where θ is a scalar parameter.

Definition 11.1 (Coverage of a random interval)**.** Let $\mathbf{X} \sim f_n(\cdot;\theta)$, $\theta \in \Theta = \mathbb{R}$ and let $L(\cdot) : \mathcal{X}^n \to \Theta$ and $U(\cdot) : \mathcal{X}^n \to \Theta$, where for any $\mathbf{x} \in \mathcal{X}^n$, $L(\mathbf{x}) < U(\mathbf{x})$. For any $\theta \in \Theta$, the coverage $C_{\mathcal{I}}(\theta)$ of the **random interval** $\mathcal{I}(\mathbf{X}) := [L(\mathbf{X}), U(\mathbf{X})]$ at θ is

$$C_{\mathcal{I}}(\theta) := \mathbb{P}(\theta \in [L(\mathbf{X}), U(\mathbf{X})]; \theta) = \mathbb{P}(L(\mathbf{X}) \leq \theta \leq U(\mathbf{X}); \theta) \quad .$$

In words, $C_{\mathcal{I}}(\theta)$ is the probability that the **deterministic** quantity θ falls into the random interval $\mathcal{I}(\mathbf{X})$ under the probability distribution $\mathbb{P}(\cdot;\theta)$, i.e. when $\mathbf{X} \sim f_n(\cdot;\theta)$.

Remark 11.1. This notion of coverage can be naturally extended to the case where θ is multidimensional and one then talks about confidence sets (e.g. $\mathcal{I}(\mathbf{x})$ can be taken to be a rectangle) and consider the probabilities $\mathbb{P}(\theta \in \mathcal{I}(\mathbf{X}); \theta)$.

We can now define confidence intervals.

Definition 11.2 ($1-\alpha$ confidence interval)**.** For any $\alpha \in [0,1]$, we say that an interval $\mathcal{I}(\mathbf{X}) := [L(\mathbf{X}), U(\mathbf{X})]$ is a $1-\alpha$ confidence interval (or confidence interval of level $1-\alpha$) if for all $\theta \in \Theta$ its coverage is at least $1-\alpha$, or more formally $\inf_{\theta \in \Theta} C_{\mathcal{I}}(\theta) \geq 1-\alpha$.

Remark 11.2. If $C_{\mathcal{I}}(\theta) = 1-\alpha$ for every $\theta \in \Theta$ then $\mathcal{I}$ is an exact $1-\alpha$ confidence interval.

Informally a, say, 95% confidence interval $\mathcal{I}(\mathbf{X})$ for $\mathbf{X} \sim f_n(\cdot;\theta)$ is such that for any possible value of the parameter $\theta \in \Theta$ if we observe a large number of realizations $\{\mathbf{x}_i, i = 1, \ldots, m\}$ of independent $\mathbf{X}_i \sim f_n(\cdot;\theta)$, then we expect an approximate proportion of 95% of the observed intervals $\{\mathcal{I}(\mathbf{x}_i)\}$ to contain the parameter θ.

Here is some R code to illustrate this fact, together with a figure that summarizes the results (we will see later how one theoretically constructs these confidence intervals).

```
p <- 0.5
m <- 100 # no. of experiments
n <- 20 # no. of Bernoulli trials in each experiment

# simulate m different Binomial(n,p) realizations
experiments <- rbinom(m, n, p)
phat <- experiments/n

alpha <- 0.05
z.value <- qnorm(1-alpha/2)

U <- phat + z.value*sqrt(p*(1-p))/sqrt(n)
L <- phat - z.value*sqrt(p*(1-p))/sqrt(n)
experiment.numbers <- seq(1, m, 1)
plot(experiment.numbers, phat, xlim=c(0, 100), ylim=c(0, 1))
for (i in experiment.numbers) {
  lines(c(i, i), c(L[i],U[i]), col=2, lty='dashed')
  abline(p,0)
}
```

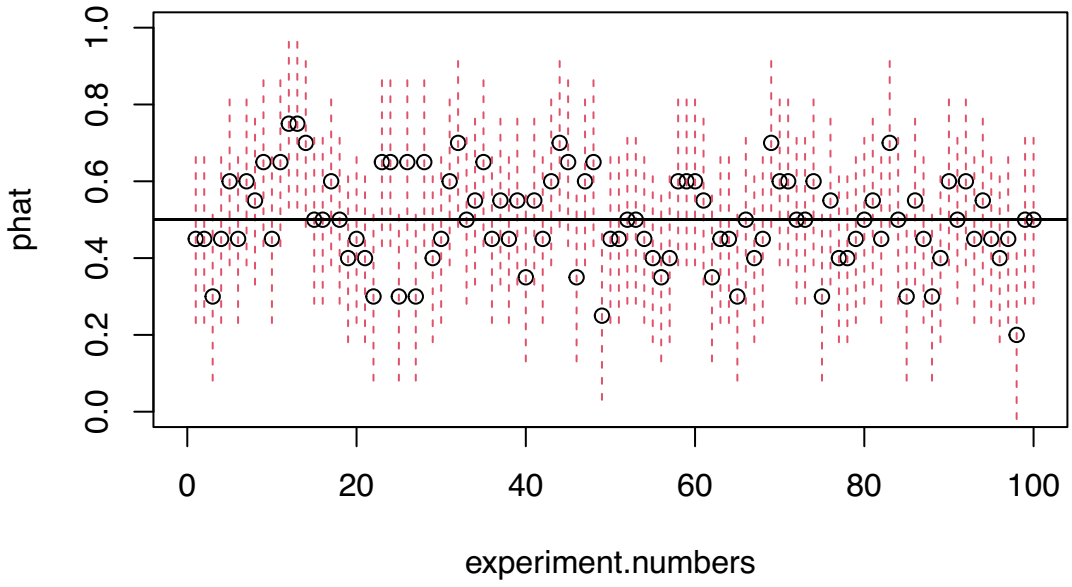


Figure 11.1: Observed confidence intervals for $m = 100$ realizations of the experiment "Toss a coin $n = 20$" times in order to estimate $p^* = 0.5$. The circles represent the $\hat{p}_{20}$'s and are the centres of the confidence intervals.

Remark 11.3. As above one can generalize the notion of confidence interval to the case where θ is multivariate and talks about confidence sets with level $1-\alpha$.

Definition 11.3 (Observed confidence interval)**.** For an interval $\mathcal{I}(\cdot) = [L(\cdot), U(\cdot)]$ with $L : \mathcal{X}^n \to \Theta$ and $U : \mathcal{X}^n \to \Theta$, and a realization $\mathbf{x}$, the corresponding *observed* confidence interval is $\mathcal{I}(\mathbf{x})$.

Remark 11.4. Notice that absolutely nothing interesting can be said about the probability that $\theta \in \mathcal{I}(\mathbf{x})$, since θ and $\mathcal{I}(\mathbf{x})$ are deterministic.

We now consider an example, which turns out to be central in what follows. It is useful to use a shorthand for the $1-\beta$ quantile of the standard normal distribution: for any $\beta \in (0,1)$, let $z_\beta \in \mathbb{R}$ be such that for $Z \sim \mathcal{N}(0,1)$, $1-\Phi(z_\beta) = \mathbb{P}(Z > z_\beta) = \beta$, i.e. $\Phi(z_\beta) = \mathbb{P}(Z \leq z_\beta) = 1-\beta$. Alternatively, $z_\beta := \Phi^{-1}(1-\beta)$. [in R, we could compute this as `qnorm(1-`β`)`]

Example 11.1 (Confidence interval for the mean of a normal distribution)**.** Assume that $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ for $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_+$ and where σ^2 is assumed known. Consider the estimator $\hat{\mu}_n = \hat{\mu}_n(\mathbf{X}) = \sum_{i=1}^n X_i/n$ of μ. Then we know that the following non-asymptotic result holds

Now imagine what we might do if we did not know σ^2. We could define $\{\hat{\sigma}_n^2\}$ to be a consistent sequence of estimators of σ^2, e.g. via the sample variance:

$$\hat{\sigma}_n^2(\mathbf{X}) = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \hat{\mu}_n(\mathbf{X})\right)^2,$$

and in the next section we will see that this implies that the random interval

$$\left[\hat{\mu}_n(\mathbf{X}) - z_{\alpha/2}\sqrt{\frac{\hat{\sigma}_n^2(\mathbf{X})}{n}}, \hat{\mu}_n(\mathbf{X}) + z_{\alpha/2}\sqrt{\frac{\hat{\sigma}_n^2(\mathbf{X})}{n}}\right]$$

is an *asymptotically exact* $1-\alpha$ confidence interval.

As already mentioned several times, it is only in rare situations that one can obtain a tractable expression for the distribution of an estimator. We hence turn to asymptotic results which exploit the implications of the central limit theorem for sequences of ML estimators $\{\hat{\theta}_n\}$. It should come as no surprise that the particular example treated above is very relevant in the latter scenario.

Final note: the fact that one cannot assign an interesting probability to the event that θ lies in an observed confidence interval, $\theta \in \mathcal{I}(\mathbf{x})$, is essentially the same as the fact that we cannot make probabilistic statements about realizations of random variables. For example, if $X \sim \text{Bernoulli}(1/2)$ then $\mathbb{P}(X=0) = 1/2$, but if x is a realization of X then $\mathbb{P}(x=0) = \mathbb{I}(x=0) \in \{0,1\}$; the probability is certainly not $1/2$. Similarly, imagine θ takes values in $\Theta = [0,1]$ and we have $X \sim f(\cdot;\theta)$ and $U \sim \text{Uniform}(0,1)$. A totally useless but technically exact 95% CI for θ can be constructed as

$$\mathcal{I}(X,U) = \begin{cases} [0,1] & U \leq 0.95, \\ \{2\} & U > 0.95. \end{cases}$$

For a realization (x,u) we have $\mathbb{P}(\theta \in \mathcal{I}(x,u)) = \mathbb{I}(u \leq 0.95) \in \{0,1\}$.

12 Asymptotic approximations of CIs

Theorem 12.1. *Assume* $\mathbf{X} \sim f(\cdot;\theta^*)$*, let* $\{\hat{\theta}_n\}$ *be a consistent sequence of estimators of* θ^* *(not necessarily ML estimators) and assume that* $\{\hat{\theta}_n\}$ *is asymptotically normal in the sense that for some* $\sigma^2 > 0$*,*

$$\frac{\hat{\theta}_n(\mathbf{X}) - \theta^*}{\sqrt{\sigma^2/n}} \to_{\mathcal{D}(\cdot;\theta^*)} Z \sim \mathcal{N}(0,1) \quad .$$

Then for any $\alpha \in (0,1)$*,* $\mathcal{I}_n(\mathbf{X}) = [L_n(\mathbf{X}), U_n(\mathbf{X})]$ *is an asymptotically exact* $1-\alpha$ *confidence interval, where* $L_n(\mathbf{x}) := \hat{\theta}_n(\mathbf{x}) - z_{\alpha/2}\sigma/\sqrt{n}$ *and* $U_n(\mathbf{x}) := \hat{\theta}_n(\mathbf{x}) + z_{\alpha/2}\sigma/\sqrt{n}$*.*

Proof.

Remark 12.1. The confidence interval is only asymptotically exact. For finite n, the coverage of the confidence interval will be different from $1-\alpha$ but the difference will converge to 0 as n increases.

In practice σ^2 may be (and often will be) unknown. Fortunately, it is possible to substitute for σ^2 a consistent sequence of estimators.

Theorem 12.2. *Assume $\mathbf{X} \sim f(\cdot;\theta^*)$, let $\{\hat{\theta}_n\}$ be a consistent sequence of estimators of θ^* and assume that $\{\hat{\theta}_n\}$ is asymptotically normal in the sense that for some $\sigma^2 > 0$,*

$$\frac{\hat{\theta}_n(\mathbf{X}) - \theta^*}{\sqrt{\sigma^2/n}} \to_{\mathcal{D}(\cdot;\theta^*)} Z \sim \mathcal{N}(0,1) \quad .$$

Assume also that $\{\hat{\sigma}_n^2\}$ is a consistent sequence of estimators of σ^2. Then for any $\alpha \in (0,1)$, $\mathcal{I}_n(\mathbf{X}) = [L_n(\mathbf{X}), U_n(\mathbf{X})]$ is an asymptotically exact $1-\alpha$ confidence interval, where $L_n(\mathbf{x}) := \hat{\theta}_n(\mathbf{x}) - z_{\alpha/2}\sqrt{\hat{\sigma}_n^2(\mathbf{x})/n}$ and $U_n(\mathbf{x}) := \hat{\theta}_n(\mathbf{x}) + z_{\alpha/2}\sqrt{\hat{\sigma}_n^2(\mathbf{x})/n}$.

Proof.

Remark 12.2. For a given n, the quality of the normal approximation will be affected by this additional approximation. One may also find that for less accurate estimators of σ^2, the n required for the confidence interval to have almost the right coverage will be higher.

13 Estimating the information for ML estimators

Now we turn to the application of this result to sequences of ML estimators for regular statistical models (i.e. asymptotic normality holds). In this case, we have $\sigma^2 = 1/I\big(\theta^*\big)$ so

$$\sqrt{nI(\theta^*)}(\hat{\theta}_n - \theta^*) \to_{\mathcal{D}(\cdot;\theta^*)} Z$$

with $Z \sim \mathcal{N}(0,1)$. The Fisher information is unknown [even if $I(\theta)$ can be computed for any θ, θ^* is unknown]. One can therefore suggest replacing θ^* by $\hat{\theta}_n$ in $I(\theta^*) = -\mathbb{E}(\ell''(\theta^*; X_1); \theta^*)$ when the expectation can be calculated, or when the expectation is intractable one can invoke a WLLN and consider the following sequence of estimators

$$J_n(\hat{\theta}_n) := -\frac{1}{n}\sum_{i=1}^{n} \ell''(\hat{\theta}_n; X_i),$$

for which the question of consistency might not be obvious as there are *two* approximations. We treat these cases separately.

Theorem 13.1. *Assume $\{\hat{\theta}_n\}$ is a sequence of ML estimators such that $\hat{\theta}_n \to_{\mathbb{P}(\cdot;\theta^*)} \theta^*$ and I is a continuous function of θ. Then $I(\hat{\theta}_n) \to_{\mathbb{P}(\cdot;\theta^*)} I(\theta^*)$.*

Proof. This follows directly from the Continuous Mapping Theorem. □

Remark 13.1. It is only really necessary that I be continuous in a neighborhood of θ^*. This is due to an extension of the CMT: if $X_n \to_{\mathbb{P}} X$ and g is a function with discontinuity set D then $\mathbb{P}(X \in D) = 0$ implies $g(X_n) \to_{\mathbb{P}} g(X)$.

The second case is more complex: we are dealing with two different types of convergence. First, we shall assume that $\hat{\theta}_n \to_{\mathbb{P}(\cdot;\theta^*)} \theta^*$ and also that $J_n(\theta) \to_{\mathbb{P}(\cdot;\theta)} I(\theta)$ for any $\theta \in \Theta$ by the WLLN [notice however that this is convergence in probability w.r.t. $\mathbb{P}(\cdot;\theta)$]. In order to make progress, we need additional assumptions to ensure that $J_n(\hat{\theta}_n) \to_{\mathbb{P}(\cdot;\theta^*)} I(\theta^*)$.

Theorem 13.2. *Assume that $\{\hat{\theta}_n\}$ is a sequence of ML estimators such that*

1. $\hat{\theta}_n \to_{\mathbb{P}(\cdot;\theta^*)} \theta^*$,
2. $I(\theta) = -\mathbb{E}[\ell''(\theta; X); \theta]$ for every $\theta \in \Theta$
3. there exist $C : \mathcal{X} \to [0, \infty)$ such that $\mathbb{E}(C(X_1); \theta^*) < \infty$, $\Xi \subset \Theta$ an open set containing θ^* and $\Delta(\cdot) : \Xi \to [0, \infty)$ continuous at 0 such that $\Delta(0) = 0$, and such that for any $\theta, \theta', x \in \Xi^2 \times \mathcal{X}$

$$|\ell''(\theta; x) - \ell''(\theta'; x)| \leq C(x)\Delta(\theta - \theta') \quad .$$

Then

$$J_n(\hat{\theta}_n) \to_{\mathbb{P}(\cdot;\theta^*)} I(\theta^*).$$

Proof. We are going to use the following decomposition

$$\begin{aligned} J_n(\hat{\theta}_n) - I(\theta^*) &= -\frac{1}{n}\sum_{i=1}^{n} \ell''(\hat{\theta}_n; X_i) - I(\theta^*) \\ &= T_1 + T_2, \end{aligned}$$

where

$$T_1 = -\frac{1}{n}\sum_{i=1}^{n} \ell''(\hat{\theta}_n; X_i) + \frac{1}{n}\sum_{i=1}^{n} \ell''(\theta^*; X_i)$$
$$T_2 = -\left\{\frac{1}{n}\sum_{i=1}^{n} \ell''(\theta^*; X_i)\right\} - I(\theta^*)$$

Now the first term can be upper bounded as follows (for n sufficiently large, with arbitrary large probability the second inequality holds)

$$\begin{aligned}
|T_1| &= \left| -\frac{1}{n}\sum_{i=1}^{n} \ell''(\hat{\theta}_n, X_i) + \frac{1}{n}\sum_{i=1}^{n} \ell''(\theta^*, X_i) \right| \\
&\leq \frac{1}{n}\sum_{i=1}^{n} \left| \ell''(\hat{\theta}_n, X_i) - \ell''(\theta^*, X_i) \right| \\
&\leq \Delta(\hat{\theta}_n - \theta^*) \frac{1}{n}\sum_{i=1}^{n} C(X_i).
\end{aligned}$$

Now from the WLLN,

$$\frac{1}{n}\sum_{i=1}^{n} C(X_i) \to_{\mathbb{P}(\cdot;\theta^*)} \mathbb{E}\left[C(X_1); \theta^*\right],$$

and by the assumed consistency of $\{\hat{\theta}_n\}$ and continuity of Δ we have that

$$\Delta(\hat{\theta}_n - \theta^*) \to_{\mathbb{P}(\cdot;\theta^*)} 0$$

and consequently $T_1 \to_{\mathbb{P}(\cdot;\theta^*)} 0$ as $n \to \infty$. Now by the WLLN we have

$$-\frac{1}{n}\sum_{i=1}^{n} \ell''(\theta^*; X_i) \to_{\mathbb{P}(\cdot;\theta^*)} I(\theta^*)$$

and hence

$$T_2 = -\frac{1}{n}\sum_{i=1}^{n} \ell''(\theta^*; X_i) - I(\theta^*) \to_{\mathbb{P}(\cdot;\theta^*)} 0$$

Now since $T_1 \to_{\mathbb{P}(\cdot;\theta^*)} 0$ and $T_2 \to_{\mathbb{P}(\cdot;\theta^*)} 0$ as $n \to \infty$ we deduce from the earlier decomposition that

$$J_n(\hat{\theta}_n) \to_{\mathbb{P}(\cdot;\theta^*)} I(\theta^*).$$

□

To summarize our results, whenever Theorem 10.1 holds for a sequence of MLE estimators, i.e.

$$\sqrt{nI(\theta^*)}(\hat{\theta}_n - \theta^*) \to_{\mathcal{D}(\cdot;\theta^*)} Z$$

one can replace $I(\theta^*)$

1. with $I(\hat{\theta}_n)$ above whenever $I(\theta)$ is continuous in a neighborhood of θ^* and the interval $[L(\mathbf{X}), U(\mathbf{X})]$ with $L(\mathbf{x}) := \hat{\theta}_n - z_{\alpha/2}/\sqrt{nI(\hat{\theta}_n)}$ and $U(\mathbf{x}) := \hat{\theta}_n + z_{\alpha/2}/\sqrt{nI(\hat{\theta}_n)}$ is an asymptotically exact $1-\alpha$ confidence interval for θ^*.

2. with $J_n(\hat{\theta}_n) := -\frac{1}{n}\sum_{i=1}^n \ell''(\hat{\theta}_n; X_i)$ under the assumptions of the last theorem above, and the interval $[L(\mathbf{X}), U(\mathbf{X})]$ with $L(\mathbf{x}) := \hat{\theta}_n - z_{\alpha/2}/\sqrt{nJ_n(\hat{\theta}_n)}$ and $U(\mathbf{x}) := \hat{\theta}_n + z_{\alpha/2}/\sqrt{nJ_n(\hat{\theta}_n)}$ is an asymptotically exact $1-\alpha$ confidence interval for θ^*.

Example 13.1 (Coin example again)**.** We have seen earlier a method to find $L(\mathbf{x})$ and $U(\mathbf{x})$ to ensure that θ^* (the probability of observing a head H) is in $[L(\mathbf{X}), U(\mathbf{X})]$ with at least a probability of 95% by using Markov's inequality and the upper bound $\theta^*(1-\theta^*) \leq 1/4$. We also saw that one could obtain an asymptotically exact confidence interval can be constructed using a careful combination of the CLT, CMT and Slutsky's Theorem.

Now, however, we can apply our new results to simplify the situation considerably. The sequence of estimators $\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^n X_i$ is consistent by the WLLN and the conditions for asymptotic normality to hold are satisfied. Hence

$$\sqrt{nI(\theta^*)}(\hat{\theta}_n - \theta^*) \to_{\mathcal{D}(\cdot;\theta^*)} Z \sim \mathcal{N}(0,1).$$

The numerical illustration of the meaning of confidence interval presented at the beginning of Section 11 relies on this result and the fact that $\theta^* = 1/2$ was known in this case. Indeed the Fisher information can be computed for any $\theta \in \Theta$. We have

$$\ell'(\theta; x) = x/\theta - (1-x)/(1-\theta)$$

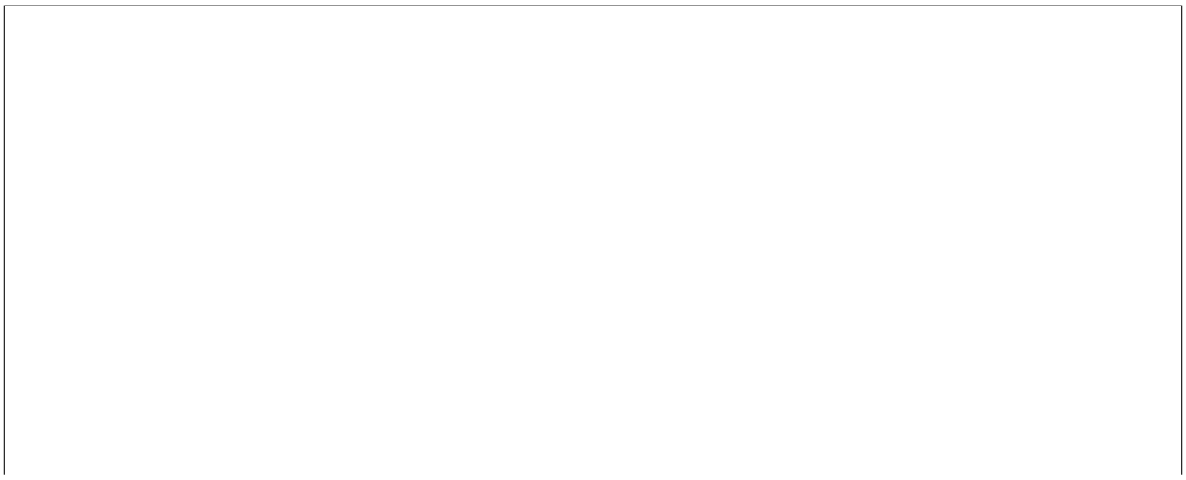
and

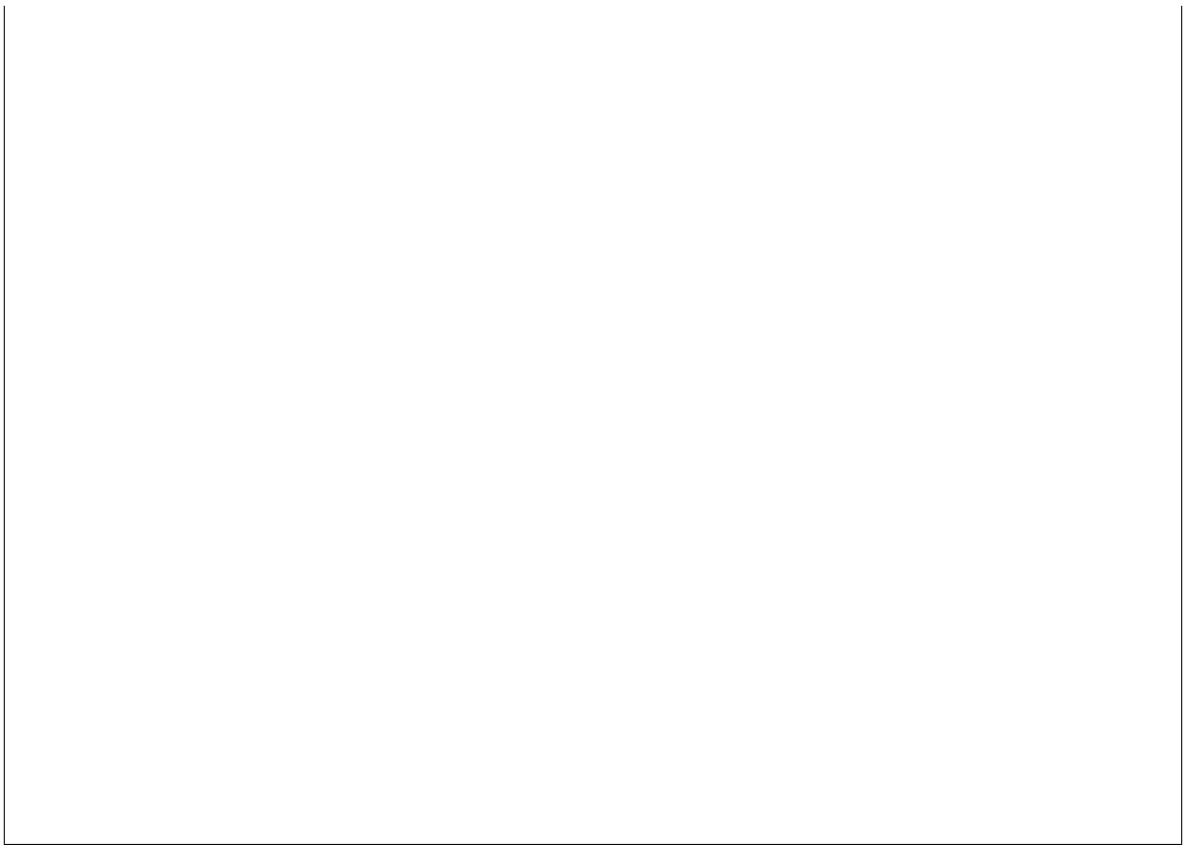
$$\ell''(\theta; x) = -x/\theta^2 - (1-x)/(1-\theta)^2$$

and therefore

$$I(\theta) = 1/\theta + 1/(1-\theta) = 1/[\theta(1-\theta)]$$

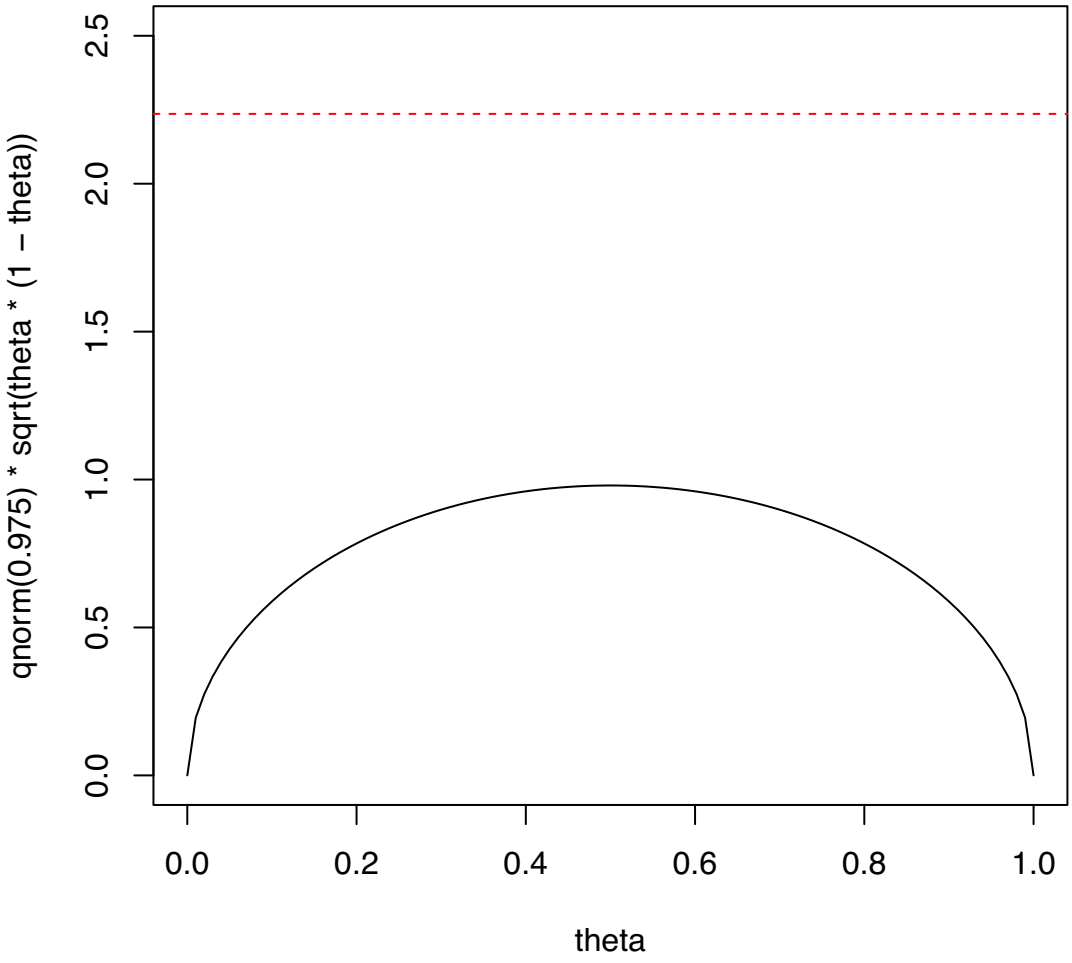
(you can now check the R code of the aforementioned example and in particular the expressions for $L(\mathbf{x})$ and $U(\mathbf{x})$).





Remark 13.2. We notice that the approximation made earlier in the course led to a 95% confidence interval of the form $[\hat{\theta}_n - \frac{1}{2\sqrt{n\alpha}}, \hat{\theta}_n + \frac{1}{2\sqrt{n\alpha}}]$ for any $n \geq 1$ and θ^*. Now, taking $\alpha = 0.05$ one can plot $z_{0.025}\sqrt{\theta(1-\theta)}$ and $\frac{1}{2\sqrt{\alpha}}$ for $\theta \in [0,1]$ and we confirm that the cruder confidence interval is very pessimistic, at least for n sufficiently large.

```
theta <- seq(0,1,.01)
plot(theta, qnorm(.975)*sqrt(theta*(1-theta)), xlim=c(0, 1), ylim=c(0, 2.5), type='l')
abline(1/2/sqrt(0.05),0,col='red',lty='dashed')
```



Remark 13.3. Let $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot; \theta^*)$ be a vector of n independent random variables. The quantity

$$nJ_n(\theta) = -\ell''(\theta; \mathbf{X}) = -\sum_{i=1}^{n} \ell''(\theta; X_i),$$

is known as the *observed Fisher information* at θ. Notice that $\mathbb{E}\left[J_n(\theta^*); \theta^*\right] = I(\theta^*)$, and that it differs from the Fisher information (under the FIRCs) by not being an expectation.

14 Transformations and confidence intervals

What we have seen so far is the **Wald approach** to the specification of CIs for θ^* using the asymptotic normality of the ML estimator: if $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot;\theta^*)$ where $\theta^* \in \Theta \subset \mathbb{R}$ then one can define a CI for θ^* using the asymptotic distribution of the ML estimator, i.e. [we can also plug-in appropriate estimates of $I(\theta^*)$]

$$L(\mathbf{x}) = \hat{\theta}_n - z_{\alpha/2}/\sqrt{nI(\theta^*)}$$
$$U(\mathbf{x}) = \hat{\theta}_n + z_{\alpha/2}/\sqrt{nI(\theta^*)},$$

which ensure that, as $n \to \infty$,

$$\mathbb{P}(L(\mathbf{X}) \leq \theta^* \leq U(\mathbf{X}); \theta^*) \to 1 - \alpha.$$

Now assume that we parameterize the problem in terms of $\tau = g(\theta)$ where g is bijective and continuously differentiable. What is a CI for τ? We investigate two approaches, which do not lead to the same result.

The first is a transformed CI. When g is increasing, a direct transformation of the confidence interval $[L(\mathbf{x}), U(\mathbf{x})]$ for θ^* leads to the interval $[g(L(\mathbf{x})), g(U(\mathbf{x}))]$. That is, we notice that

$$\{\mathbf{x} \in \mathcal{X}^n \,:\, L(\mathbf{x}) \leq \theta^* \leq U(\mathbf{x})\} = \{\mathbf{x} \in \mathcal{X}^n \,:\, g(L(\mathbf{x})) \leq \tau^* = g(\theta^*) \leq g(U(\mathbf{x}))\},$$

and consequently we have that

$$\mathbb{P}(L(\mathbf{X}) \leq \theta^* \leq U(\mathbf{X}); \theta^*) = \mathbb{P}\left(g(L(\mathbf{X})) \leq \tau^* = g(\theta^*) \leq g(U(\mathbf{X})); \theta^*\right) \to 1 - \alpha,$$

as $n \to \infty$. That is, $[g(L(\mathbf{X})), g(U(\mathbf{X}))]$ is an asymptotically exact $1 - \alpha$ CI for τ^*. When g is decreasing we simply change the direction of the inequalities above and $[g(U(\mathbf{X})), g(L(\mathbf{X}))]$ is then an asymptotically exact $1 - \alpha$ CI for τ^*.

The second CI is obtained by working with the reparameterized likelihood, that is with

$$\tilde{f}(\mathbf{x}; \tau) := f(\mathbf{x}; g^{-1}(\tau)).$$

We can find $\tilde{L}(\mathbf{x})$ and $\tilde{U}(\mathbf{x})$ directly. Clearly, we have to consider the ML estimator of τ and the Fisher information corresponding to τ using $\tilde{f}(\mathbf{x}; \tau)$. The former is easy since we have already established that the ML estimates/estimators satisfy $\hat{\tau} = g(\hat{\theta})$ while the latter motivates the following lemma.

Lemma 14.1. *Assume $X \sim f(\cdot;\theta)$ for $\theta \in \Theta \subseteq \mathbb{R}$, and let $\tau = g(\theta)$. Then, with g bijective and continuously differentiable, the Fisher information for the parameterization $\tilde{f}(x;\tau) := f(x; g^{-1}(\tau))$ is*

$$\tilde{I}(\tau) = \frac{I(\theta)}{g'(\theta)^2}.$$

Proof.



Corollary 14.1. *As a consequence for regular statistical models*

$$\sqrt{n\tilde{I}(\tau^*)}\,(\hat{\tau}_n - \tau^*) \to_{\mathcal{D}(\cdot;\tau^*)} Z \sim \mathcal{N}(0,1),$$

is equivalent to

$$\sqrt{nI(\theta^*)/g'(\theta^*)^2}\,(\hat{\tau}_n - \tau^*) \to_{\mathcal{D}(\cdot;\theta^*)} Z \sim \mathcal{N}(0,1),$$

which leads to the CI boundaries [we can also plug-in appropriate estimates of $I(\theta^*)$]

$$\tilde{L}(\mathbf{x}) = \hat{\tau}_n - z_{\alpha/2}\sqrt{\frac{g'(\theta^*)^2}{nI(\theta^*)}}$$

$$\tilde{U}(\mathbf{x}) = \hat{\tau}_n + z_{\alpha/2}\sqrt{\frac{g'(\theta^*)^2}{nI(\theta^*)}}.$$

This does not define the same CI as that obtained by transforming $[L(\mathbf{x}), U(\mathbf{x})]$ directly, in general.

This is not completely satisfactory from the perspective of choosing a parameterization. The next section is dedicated to another way of constructing CIs which remain invariant under reparameterizations, and also have the property that they include those values of θ with the largest likelihood values.

Example 14.1. Consider $\mathbf{X} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$. We know that the ML estimator of μ is $\hat{\mu}_n = \bar{X} \sim \mathcal{N}(\mu, \frac{1}{n})$. An exact $1-\alpha$ CI for μ is

$$\mathcal{I}(\mathbf{X}) = \left[\bar{X} - \frac{z_{\alpha/2}}{\sqrt{n}}, \bar{X} + \frac{z_{\alpha/2}}{\sqrt{n}}\right].$$

Now consider the parameterization $\tau = 1/\mu$, corresponding to the function $g = x \mapsto 1/x$, which is bijective and continuously differentiable except at 0, and decreasing. Hence, if $0 \notin \mathcal{I}(\mathbf{X})$ an exact $1-\alpha$ CI for τ is

$$\left[\frac{1}{\bar{X} + z_{\alpha/2}/\sqrt{n}}, \frac{1}{\bar{X} - z_{\alpha/2}/\sqrt{n}}\right].$$

Now consider the two ways to find an asymptotically exact $1-\alpha$ CI for τ. After direct calculations, one can find that

$$\tilde{\ell}''(\tau; x) = -\frac{3}{\tau^4} + \frac{2x}{\tau^3}.$$

So

$$\tilde{I}(\tau) = -\mathbb{E}\left[\tilde{\ell}''(\tau; X); \tau\right] = \frac{3}{\tau^4} - \frac{2}{\tau^4} = \frac{1}{\tau^4},$$

and noting that the ML estimator for τ is $\hat{\tau}_n = 1/\bar{X}$, we find that

$$\sqrt{n/\tau^4}\,(\hat{\tau}_n - \tau) \to_{\mathcal{D}(\cdot;\tau)} Z \sim \mathcal{N}(0,1),$$

so an asymptotically exact $1-\alpha$ CI is

$$\left[\hat{\tau}_n - z_{\alpha/2}\frac{\hat{\tau}_n^2}{\sqrt{n}}, \hat{\tau}_n + z_{\alpha/2}\frac{\hat{\tau}_n^2}{\sqrt{n}}\right].$$

Alternatively, instead of working out $\tilde{I}(\tau)$ as above, we could use the Theorem above to find that

$$\tilde{I}(\tau) = \frac{I(\mu)}{g'(\mu)^2}, \qquad \mu = g^{-1}(\tau) = \frac{1}{\tau}.$$

Since $I(\mu) = 1$ and $g(\mu) = 1/\mu \Rightarrow g'(\mu) = -1/\mu^2 = -\tau^2$, we have

$$\tilde{I}(\tau) = \frac{1}{[-1/\mu^2]^2} = \frac{1}{[-\tau^2]^2} = \frac{1}{\tau^4}.$$

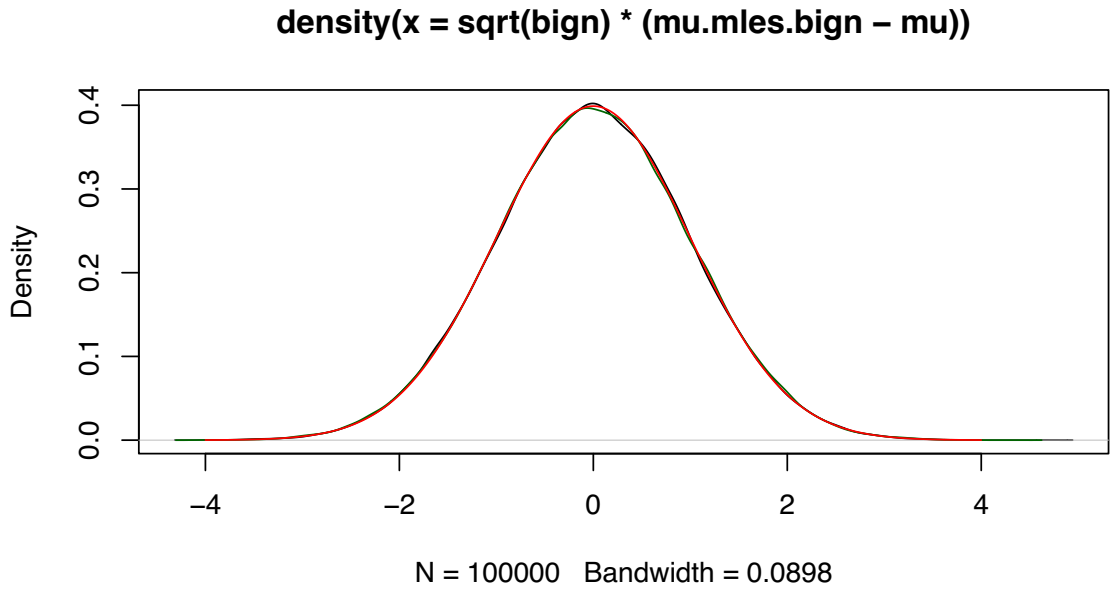
Remark 14.1. Notice that

1. the transformed CI is exact, while the second CI is not, since $\sqrt{n/\tau^4}\left(\frac{1}{\bar{X}} - \tau\right)$ is not exactly normally distributed, but only asymptotically so.
2. the transformed CI is not generally centred at $\hat{\tau}$.
3. this also serves as an example that convergence in distribution says nothing about convergence of moments. In particular, you can derive that $1/\bar{X}$ does not have a mean for any $\mu \in \mathbb{R}$.

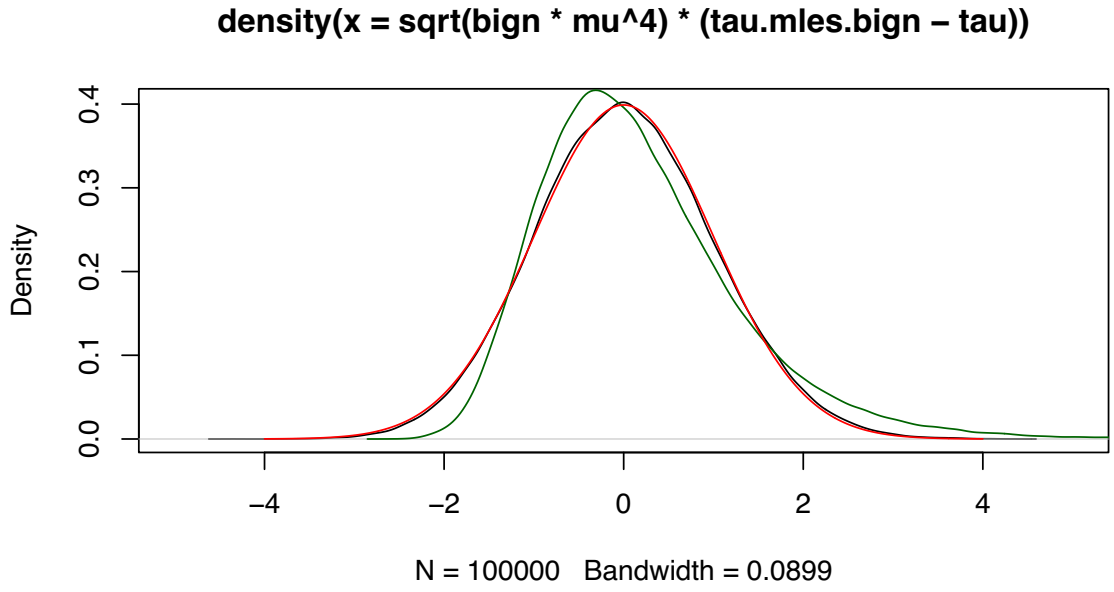
```
m <- 1e5
bign <- 1000; smalln <- 10
mu <- 2; tau <- 1/mu

mu.mles.bign <- rnorm(m, mean=mu, sd=1/sqrt(bign))
mu.mles.smalln <- rnorm(m, mean=mu, sd=1/sqrt(smalln))
tau.mles.bign <- 1/mu.mles.bign
tau.mles.smalln <- 1/mu.mles.smalln
zs <- seq(-4,4,.01)

# compare sqrt(n)*(mu.mle - mu) with a standard normal density
plot(density(sqrt(bign)*(mu.mles.bign-mu)))
lines(density(sqrt(smalln)*(mu.mles.smalln-mu)), col="darkgreen")
lines(zs, dnorm(zs), col="red")
```




```
# compare sqrt(n*mu^4)*(tau.mle - tau) with a standard normal density
plot(density(sqrt(bign*mu^4)*(tau.mles.bign-tau)), xlim=c(-5,5))
lines(density(sqrt(smalln*mu^4)*(tau.mles.smalln-tau)), col="darkgreen")
lines(zs,dnorm(zs), col="red")
```




```
n <- 10
zval <- qnorm(1-0.025)

mu.mle <- 2
CI.mu.L <- mu.mle - zval/sqrt(n)
CI.mu.R <- mu.mle + zval/sqrt(n)

tau.mle <- 1/2
CI.transformed.tau.L <- 1/CI.mu.R
CI.transformed.tau.R <- 1/CI.mu.L

CI.wald.tau.L <- tau.mle - zval*sqrt(tau.mle^4/n)
CI.wald.tau.R <- tau.mle + zval*sqrt(tau.mle^4/n)

CI.mu <- c(CI.mu.L, CI.mu.R)
CI.transformed.tau <- c(CI.transformed.tau.L, CI.transformed.tau.R)
CI.wald.tau <- c(CI.wald.tau.L, CI.wald.tau.R)

CI.mu

## [1] 1.380205 2.619795

CI.transformed.tau

## [1] 0.3817093 0.7245301

CI.wald.tau

## [1] 0.3450512 0.6549488
```

15 Likelihood ratio confidence sets: Wilks' approach

In this section, we assume that θ is one-dimensional. We introduce another type of confidence set which addresses some of the issues in Wald's approach:

1. It is possible for some θ not in the Wald CI to have a larger likelihood value than some other θ' that is in the Wald CI. Wald CIs can even contain values of θ such that $L(\theta; \mathbf{x}) = 0$.
2. Wald CIs are not invariant under reparameterization.

The main reason the approach works is the following Theorem.

Theorem 15.1. *Assume that $\mathbf{X} \overset{\text{iid}}{\sim} f(\cdot; \theta^*)$ for some $\theta^* \in \Theta$ and let $\{\hat{\theta}_i\}$ be a sequence of consistent ML estimators of $\theta^* \in \Theta$ and assume that the conditions of Theorem 10.1 hold (implying asymptotic normality). Define for any $\mathbf{x} \in \mathcal{X}^n$ the likelihood ratio*

$$\Lambda_n(\mathbf{x}) := \frac{L(\theta^*; \mathbf{x})}{L(\hat{\theta}_n; \mathbf{x})},$$

which takes values in $[0, 1]$. Then

$$-2 \log \Lambda_n(\mathbf{X}_n) \to_{\mathcal{D}(\cdot;\theta^*)} Z^2$$

where $Z \sim \mathcal{N}(0, 1)$ and therefore $Z^2 \sim \chi^2(1)$, a chi-squared distribution with one degree of freedom.

Remark 15.1. We observe that

$$\begin{aligned} -2 \log \Lambda_n(\mathbf{x}) &= -2 \left[\ell(\theta^*; \mathbf{x}) - \ell(\hat{\theta}_n; \mathbf{x}) \right] \\ &= 2 \left[\ell(\hat{\theta}_n; \mathbf{x}) - \ell(\theta^*; \mathbf{x}) \right], \end{aligned}$$

i.e. this is twice the difference of the log-likelihoods for $\hat{\theta}_n$ and θ^*.

Corollary 15.1. *With $\chi^2_{1,\alpha}$ the number such that $\mathbb{P}(W \leq \chi^2_{1,\alpha}) = 1 - \alpha$ for $W \sim \chi^2(1)$, the confidence sets*

$$C(\mathbf{X}_n) := \left\{ \theta \in \Theta : 2 \left[\ell(\hat{\theta}_n; \mathbf{X}_n) - \ell(\theta; \mathbf{X}_n) \right] \leq \chi^2_{1,\alpha} \right\} \subseteq \Theta,$$

are asymptotically exact $1 - \alpha$ confidence sets for θ^* since

$$\mathbb{P}\left(\theta^* \in C(\mathbf{X}_n); \theta^*\right) = \mathbb{P}\left(-2 \log \Lambda_n(\mathbf{X}_n) \leq \chi^2_{1,\alpha}; \theta^*\right) \to 1 - \alpha,$$

as $n \to \infty$.

These confidence sets have a nice interpretation: $C(\mathbf{x}_n)$ contains those values θ such that $\ell(\theta; \mathbf{x}_n)$ is not too much smaller than $\ell(\hat{\theta}_n; \mathbf{x}_n)$, the maximum value of $\ell(\cdot; \mathbf{x}_n)$. Hence, these confidence intervals contain those values of θ with the largest likelihood values.

Remark 15.2. The observed confidence set $C(\mathbf{x})$ is defined implicitly, and finding an explicit representation of such sets might not be easy (boundaries of an interval in the univariate scenario?). This difficulty can explain why Wald's approach has been historically popular, despite its shortcomings. However, with the help of a computer, it is often easy to determine $C(\mathbf{x})$ numerically.

Sketch of proof.

Proposition 15.1 (A rule of thumb)**.** *Under the assumptions of the theorem above, the set*

$$\left\{\theta \in \Theta : \ell(\theta; \mathbf{x}) \geq \ell(\hat{\theta}_n; \mathbf{x}) - 2\right\}$$

is an asymptotically approximate 95% *confidence set for* θ^**.*

Proof. $\chi^2_{0.05} = 3.84$ and the result follows from the approximation $1.92 \approx 2$. □

16 Transformation invariant confidence sets

We investigate whether the likelihood ratio approach to determining confidence sets is invariant to transformations, in contrast to Wald's approach. For a reparameterization of the likelihood in terms of $\tau = g(\theta)$ for bijective $g : \Theta \to G$, we have

$$\tilde{f}(\mathbf{x}; \tau) := f(\mathbf{x}; g^{-1}(\tau)) = f(\mathbf{x}; \theta).$$

Now we define

$$C(\mathbf{x}) := \left\{\theta \in \Theta : -2\left[\ell(\theta; \mathbf{x}) - \ell(\hat{\theta}_n; \mathbf{x})\right] \leq \chi^2_{1,\alpha}\right\}$$

and

$$\tilde{C}(\mathbf{x}) := \left\{\tau \in G : -2\left[\tilde{\ell}(\tau; \mathbf{x}) - \tilde{\ell}(\hat{\tau}_n; \mathbf{x})\right] \leq \chi^2_{1,\alpha}\right\}.$$

We can now ask whether these sets satisfy $\theta \in C(\mathbf{x}) \iff g(\theta) \in \tilde{C}(\mathbf{x})$ for every $\mathbf{x} \in \mathcal{X}^n$. That is, they define the same sets up to reparameterization.

Theorem 16.1. *Let $\mathbf{X} \sim f(\cdot; \theta^*)$, C and $\tilde{C}$ be defined as above, and assume that $g : \Theta \to G$ is bijective. Then for any $\mathbf{x} \in \mathcal{X}^n$, and any $\theta^* \in \Theta$, $\theta \in C(\mathbf{x}) \iff g(\theta) \in \tilde{C}(\mathbf{x})$ and so*

$$\mathbb{P}\left(\theta^* \in C(\mathbf{X}); \theta^*\right) = \mathbb{P}\left(g(\theta^*) \in \tilde{C}(\mathbf{X}); \tau = g(\theta^*)\right).$$

Proof. Let $\mathbf{x} \in \mathcal{X}^n$ be arbitrary. Everything rests on the observation that for any $\theta \in \Theta$,

$$\ell(\theta; \mathbf{x}) = \log f(\mathbf{x}; \theta) = \log \tilde{f}(\mathbf{x}; g(\theta)) = \tilde{\ell}(g(\theta); \mathbf{x}),$$

and similarly for any $\tau \in G$,

$$\tilde{\ell}(\tau; \mathbf{x}) = \log \tilde{f}(\mathbf{x}; \tau) = \log f(\mathbf{x}; g^{-1}(\tau)) = \ell(g^{-1}(\tau); \mathbf{x}),$$

and that $g(\hat{\theta}_n)$ is the ML estimate of τ. First, assume $\theta \in C(\mathbf{x})$. This implies

$$-2\left[\ell(\theta; \mathbf{x}) - \ell(\hat{\theta}_n; \mathbf{x})\right] \leq \chi^2_{1,\alpha}$$

which in turn implies

$$-2\left[\tilde{\ell}(g(\theta); \mathbf{x}) - \tilde{\ell}(g(\hat{\theta}_n); \mathbf{x})\right] \leq \chi^2_{1,\alpha}$$

and so $g(\theta) \in \tilde{C}(\mathbf{x})$. So $\theta \in C(\mathbf{x}) \Rightarrow g(\theta) \in \tilde{C}(\mathbf{x})$.

Similarly, now assume that $g(\theta) \in \tilde{C}(\mathbf{x})$. This implies

$$-2\left[\tilde{\ell}(g(\theta); \mathbf{x}) - \tilde{\ell}(g(\hat{\theta}_n); \mathbf{x})\right] \leq \chi^2_{1,\alpha}$$

which in turn implies

$$-2\left[\ell(\theta; \mathbf{x}) - \ell(\hat{\theta}_n; \mathbf{x})\right] \leq \chi^2_{1,\alpha}$$

and so $\theta \in C(\mathbf{x})$. Hence $\theta \in C(\mathbf{x}) \iff g(\theta) \in \tilde{C}(\mathbf{x})$.

For the last part, this correspondence implies that

$$\{\mathbf{x} \in \mathcal{X}^n : \theta^* \in C(\mathbf{x})\} = \{\mathbf{x} \in \mathcal{X}^n : g(\theta^*) \in \tilde{C}(\mathbf{x})\},$$

and so we can conclude from the equivalence of the events

$$\{\theta^* \in C(\mathbf{X})\} = \left\{g(\theta^*) \in \tilde{C}(\mathbf{X})\right\}.$$

□

17 Case Studies

Since statistics is more than just deriving estimators, confidence intervals, credible intervals and hypthosis tests on paper, here are a number of case studies in R where we can explore the content of the course concepts in more detail. RMkdown source files for these case studies can be found on Blackboard.

1. Likelihood and invariance
2. Distribution of an estimator, some inequalities
3. Probabilistic convergence and the Cramer-Rao lower bound
4. Asymptotic normality of the MLE and normal mean CIs
5. Wald Confidence Intervals in Practice

Case study 1: Likelihood and invariance

Juliette Unwin

The half-normal distribution

Let $Z \sim N(0,1)$. Then for $\sigma > 0$, $X = \sigma|Z|$ is a Half-Normal(σ) random variable.

Notice that X is a non-negative random variable. One can use a symmetry argument to deduce that its PDF is

$$f_X(x;\sigma) = \frac{\sqrt{2}}{\sigma\sqrt{\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \mathbb{I}(x \geq 0).$$

Observations of independent half-normal random variables

Given data $(x_1, \ldots, x_n)$, we adopt the model that this is a realization of a vector of independent Half-Normal(σ) random variables. The likelihood function is

$$L(\sigma; \mathbf{x}) = \prod_{i=1}^{n} f_X(x_i;\sigma),$$

so one can use the log-likelihood function (additive constants are irrelevant)

$$\ell(\sigma; \mathbf{x}) = -n\log(\sigma) - \sum_{i=1}^{n} \frac{x_i^2}{2\sigma^2},$$

as long as $\min\{x_1, \ldots, x_n\} \geq 0$.

Maximum likelihood estimator

This is a regular model, and the ML estimate can be obtained by equating the derivative of the log-likelihood to 0 and checking whether any candidate points are maximizers. We find that

$$\ell'(\sigma; \mathbf{x}) = -\frac{n}{\sigma} + \sum_{i=1}^{n} \frac{x_i^2}{\sigma^3} = 0$$

if and only if

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2.$$

We notice also that $\ell(\sigma; \mathbf{x}) \to -\infty$ as $\sigma \to 0$ or $\sigma \to \infty$, so

$$\hat{\sigma}_{\mathrm{ML}}(\mathbf{x}) = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^2\right)^{1/2},$$

is the ML estimate. The ML estimator is therefore

$$\hat{\sigma}_{\mathrm{ML}}(\mathbf{X}) = \left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right)^{1/2}.$$

It's never a bad idea to quickly check that the estimator makes sense via a quick simulation.

```r
ml.estimate <- function(xs) {
  sqrt(mean(xs^2))
}

sigma.true <- 2.3 # a possible value of the statistical parameter
n <- 1000 # sample size
# generate a simulated dataset by the definition of Z and X
zs <- rnorm(n)
xs <- sigma.true*abs(zs)

ml.estimate(xs)
```

```
## [1] 2.275766
```

Looks reasonably good, but this is just one simulated experiment with fairly large n.

Method of moments estimator

In order to apply the method of moments (MoM), we can equate the first moment of X (i.e. the mean) to its empirical first moment (empirical mean). We find that

$$\mathbb{E}[X;\sigma] = \int_0^\infty x f_X(x;\sigma)\mathrm{d}x = \sigma\sqrt{\frac{2}{\pi}}.$$

Hence, the MoM estimate satisfies

$$\hat{\sigma}_{\mathrm{MoM}} = \sqrt{\frac{\pi}{2}}\bar{x},$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i$. The corresponding MoM estimator is

$$\hat{\sigma}_{\mathrm{MoM}} = \sqrt{\frac{\pi}{2}}\bar{X}.$$

As before, we can quickly check if the estimator is any good in a simulated experiment

```r
mom.estimate <- function(xs) {
  sqrt(pi/2)*mean(xs)
}

sigma.true <- 2.3
n <- 1000
zs <- rnorm(n)
xs <- sigma.true*abs(zs)

mom.estimate(xs)
```

```
## [1] 2.263243
```

Not too shabby, although again it's just one simulated experiment and a fairly large value of n.

Empirical mean-squared errors

We can always simulate several realizations of the estimator for a given σ^* and n, e.g. to approximate the mean-squared error.

```r
trials <- 10000
n <- 25
sigma.true <- 3.6
```

```r
sigmas.ml <- rep(0,trials)
sigmas.mom <- rep(0,trials)

for (i in 1:trials) {
  zs <- rnorm(n)
  xs <- sigma.true*abs(zs)
  sigmas.ml[i] <- ml.estimate(xs)
  sigmas.mom[i] <- mom.estimate(xs)
}

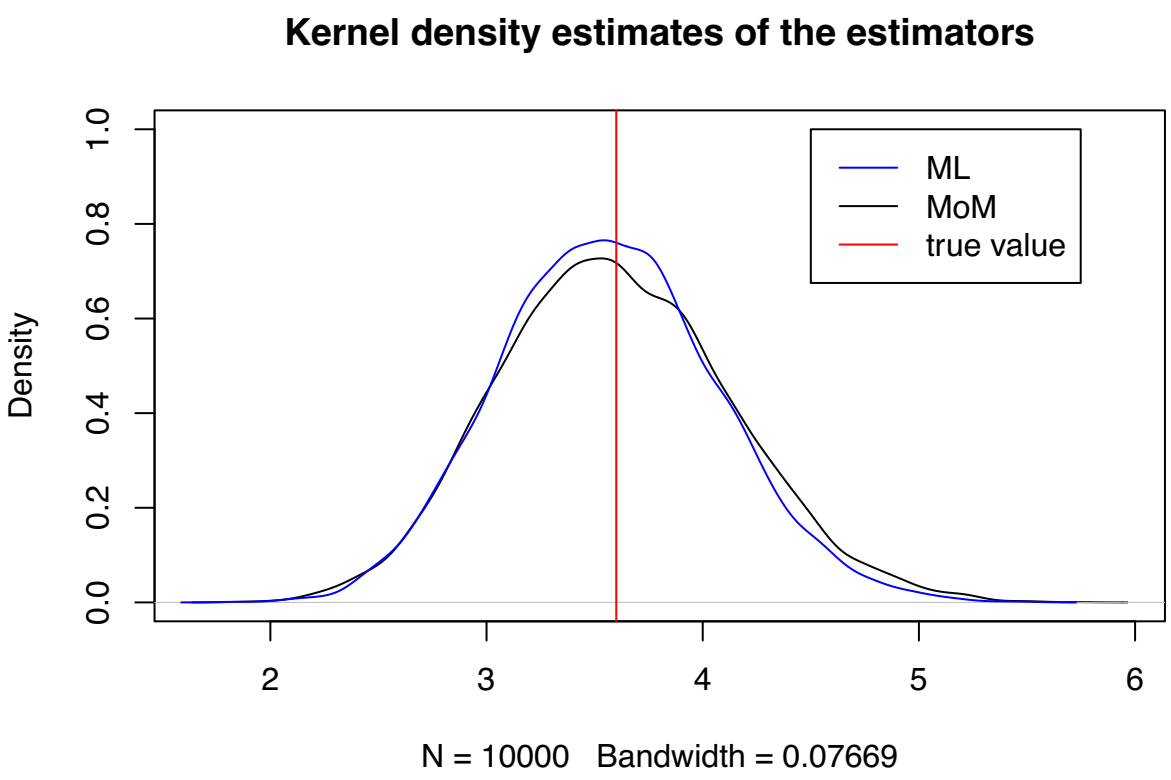
mean((sigmas.ml-sigma.true)^2) # empirical MSE for the ML estimator
```

```
## [1] 0.2532131
```

```r
mean((sigmas.mom-sigma.true)^2) # empirical MSE for the MoM estimator
```

```
## [1] 0.2890885
```

```r
plot(density(sigmas.mom),ylim=c(0,1),main="")
lines(density(sigmas.ml),col="blue")
abline(v=sigma.true,col="red")
legend(4.5,1,legend=c("ML","MoM","true value"),col=c("blue","black","red"),lty=1)
title("Kernel density estimates of the estimators")
```



It looks like the ML estimator is slightly better than the MoM estimator. The MSE appears to be slightly smaller, and the distribution is slightly more peaked around the true value of σ.

What about transformations of σ?

First consider a transformed parameter $\tau = \sigma^2$. By the invariance of the MLE to reparametrizations of the statistical parameter, we have that $\hat{\tau}_{\mathrm{ML}} = \hat{\sigma}^2_{\mathrm{ML}}$. Similarly, for the MoM estimator we would solve

$$\sqrt{\frac{2\tau}{\pi}} = \bar{X},$$

which also gives $\hat{\tau}_{\mathrm{MoM}} = \hat{\sigma}^2_{\mathrm{MoM}}$.

However, now consider $\tau = g(\sigma) = \mathbb{I}(\sigma > 2)$. Here we obtain that $\hat{\tau}_{\mathrm{ML}} = g(\hat{\sigma}_{\mathrm{ML}}) = \mathbb{I}(\hat{\sigma}_{\mathrm{ML}} > 2)$. On the other hand, it is not obvious how to even define the MoM estimator in this case.

What about a transformed X?

Now consider observations of $Y = X^2$. Because the likelihood function is invariant to reparametrization of the data by a bijective function (independent of the statistical parameter), the ML estimator of σ is not changed. Note that this is bijective in this case because X is a non-negative random variable.

On the other hand, we obtain

$$\mathbb{E}[Y; \sigma] = \mathbb{E}[X^2; \sigma] = \mathbb{E}[\sigma^2 Z^2; \sigma] = \sigma^2,$$

so that the MoM estimator is given by

$$\hat{\sigma}^2_{\mathrm{MoM}} = \frac{1}{n}\sum_{i=1}^{n} Y_i = \frac{1}{n}\sum_{i=1}^{n} X_i^2 = \hat{\sigma}^2_{\mathrm{ML}}.$$

In other words, after reparametrization of X (i.e. the data) in this way the MoM estimator and the ML estimator are identical. So clearly the MoM estimator is not invariant to reparametrization.

Case study 2: Distribution of an estimator, some inequalities

Juliette Unwin

Cauchy random variables

Let $X_1, \ldots, X_n$ be independent Cauchy(θ) random variables. The PDF of X_1 is

$$f(x;\theta) = \frac{1}{\pi\{1+(x-\theta)^2\}},$$

The statistical parameter $\theta \in \mathbb{R}$ is a location parameter, but in this case it is not the mean of X_1. In fact, X_1 does not have a mean, since $\mathbb{E}[|X_1|;\theta] = \infty$. Hence one cannot define the simple method of moments estimator. What's perhaps more remarkable, is that the sample mean of a Cauchy(θ) random variable is also a Cauchy(θ) random variable, irrespective of the value of n.

The maximum likelihood estimator is well-defined, but it is a complicated function of $X_1, \ldots, X_n$ because the likelihood function has several local maxima. We will not consider the ML estimator here.

Sample median as an estimator of θ

It can be shown, however, that

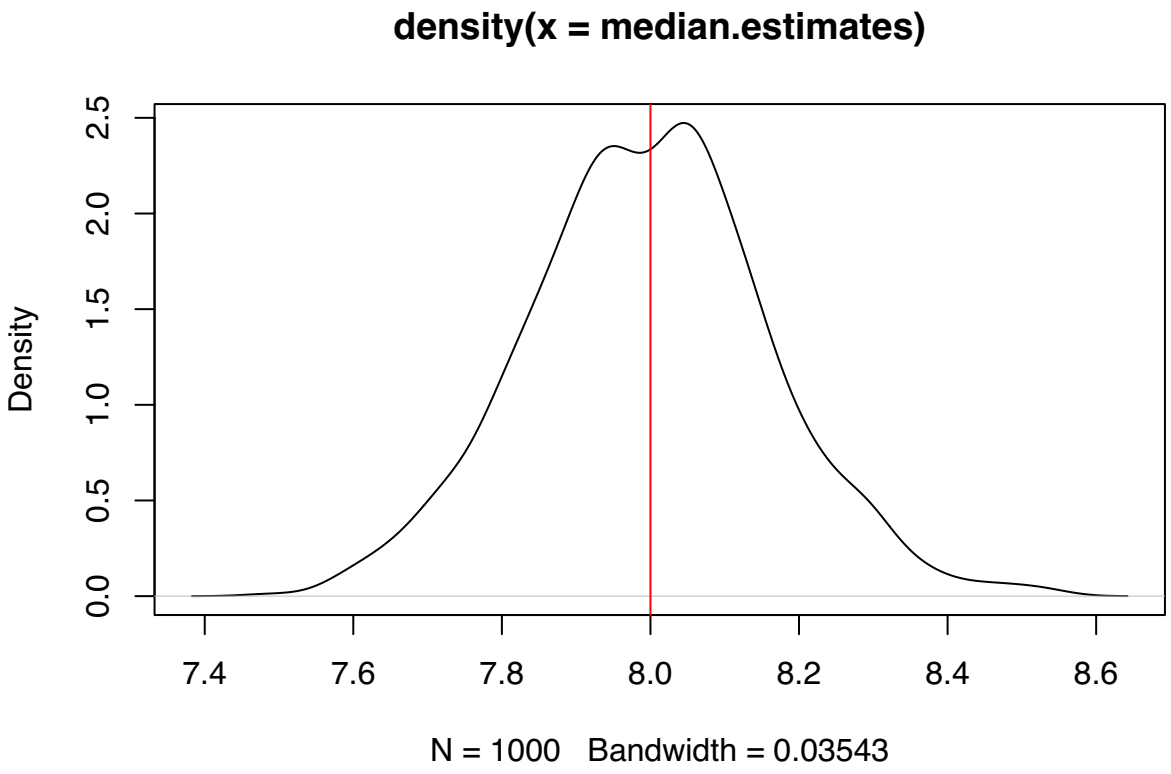
$$\mathbb{P}(X_1 \leq \theta;\theta) = \mathbb{P}(X_1 > \theta;\theta),$$

so that the median of X_1 is indeed θ.

We consider here the sample median $\tilde{\theta}_n(X_1, \ldots, X_n)$ as an estimator of θ.

We do not derive here the distribution of the sample median, but we can investigate its properties empirically. Naturally, this involves performed repeated experiments or simulations.

```
trials <- 1000
n <- 100
theta.star <- 8
median.estimates <- rep(0,trials)
for (i in 1:trials) {
  xs <- rcauchy(n,location=theta.star)
  median.estimates[i] <- median(xs)
}
plot(density(median.estimates))
abline(v=theta.star,col="red")
```



We see that the estimator is typically not too far from the true value when $n = 100$.

We can approximate the bias, the variance and the MSE.

```
# empirical bias at theta.star
mean(median.estimates)-theta.star
```

```
## [1] -0.003084038
```

```
# empirical (sample) variance at theta.star
var(median.estimates)
```

```
## [1] 0.02640069
```

```
# empirical MSE
mean((median.estimates-theta.star)^2)
```

```
## [1] 0.0263838
```

The bias is quite small (it appears that the estimator may even be unbiased) and the MSE is dominated by the variance of the estimator.

Method of moments estimator for $\log(|X|)$ as an estimator of θ^2

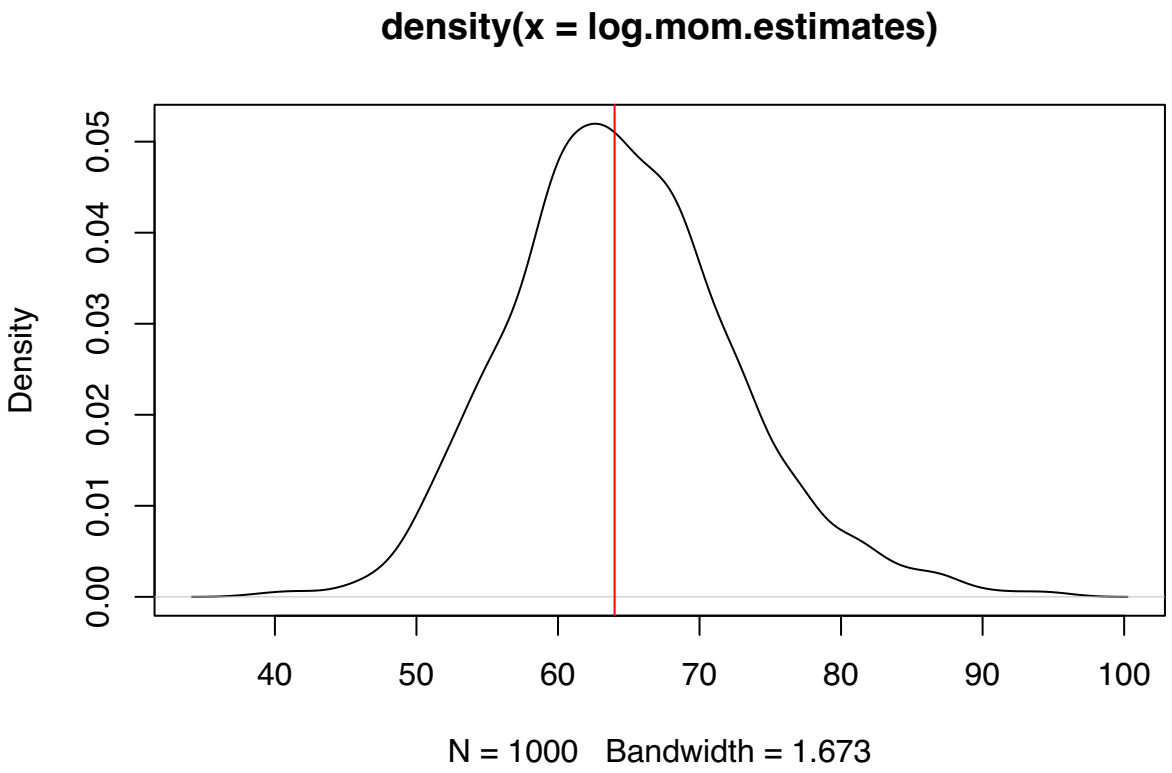
We can also consider a transformation $Y_i = \log(|X_i|)$, and the corresponding Y_i do have a mean. In particular, it can be shown (but I do not recommend that you try this one) that $\mathbb{E}[Y_1; \theta] = \log(\theta^2 + 1)/2$. It follows that one can approximate θ^2 as

$$\breve{\theta}^2 = \exp(2\bar{Y}) - 1,$$

where the sample mean $\bar{Y}$ is the mean of $Y_1, \ldots, Y_n$.

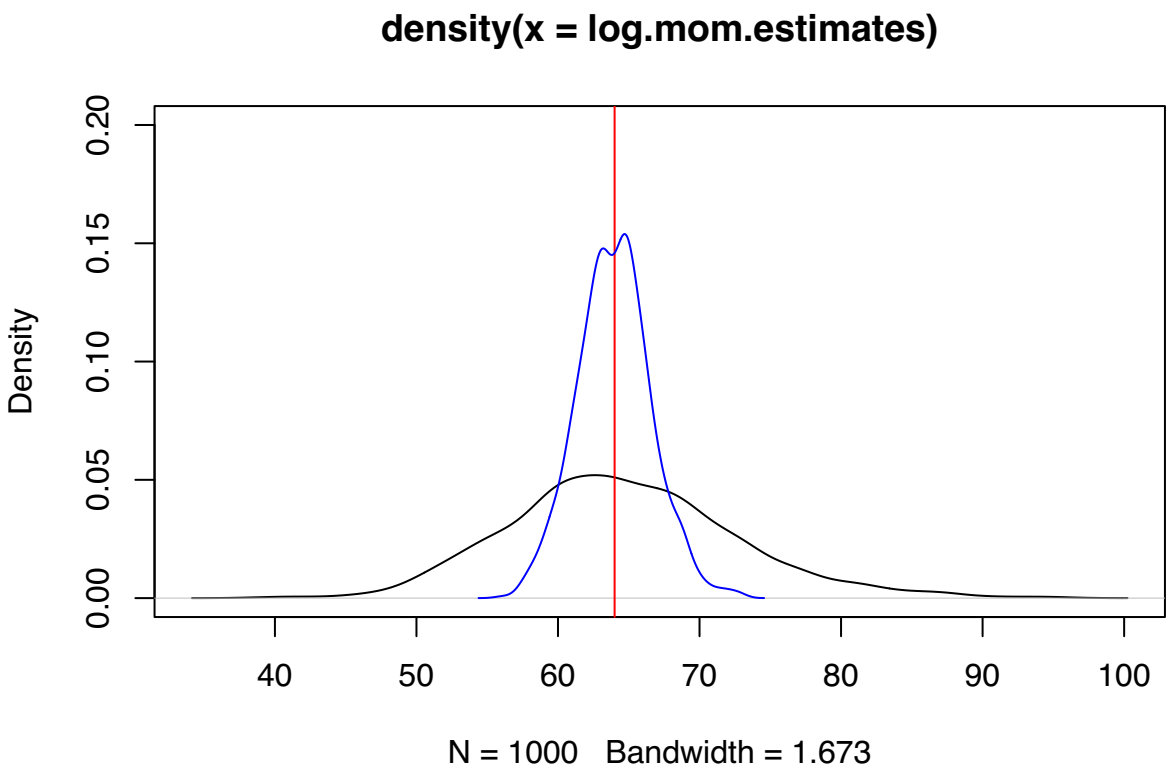
```
trials <- 1000
n <- 100
theta.star <- 8
log.mom.estimates <- rep(0,trials)
```

```
for (i in 1:trials) {
  xs <- rcauchy(n,location=theta.star)
  ys <- log(abs(xs))
  log.mom.estimates[i] <- exp(2*mean(ys))-1
}
plot(density(log.mom.estimates))
abline(v=theta.star^2,col="red")
```



Just for fun, we can superimpose the density estimate plot for the squared median estimator.

```
plot(density(log.mom.estimates),ylim=c(0,0.2))
abline(v=theta.star^2,col="red")
lines(density(median.estimates^2),col="blue")
```



It appears that the squared median estimator is "better" than this transformed method of moments estimator, in the sense that its distribution is more concentrated around the true value of the statistical parameter. At least for this particular value of θ^*.

Markov and Chebyshev inequalities

Markov's inequality

Markov's inequality states that for a non-negative random variable X,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

A very good way to supplement your understanding is to construct a simple example, if possible. Since this inequality concerns the probability of events of the form $\{X \geq a\}$ it makes sense to consider a simple distribution whose CDF we know. For example, the Exponential(1) distribution has CDF $F(x) = 1 - \exp(-x)$. Hence, we find that for $X \sim$ Exponential(1),

$$\mathbb{P}(X \geq a) = 1 - \mathbb{P}(X < a) = 1 - 1 + \exp(-a) = \exp(-a),$$

while $\mathbb{E}[X] = 1$. Indeed, the inequality then states that $\exp(-a) = 1/\exp(a) \leq 1/a$, which is true. You can also see that the inequality can be quite weak, in the sense that for large a, $\exp(-a) \ll 1/a$.

It is natural to ask if the inequality is tight, i.e. if it is an equality for some X and a. In fact, here for any a one can find an X such that the inequality is an equality. Let $X_a = B_a a$, where $B_a \sim$ Bernoulli$(1/a^2)$. It follows that $\mathbb{E}(X_a) = 1/a$ and

$$\mathbb{P}(X_a \geq a) = \frac{1}{a^2} = \frac{\mathbb{E}[X]}{a}.$$

Chebyshev's inequality

Chebyshev's inequality states that for a random variable X with mean μ and variance σ^2,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

One way to rewrite this inequality is to take $k = a\sigma$ for some $a > 0$ in which case we obtain

$$\mathbb{P}(|X - \mu| \geq a\sigma) \leq \frac{1}{a^2}.$$

We can similarly look for examples where the inequality is an equality. Again, it is possible to construct examples for any a. Define X_a to have PMF

$$f(x) = \frac{1}{2a^2}\mathbb{I}(x \in \{-1, 1\}) + \left(1 - \frac{1}{a^2}\right)\mathbb{I}(x = 0).$$

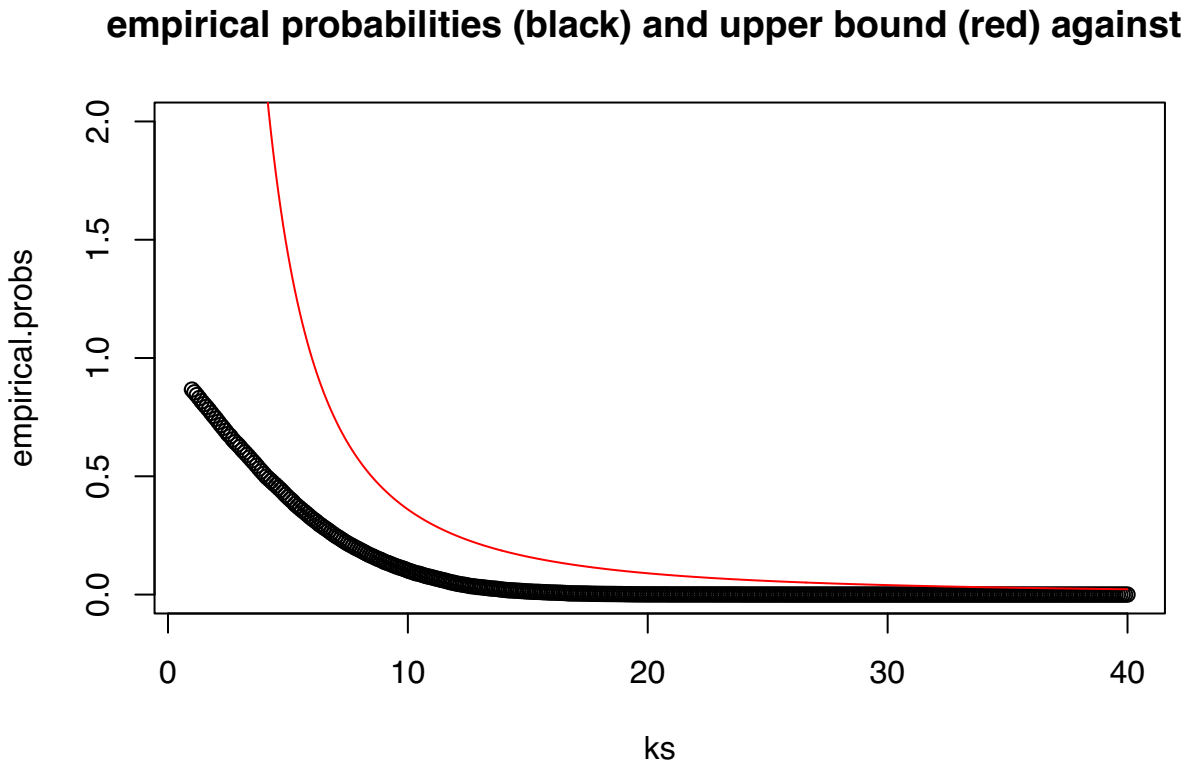
One can deduce that $\mathbb{E}[X_a] = 0$ while $\text{var}(X_a) = 1/a^2$. It follows that

$$\mathbb{P}(|X - \mu| \geq a\sigma) = \mathbb{P}(|X| = 1) = \frac{1}{a^2}.$$

If it's difficult or too time consuming to verify an inequality theoretically on an example, one can always resort to simulations. Here we verify Chebyshev's inequality for the normal distribution:

```
mu <- 3
sigma <- 6
xs <- rnorm(10000,mean=mu,sd=sigma)

ks <- seq(1,40,0.1)
empirical.probs <- rep(0,length(ks))
for (i in 1:length(ks)) {
  k <- ks[i]
  empirical.probs[i] <- mean(abs(xs-mu) > k)
}
plot(ks,empirical.probs,ylim=c(0,2))
lines(ks,sigma^2/ks^2,col="red")
title("empirical probabilities (black) and upper bound (red) against k")
```



Simulations are necessarily less satisfying, but they are often easier to do and sufficient to get some idea of what is going on.

Case study 3: Probabilistic convergence and the Cramer–Rao lower bound

Juliette Unwin

Empirical investigation of the WLLN and CLT

The weak law of large numbers

The WLLN states that if $X_1, X_2, \ldots$ are i.i.d. random variables with finite mean $\mathbb{E}[X_1]$, then the sequence of random variables

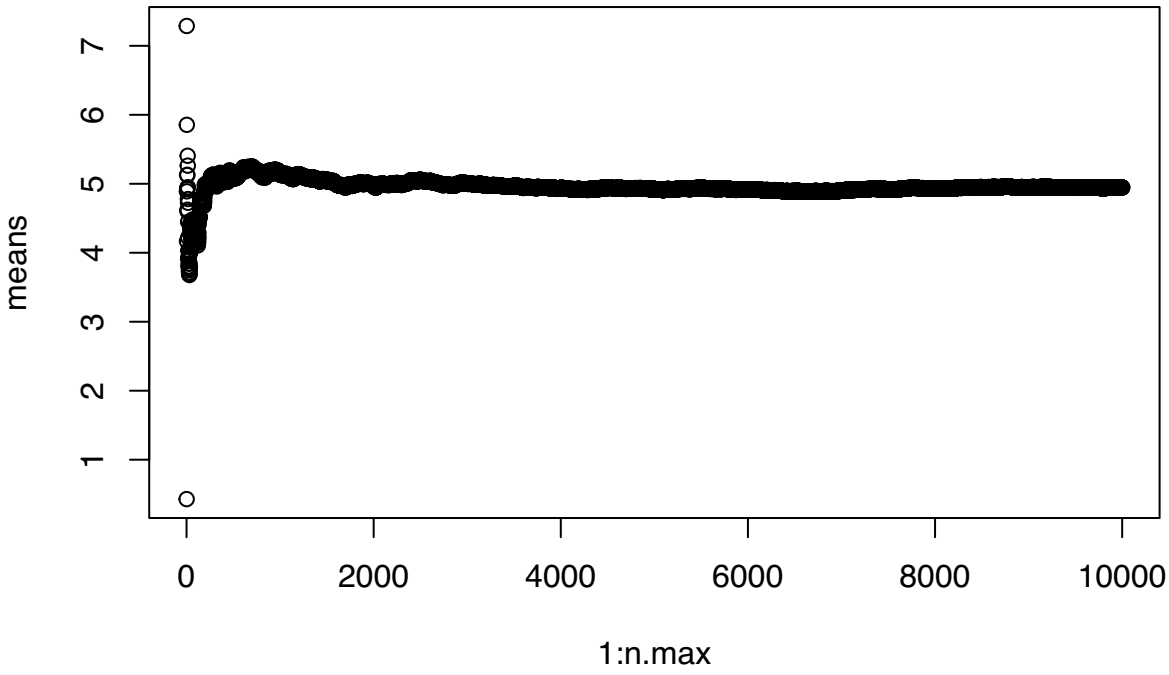
$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

converges in probability to $\mu := \mathbb{E}[X_1]$.

It is often a good idea to test your understanding empirically, a bit like running a scientific experiment to check if empirical data is supported by the theory.

First we can plot a "path" of sample means, with increasing n.

```
n.max <- 10000
mu <- 5
xs <- rexp(n.max, rate = 1/mu)
# cumsum means cumulative sum, e.g. cumsum(c(1,2,1)) = [1,3,4]
means <- cumsum(xs)/(1:n.max)
plot(1:n.max,means)
```



Next, we can fix an epsilon and see how the empirical probability that the sample mean is further than epsilon from the mean is affected by increasing n.

```
ns <- 10^(1:5)
epsilon <- 0.05
trials <- 1000
empirical.probs <- rep(0,length(ns))
for (i in 1:length(ns)) {
  n <- ns[i]
  greater.than.epsilon <- 0
  for (j in 1:trials) {
    xs <- rexp(n,rate=1/mu)
    greater.than.epsilon <- greater.than.epsilon + (abs(mean(xs)-mu) > epsilon)
  }
  empirical.probs[i] <- greater.than.epsilon/trials
}
empirical.probs
```

```
## [1] 0.967 0.921 0.740 0.317 0.002
```

The central limit theorem

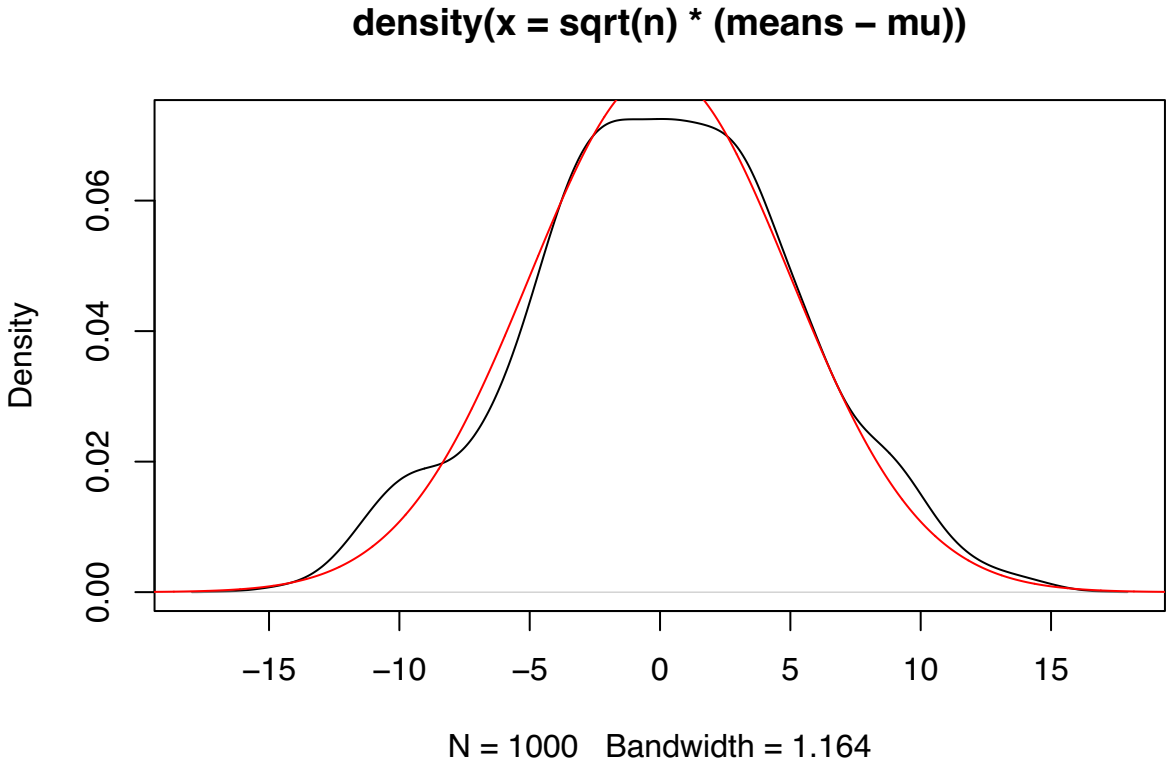
The CLT states that if the conditions of the WLLN hold and also $\text{var}(X_1) = \sigma^2 < \infty$ then we have

$$\sqrt{n}(\bar{X}_n - \mu) \to_{\mathcal{D}} \sigma Z,$$

where $Z \sim N(0, 1)$.

Again, we can do an experiment to see how the CLT works on a particular example.

```
n <- 1000
mu <- 5
sigma <- mu
trials <- 1000
means <- rep(0,trials)
for (i in 1:trials) {
  xs <- rexp(n, rate=1/mu)
  means[i] <- mean(xs)
}
plot(density(sqrt(n)*(means-mu)))
vs <- seq(-5*mu,5*mu,0.01)
lines(vs,dnorm(vs,mean = 0,sd=sigma),col="red")
```



Of course you can do similar experiments to investigate other results, e.g. the CMT and Slutsky's Theorem.

Fisher information and the Cramer-Rao lower bound

The Cramer-Rao lower bound states that

$$\text{var}(\hat{\theta}_n; \theta) \geq \frac{[m'(\theta)]^2}{nI(\theta)},$$

where $m(\theta) = \mathbb{E}[\hat{\theta}; \theta]$.

Consider again the Exponential distribution, and assume that the rate θ is the statistical parameter. The Fisher information is $I(\theta) = 1/\theta^2$.

The ML estimator $1/\bar{X}_n$ in this case is biased, and with some work[1] it can be shown that

$$\mathbb{E}[\hat{\theta}_n; \theta] = \frac{n}{n-1}\theta.$$

It follows that the Cramer-Rao lower bound is

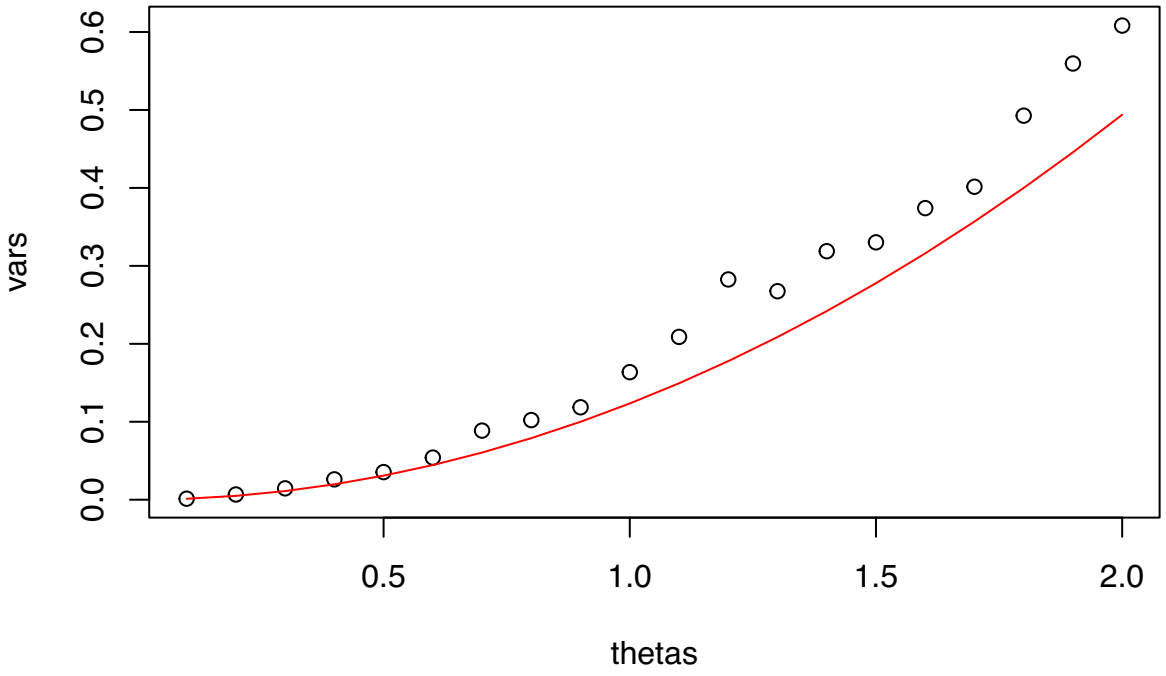
$$\text{var}(\hat{\theta}_n; \theta) \geq \frac{n^2}{(n-1)^2}\frac{\theta^2}{n} = \frac{n}{(n-1)^2}\theta^2.$$

We can plot empirical variances of the ML estimator for different values of theta, as well as the Cramer-Rao lower bound.

```
n <- 10
thetas <- seq(0.1,2,0.1)
vars <- rep(0,length(thetas))
trials <- 1000
```

[1] In particular, one can start by showing that the sum of n independent Exponential random variables is a Gamma random variable

```r
for (i in 1:length(thetas)) {
  mles <- rep(0,trials)
  for (j in 1:trials) {
    theta <- thetas[i]
    xs <- rexp(n, theta)
    mles[j] <- 1/mean(xs)
  }
  vars[i] <- var(mles)
}
plot(thetas,vars)
lines(thetas,n/(n-1)^2*thetas^2,col="red")
```



An interesting example

Consider a simple, but slightly unusual example. Let $X_1, \ldots, X_n$ be independent $N(\theta, 1)$ random variables where $\theta \in \Theta = \mathbb{R}$. It is possible to show that $\hat{\theta}_n = \bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ is unbiased and efficient since $\mathbb{E}[\hat{\theta}_n; \theta] = \theta]$ for all $\theta \in \Theta$ and $\text{var}(\hat{\theta}_n; \theta) = n^{-1} = \{nI(\theta)\}^{-1}$.

Now we consider the alternative estimator

$$\tilde{\theta}_n = \frac{n}{n+1} \bar{X}_n.$$

For $n = 10$, we can plot the estimated mean squared error of $\tilde{\theta}_n$ against θ. To make the computation quick, we make use of the fact that for every θ, $\tilde{\theta}_n = \frac{n}{n+1}(\theta + M_n)$, where $M_n \sim N(0, 1/n)$. Hence,

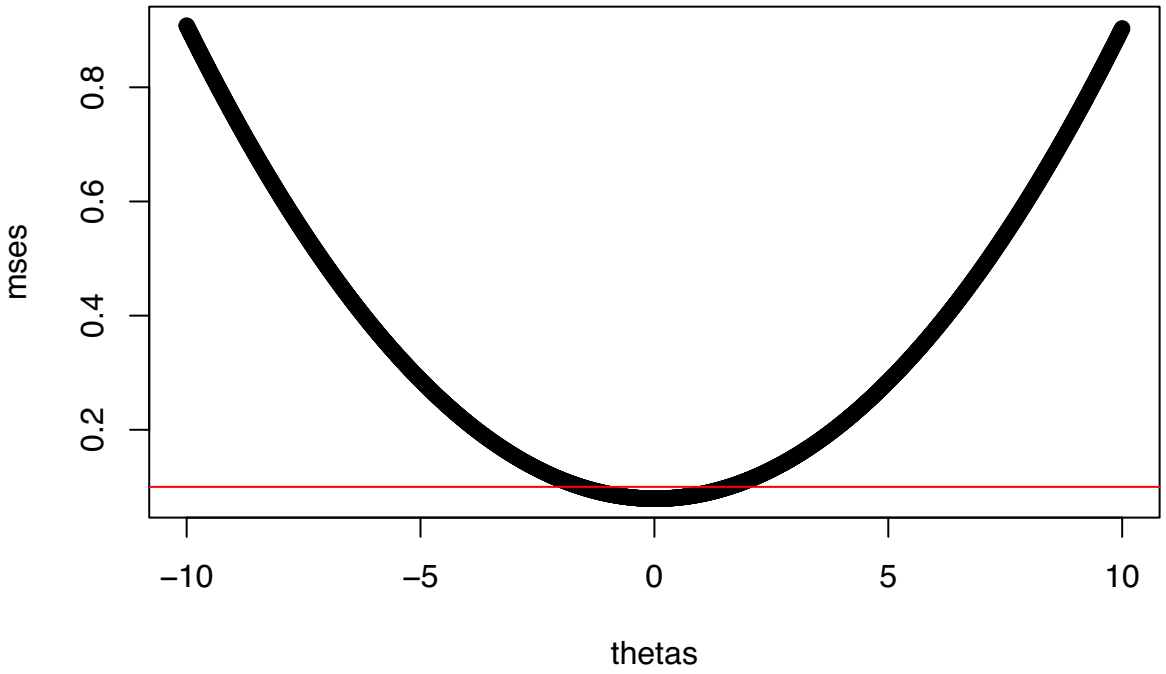
$$\tilde{\theta}_n - \theta = \frac{n}{n+1} M_n - \frac{1}{n+1} \theta.$$

This is a special feature of means of independent normal random variables.

```r
n <- 10
trials <- 1000
ms <- rnorm(trials,mean=0,sd=1/sqrt(n))
```

```r
thetas <- seq(-10,10,0.01)
mses <- rep(0,length(thetas))
for (i in 1:length(thetas)) {
  theta <- thetas[i]
  diffs <- n/(n+1)*ms - 1/(n+1)*theta
  mses[i] <- mean(diffs^2)
}
plot(thetas,mses)

unbiased.crlb <- 1/n
abline(h=unbiased.crlb, col="red")
```



We see that the mean-squared error near 0 appears to be better than the Cramer-Rao lower bound for an unbiased estimator, but it is worse if theta is sufficiently far from 0.

If this is surprising to you (perhaps it should be), you can take a look at the supplementary information sheet "Mean-squared error and the Cramer-Rao lower bound".

Case study 4: Asymptotic normality of the MLE and normal mean CIs

Juliette Unwin

Asymptotic normality of the ML estimator

We can consider the same model used in the "Using the `optim` function" supplementary information handout, a Gumbel.

To simulate realizations of n independent Gumbel(μ, β) random variables we can use the following function:

```
rgumbel <- function(n, mu, beta) {
  us <- runif(n)
  mu - beta*log(-log(us))
}
```

As covered in the handout, the PDF of a Gumbel(μ, β) random variable is given by the following function:

```
dgumbel <- function(x, mu, beta, log.p=FALSE) {
  stopifnot(all(beta > 0))
  z <- (x-mu)/beta
  logf <- -(z + exp(-z) + log(beta))
  if (!log.p) return(exp(logf)) else return(logf)
}
```

We are now in a position to investigate empirically whether the sequence of ML estimators (indexed by sample size) for the Gumbel distribution is asymptotically normal. One way to do this is to look at the distribution of the ML estimator for a large sample size n and some value of the statistical parameter, and plot its density estimate. To keep things simple, we will assume that $\mu = 3$ is known and the statistical parameter is β.

First, we need to define a function that numerically computes the ML estimate using the `optim` function.

```
ml.estimate <- function(xs) {
  ell <- function(beta) {
    sum(dgumbel(xs, mu, beta, log.p=TRUE))
  }
  neg.ell <- function(beta) {
    -ell(beta)
  }
  init <- 1
  optim.out <- optim(init,neg.ell,method="L-BFGS-B",lower=1e-5,upper=Inf)
  optim.out$par
}
```

Let's just test that it seems to work on a large simulated dataset

```
mu <- 3
beta.star <- 6
xs <- rgumbel(1e6,mu,beta.star)
ml.estimate(xs)
```

```
## [1] 5.996279
```

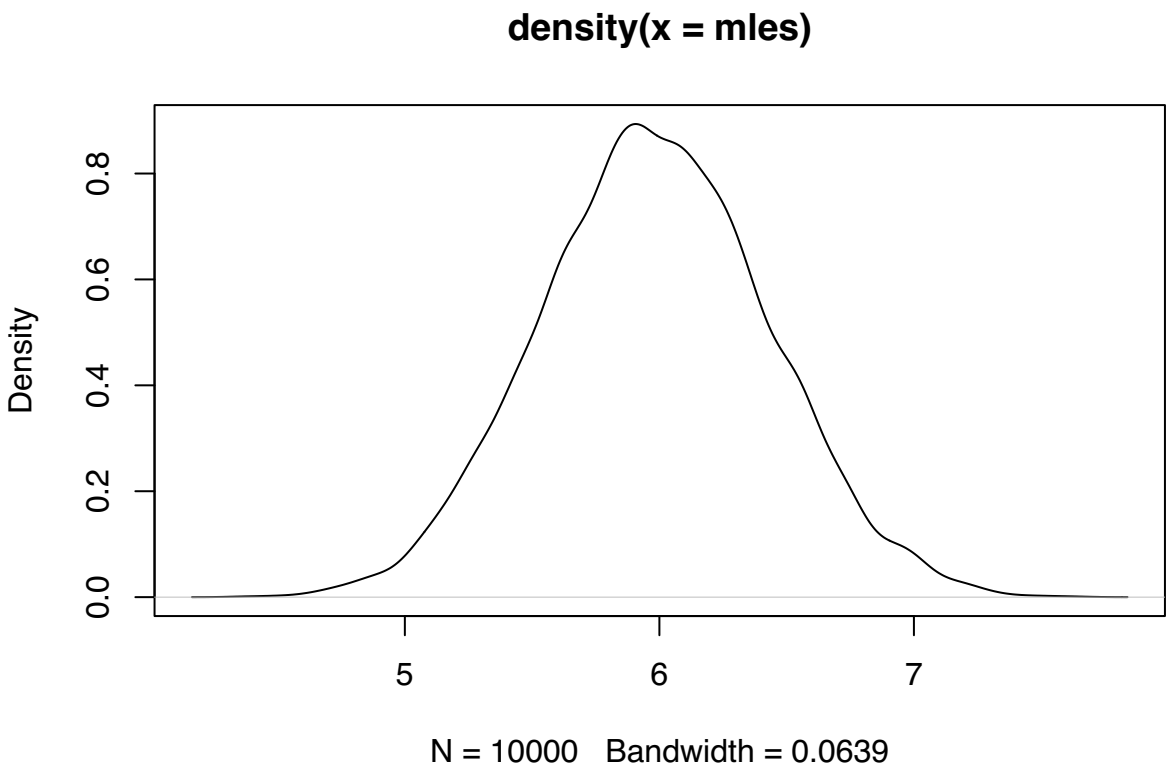
The answer is close to 6 so probably we haven't made a mistake.

Now we can perform the simulation experiment.

```
n <- 100 # reasonably large
mu <- 3
beta.star <- 6
trials <- 10000
mles <- rep(0,trials)
for (i in 1:trials) {
  xs <- rgumbel(n, mu, beta.star)
  mles[i] <- ml.estimate(xs)
}
```

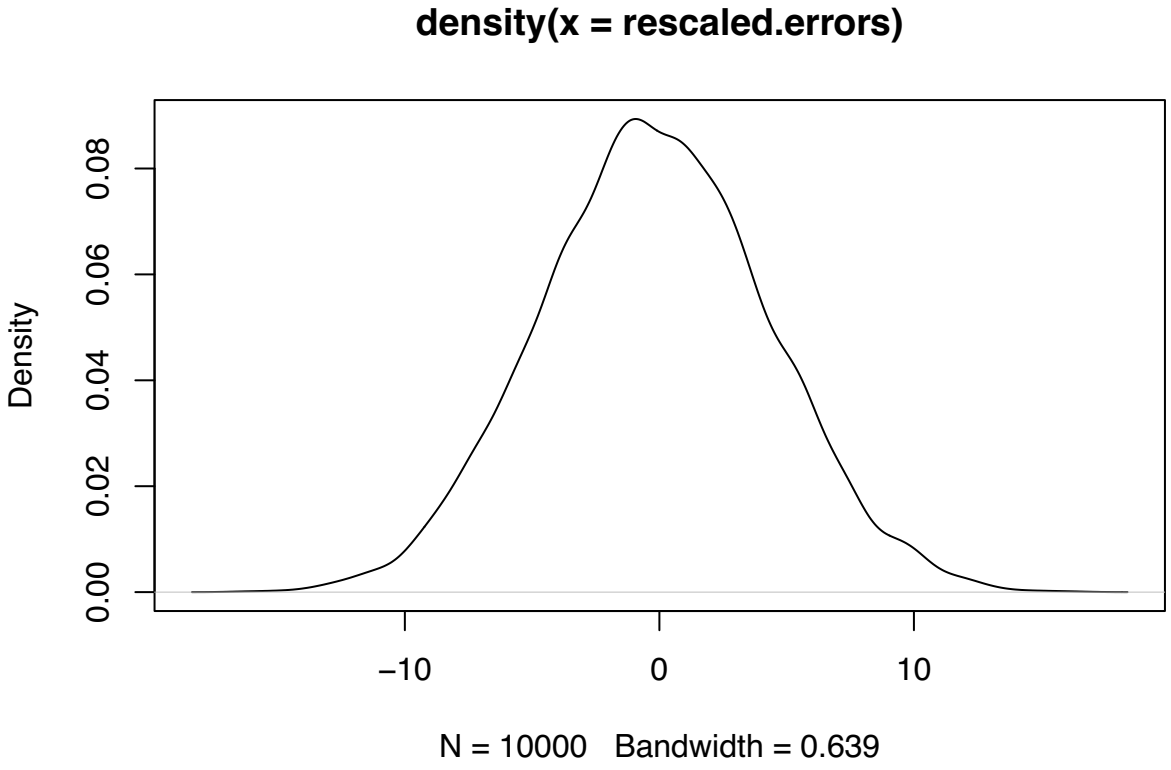
We can plot the estimated density of the ML estimator using the ML estimates.

```
plot(density(mles))
```



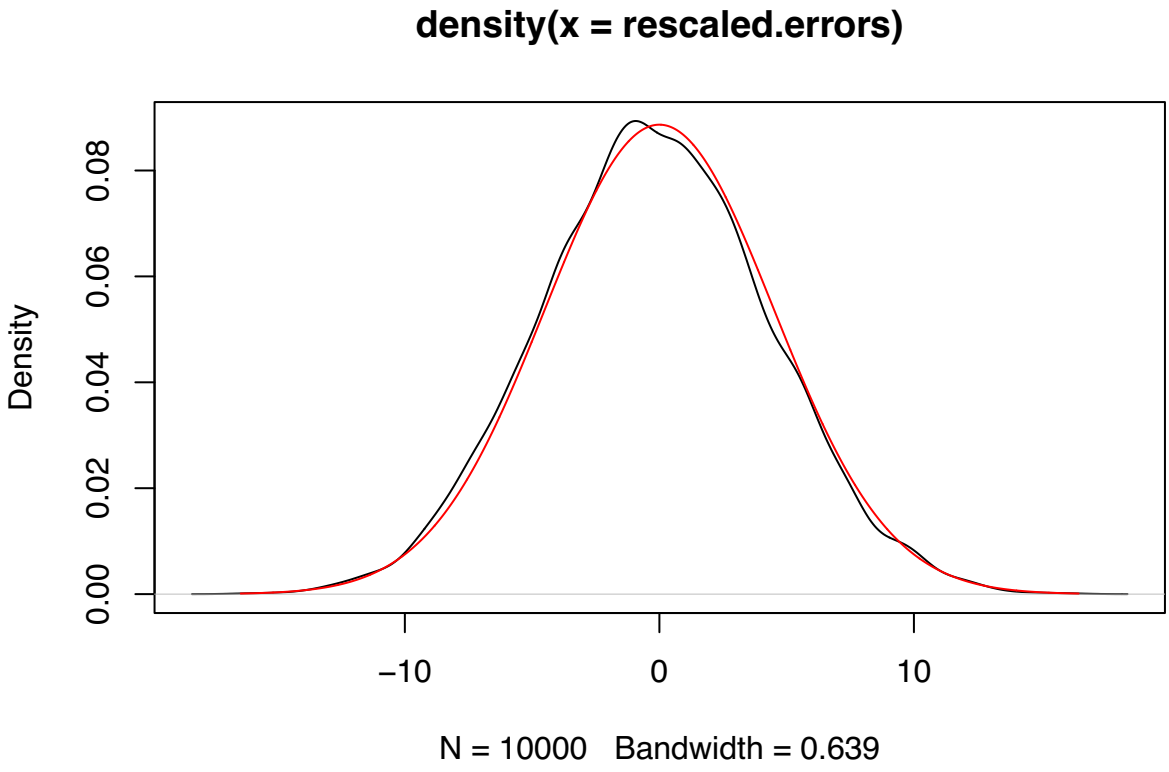
It looks like it could be close to normal. We can also centre and rescale to look at the distribution of the random error.

```
rescaled.errors <- sqrt(n)*(mles-beta.star)
plot(density(rescaled.errors))
```



Although the distribution seems like it could be close to normal, we don't know what the variance of the normal distribution is. That is, we don't know the Fisher Information function for this model. After some guessing, we can see quite a good fit with $\sigma^2 = 4.5^2 = 20.25$.

```
plot(density(rescaled.errors))
vs <- seq(min(rescaled.errors),max(rescaled.errors),0.01)
lines(vs,dnorm(vs,0,sd=4.5),col="red")
```



Confidence intervals for the normal mean

Section 10 introduces confidence intervals for the normal mean. Since this is such an important example, it is worthwhile to do some simulations to check our understanding.

In particular, we can check that for a given value of n, the stated confidence interval does indeed have the right coverage. We will construct several observed confidence intervals, by tracking the lower and upper points. Then we can plot them to see how many actually include the mean. Just to be a bit odd, we will set $\alpha = 0.07$.

We can define a function that outputs the lower and upper endpoints for an observed normal mean CI, with inputs the data, σ and α.

```
normal.mean.CI <- function(xs, sigma, alpha) {
  n <- length(xs)
  m <- mean(xs)
  z <- qnorm(1-alpha/2)
  l <- m - z*sigma/sqrt(n)
  u <- m + z*sigma/sqrt(n)
  return(c(l,u))
}
```

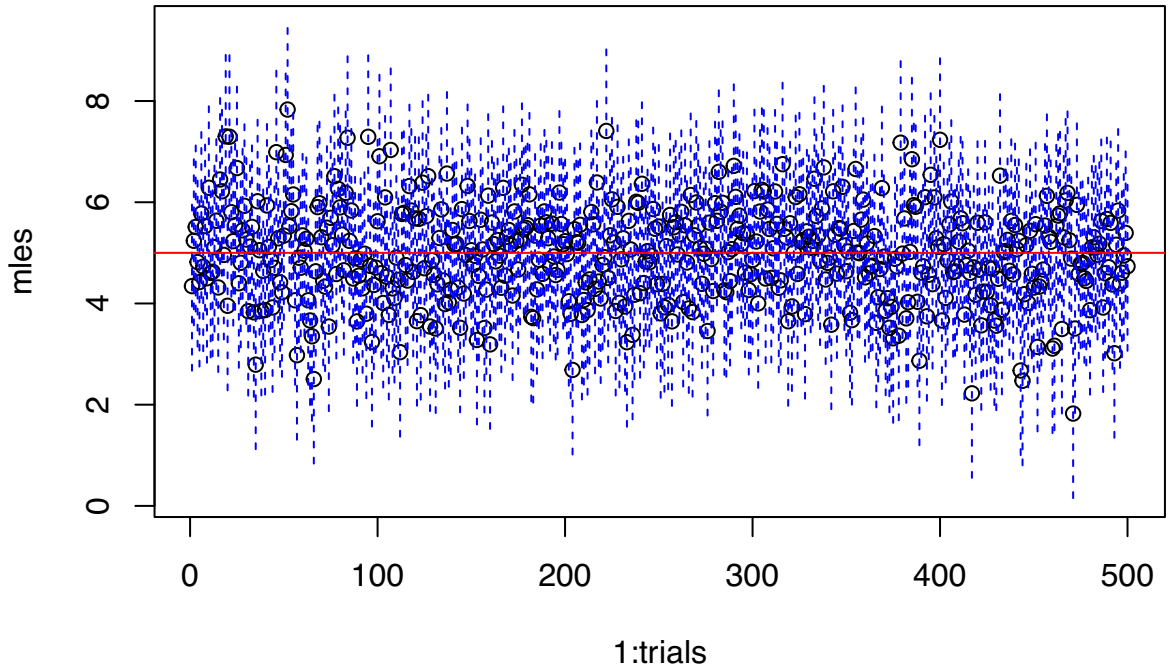
Now we can generate the observed confidence intervals.

```
n <- 20
mu <- 5
sigmaSq <- 17
sigma <- sqrt(sigmaSq)
trials <- 500
CI.L <- rep(0,trials)
CI.U <- rep(0,trials)
mles <- rep(0,trials)
```

```
for (i in 1:trials) {
  xs <- rnorm(n, mean=mu, sd=sigma)
  ci <- normal.mean.CI(xs, sigma, 0.07)
  mles[i] <- mean(xs)
  CI.L[i] <- ci[1]
  CI.U[i] <- ci[2]
}
```

We are now in a position to do some plotting.

```
plot(1:trials,mles,ylim=c(min(CI.L),max(CI.U)))
for (i in 1:trials) {
  lines(c(i,i),c(CI.L[i],CI.U[i]),col="blue",lty="dashed")
}
abline(h=mu,col="red")
```



We can see that many of the observed CIs do contain $\mu = 5$. We can get R to give us the proportion.

```
sum(CI.L < mu & CI.U > mu) / trials
```

```
## [1] 0.928
```

That's pretty close to $1 - \alpha$.

Incidentally, you know that the random variable $B = \mathbb{I}(\mu \in \mathcal{I}(\mathbf{X}))$ is a Bernoulli random variable with probability of success $p = 1 - \alpha$. Hence, you can also work out the variance of the random variable $\frac{1}{t}\sum_{i=1}^{t} B_i$, where the B_i are independent Bernoulli(p) random variables: $p(1-p)/t$. With t denoting the number of trials, this gives an indication of how accurate our estimator of the coverage is.

Case study 5: Wald Confidence Intervals in Practice

Juliette Unwin

Wald Confidence Intervals

A Wald $1-\alpha$ Confidence Interval for a statistical parameter $\theta \in \Theta = \mathbb{R}$ has the form $[L(\mathbf{X}), U(\mathbf{X})]$ where

$$L(\mathbf{x}) = \hat{\theta}_n(\mathbf{x}) - \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n(\mathbf{x}))}}, \qquad U(\mathbf{x}) = \hat{\theta}_n(\mathbf{x}) + \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n(\mathbf{x}))}},$$

or

$$L(\mathbf{x}) = \hat{\theta}_n(\mathbf{x}) - \frac{z_{\alpha/2}}{\sqrt{nJ_n(\hat{\theta}_n(\mathbf{x}))}}, \qquad U(\mathbf{x}) = \hat{\theta}_n(\mathbf{x}) + \frac{z_{\alpha/2}}{\sqrt{nJ_n(\hat{\theta}_n(\mathbf{x}))}}.$$

Weibull distribution

We consider here the Weibull distribution, which has PDF

$$f(x;\lambda,k) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} \exp\left\{\left(-\frac{x}{\lambda}\right)^{k}\right\} \mathbb{I}(x \geq 0).$$

The Weibull distribution is often used to model failure times, and is a generalization of the Exponential distribution. In particular,

- if $k = 1$ then the distribution is Exponential and the failure rate is constant over time
- if $k < 1$ then the failure rate decreases over time: this could be because if a component is defective it fails early
- if $k > 1$ then the failure rate increases over time: this could be because parts are more likely to fail the older they are.

Confidence intervals for $\theta = k$

We will assume for simplicity that $\lambda = 1$ is known and so the statistical parameter is $\theta := k \in \Theta = (0, \infty)$.

As usual, to do a simulation experiment we need to have a way to simulate Weibull random variables. We can do this with the `rweibull` function Similarly, we can use the `dweibull` function to compute (log) densities.

A function to compute the ML estimate is

```
ml.estimate <- function(xs) {
  # log-likelihood
  ell <- function(theta) {
    sum(dweibull(xs,shape=theta,log = TRUE))
  }
  # negative log-likelihood
  neg.ell <- function(theta) {
    -ell(theta)
  }
  # run the optimization algorithm
  optim(1,neg.ell,method="L-BFGS-B",lower=1e-5,upper=Inf)$par
}
```

Let's test that our estimator works.

```
test.xs <- rweibull(10000,3)
ml.estimate(test.xs)
```

```
## [1] 3.000908
```

Close enough!

In order to compute a Wald confidence interval we need either a function to compute the Fisher Information at a given θ or at least a function to compute the second derivative of the log-likelihood.

Using the Observed Information

Let us try the second approach first.

We find that the log-likelihood for one observation is

$$\ell(\theta; x) = \log f(x; \theta) = \log(\theta) + (\theta - 1)\log(x) - x^{\theta},$$

so the score is

$$\ell'(\theta; x) = \frac{1}{\theta} + \log(x) - x^{\theta}\log(x),$$

and the second derivative of the log-likelihood is

$$\ell''(\theta; x) = -\frac{1}{\theta^2} - x^{\theta}\log(x)^2.$$

Hence we have,

$$J_n(\theta; \mathbf{x}) = \frac{1}{\theta^2} + \frac{1}{n}\sum_{i=1}^{n} x_i^{\theta}\log(x_i)^2.$$

It follows that we can easily compute the observed information.

```
observed.info <- function(theta,xs) {
  # observed info is n * J_n
  n <- length(xs)
  n/theta^2 + sum(xs^theta*log(xs)^2)
}
```

Our left and right endpoints are therefore

```
compute.obs.info.CI <- function(xs,alpha) {
  theta.hat <- ml.estimate(xs) # compute the ML estimate
  o.info <- observed.info(theta.hat,xs) # compute the observed info
  z.alpha <- qnorm(1-alpha/2) # compute the appropriate N(0,1) quantile
  l <- theta.hat - z.alpha/sqrt(o.info)
  u <- theta.hat + z.alpha/sqrt(o.info)
  c(l,u)
}
```

We can see if this is working by running a simulation experiment.

```
# takes as input the function for computing observed CIs
coverage.ci <- function(ci.function,theta,n,alpha,trials) {
  inside <- 0
  for (i in 1:trials) {
    xs <- rweibull(n,theta) # simulate data
    ci <- ci.function(xs, alpha) # compute the observed CI endpoints
    # check whether theta is in the observed CI
```

```
    if (ci[1] < theta & theta < ci[2]) {
      inside <- inside + 1
    }
  }
  inside/trials
}
```

```
coverage.ci(compute.obs.info.CI, 3, 100, 0.05, 10000)
```

```
## [1] 0.9528
```

Using the Fisher Information

In general it is not necessary that the Fisher information is a function that can be expressed exactly using standard functions. In this case, it is possible but hard to derive. In particular,

$$\begin{aligned} I(\theta) &= -\int_0^\infty \ell''(\theta; x) f(x; \theta) \mathrm{d}x \\ &= \frac{1}{\theta^2} + \int_0^\infty x^\theta \log(x)^2 f(x; \theta) \mathrm{d}x \\ &= \frac{1}{\theta^2} \left(\frac{\pi^2}{6} + \gamma^2 - 2\gamma + 1 \right), \end{aligned}$$

where γ is the Euler–Mascheroni constant. I do not recommend that you spend time trying to derive the last equality!

```
gamma <- -digamma(1) # Euler--Mascheroni constant

fisher.info <- function(theta) {
  (pi^2/6 + gamma^2 - 2*gamma + 1)/theta^2
}

compute.fisher.info.CI <- function(xs,alpha) {
  theta.hat <- ml.estimate(xs) # compute the ML estimate
  fi <- fisher.info(theta.hat) # compute the Fisher info
  z.alpha <- qnorm(1-alpha/2) # compute the appropriate N(0,1) quantile
  n <- length(xs)
  l <- theta.hat - z.alpha/sqrt(n*fi)
  u <- theta.hat + z.alpha/sqrt(n*fi)
  c(l,u)
}
```

We can test again whether the coverage is decent.

```
coverage.ci(compute.fisher.info.CI, 3, 100, 0.05, 10000)
```

```
## [1] 0.9467
```

So this works, but it involved being able to find a computable expression for the Fisher Information.